

Reproducing the LHS 1140 b helium line with a solved wind: how far EXHALE gets, and what is missing

1. Goal, and the answer up front

The objective is to reproduce the measured He I 10830 Å transit line of LHS 1140 b with EXHALE, which solves the wind rather than imposing it, and to establish what (if anything) has to be added to the physics for the model to match the observation.

Everything below was re-solved on 2026-09-16, on the binary c2e9c9990b9f. The results of 2026-08-30 that earlier revisions of this memo carried are superseded twice over: those states were never stationary below $1.1 R_p$ under the residual the code now applies, and the campaign that replaced them on 2026-09-14 was itself solved with a base boundary that chose its entropy branch on the cell-1 velocity rather than on a flux, so that a “226 K base” run sat at 418 K in its first cell. Both are recorded in Sect. 2., with how far each number moved. Every EXHALE number below is read from the model tree of record, LHS1140b/models/, catalogued in LHS1140b/MODELS.md; a number that is still the archived one is marked *archived 2026-08-30* where it appears, and one from the superseded campaign *preL21*.

Result. The *amount* of absorption can be matched over a prescribed metal-free scalar base, at a composition that depends on what is assumed about the spectrum and about mixing. Over a lower atmosphere solved as a photochemical column and handed over as a profile, C, N and O included, it can no longer be matched from above: the lowest reservoir the elemental-flux closure returns, $\text{He}/\text{H} = 2.09$, already makes an equivalent width 33 per cent larger than the measured one, and the ladder rises from there. The composition that base would need lies below the ladder and has not been solved. The *width* is not matched over either base, and the velocity it asks for is the same number every time.

Two of the changes of Sect. 2. act together on the solved-column branch. The C, N and O line cooling of the six ions the photochemical column brings into the base was evaluated in the coronal limit, above the critical density of the levels it excites, and is now the CHIANTI statistical equilibrium at the cell’s own electron density. The closed-form fits it replaces also mixed theoretical with observed level energies, which at 2000 K is a factor 14 in $\exp(-\Delta E/kT)$ for O II. The base metals are therefore no longer a thermostat. We measured that at one composition, $\text{He}/\text{H} = 2.13$ over the scalar base, with and without the column’s C, N and O reservoirs: $T_{\text{max}} 5730 \rightarrow 5566$ K and the red-pair equivalent width $1.3749 \rightarrow 1.3721$ % Å, a change of 0.20 per cent. The factor 2.6 that the 2026-08-30 draft carried for this step is gone.

(i) The amount is always matchable, so what the line returns is a conditional pair rather than a measurement. Take the prescribed-base branch first. With the base a scalar state carrying no metals, every combination of assumed spectrum and assumed mixing reaches the observed red-pair equivalent width at some helium abundance, and the conditions stack inside this branch. In the well-mixed limit the crossing is $\text{He}/\text{H} = 0.4044$ (0.3877–0.4237 on the measurement’s 1σ interval) on the GJ 1132 proxy spectrum and 0.0526 (0.0510–0.0543), below solar, on the GJ 699 one: the second stand-in for LHS 1140’s unobserved XUV, 5.1 times brighter, which the paper also uses. With binary H/He element diffusion active (helium then settles out of the base unless an eddy term lifts it) it rises again, to 3.10 wherever $K_{zz} \leq 10^5 \text{ cm}^2 \text{ s}^{-1}$ does nothing against the molecular coefficient, 1.516 at the adopted $K_{zz} = 10^9$ and 0.893 at 10^{11} (Sect. 9.4). The

dependence on the eddy coefficient is weak and monotonic, as $K_{zz}^{-0.115}$ over the four decades where the eddy term acts at all (a least-squares fit to the four crossings at 10^8 to 10^{11}); the dependence on the assumed spectrum is a factor 7.7. The solved-column branch no longer brackets the measurement. Replacing that base with a photochemical column (Photochem with its climate step solved, handed to EXHALE as a profile at a $1 \mu\text{bar}$ match, at the same $K_{zz} = 10^9$ and with the same diffusion active) brings C, N and O into the base. Nine reservoirs were closed on their own elemental fluxes, and the equivalent width they return runs from $1.4765 \text{ \% } \text{\AA}$ at $\text{He}/\text{H} = 2.0924$ to $2.2289 \text{ \% } \text{\AA}$ at 12.0137 : every one of them stands above the measured 1.108 ± 0.030 , the lowest by 33 per cent (Sect. 12.7). There is no crossing on this ladder. Read against the ladder's own local slope, $d \ln EW / d \ln(\text{He}/\text{H}) = 0.49$ between the two lowest rungs, the measurement would be reached near $\text{He}/\text{H} \approx 1.1$, which is below every rung solved and is an extrapolation, not a solution. The earlier crossing at 8.212 was a consequence of the base metals cooling the wind; they no longer do. Neither branch is an independent prediction of the model. The prescribed-base number is the reservoir the observed line picks out over its own base, and the escape rate (which is an output, not a fit) barely moves along the way. The flux closure does not change that on the profile branch: it returns He/H at the match from trial fluxes a factor of ten apart, single-valued to a few parts in 10^6 (Sect. 12.), but at $K_{zz} = 10^9$ the eddy term homogenizes the column all the way to the microbar match, so the closure hands back very nearly the reservoir it was given. What the closure adds is that a composition is consistent with a photochemical lower atmosphere carrying its own C, N and O, not that it is measured. The same helium now absorbs almost equally over the two bases. On the new tree, at essentially one composition, the scalar metal-free base at $\text{He}/\text{H} = 2.13$ and the closed solution on the photochemical column at 2.0924 give: $T_{\text{max}} 5730$ against 5415 K, the metastable peak 34.6 against 35.8 cm^{-3} , the metastable column over $2\text{--}20 R_p$ 45.2 against 45.7 cm^{-2} in units of R_p , and the equivalent width 1.3749 against $1.4765 \text{ \% } \text{\AA}$. The photochemical column now makes the *stronger* line, by 7.4 per cent, where the 2026-08-30 solutions made it weaker by a factor 2.6. Raising the reservoir still reheats the wind, and far more strongly than before: at $\text{He}/\text{H} = 10.01$ the mean mass per particle at $2R_p$ is $3.17 m_H$ against 2.15, the same deposited energy is shared among fewer and heavier particles, and T_{max} rises to 7682 K (Sect. 12.6, Fig. 17). The escape rate is flat on every axis, and the crossings span a factor 59. $\log_{10}(\dot{M}/\text{g s}^{-1}) = 7.84\text{--}7.87$ on the GJ 1132 proxy across composition, diffusion and eddy coefficient alike, $7.90\text{--}7.91$ across the whole closed ladder, and $8.42\text{--}8.57$ on the GJ 699 proxy. Against that, the composition the line implies runs from 0.0526 to 3.10 over choices this measurement cannot arbitrate: strongly conditional on the assumed spectrum and on what is placed at the lower boundary, weakly and monotonically on the assumed mixing. Any composition read off this line (the paper's $\text{H}:\text{He} \lesssim 10^{-3}$ as much as any value here) inherits those conditions.

(ii) The width is not reached on any axis, and the deficit is invariant. At the equivalent-width crossing the profile is 3.3 times too deep and 3.3 times too narrow (red depth 4.085 against the measured 1.254 per cent, FWHM 0.2533 against $0.841 \text{ \AA}$), and no composition fixes it, because composition changes how much helium absorbs, not how the absorption is spread in velocity. Measured directly (convolving the solved profile with a Gaussian and asking which one reproduces the observed line) the missing spread is 22.22 km s^{-1} FWHM ($\sigma = 9.44 \text{ km s}^{-1}$) in the well-mixed limit, 22.22 km s^{-1} with diffusion and the adopted K_{zz} , 22.15 km s^{-1} for the flux-closed solution on the solved photochemical column, and 21.97 km s^{-1} on the $5.1 \times$ brighter spectrum: a spread of 1.1 per cent over four solutions that differ by a factor 40 in composition (Sects. 9.2, 9.4, 12.7 and 11.).

Supplied as one Gaussian it puts depth, blue depth and red/blue ratio inside the observed 1σ interval at once, with

nothing tuned to them; what produces it is unidentified. It is not thermal (the solved gas cools below 1000 K over the line-forming region), and not bulk outflow, which reaches 1.1 km s^{-1} at $20R_p$ on the fainter spectrum and 3.6 on the brighter. Read as turbulence it is $\mathcal{M} = 2.3\text{--}4.0$ against the sound speed these solutions themselves carry, supersonic on every convention and well above what the literature is willing to assume. The one mechanism that would supply width without a free parameter (the adiabatic-cooling metastable bump of Schulik & Owen (2025)) is present in these solutions and measured, but sits at $0.4\text{--}2.5 \text{ km s}^{-1}$ (Sect. 9.6). The shortfall is in the energy and momentum budget, not in the composition, and it is a published problem rather than a defect of this setup (Sect. 9.).

The paper's own p-winds fit does not reproduce the line either. It is broader than the wind solved here, because its temperature is imposed at 5000–6000 K where the line forms, but at FWHM $0.575 \text{ \AA}$ with its turbulence term on (0.458 without it) it is still 1.5 times narrower than the measured 0.841, and it overshoots the blue component threefold. Adding broadening does not rescue it: its absorbing power is 24% below the measurement, so no convolution can reach the observed depth and width at once. The width is a shortfall the two codes share. That is also what the audit of the measurement side (Sect. 9.3) leaves: the measurement is not at fault, and neither one-dimensional steady-state wind reproduces it.

Two caveats attach to the outer wind rather than to the line. The first is that the domain stops inside the critical point. These solutions are the subsonic part of a transonic escape whose critical point sits at $40.06R_p$, outside the $30R_p$ domain they are solved on, so the outer ghost rule and not a critical point is what closes them. It costs the mass flux almost nothing: $\log_{10} \dot{M} = 7.8737, 7.8740, 7.8735$ at 30, 45 and $60R_p$, a spread of 5×10^{-4} dex over a factor two in domain size, and the He I 10830 equivalent width moves by at most 1.7 per cent. What it does cost is the outer profile: at $28R_p$ the $30R_p$ solution is 25 per cent too hot, 29 per cent too dense and 22 per cent too slow against the converged answer, so nothing above about $10R_p$ in these profiles should be quoted (Sect. 13.).

The second is that the gas is not collisional out there either: Kn_{bulk} passes 0.1 at $7.30\text{--}7.44R_p$ and reaches 0.68–0.71 at $29R_p$, while the He 10830 layer at $1.1\text{--}3R_p$ sits at $\text{Kn} \leq 0.012$ and the line analysis is untouched.

One further caveat attaches to the composition. Above about $4R_p$ the ionization state of this wind is frozen into the flow: the recombination and photoionization Damköhler numbers fall to 10^{-2} and below, so the photoionization equilibrium the wind is *solved* with is not the composition that gas carries. The advection-corrected `_adv` profiles the transit tool consumes say so, and the difference is not small: the neutral hydrogen fraction differs by a factor 4.5 at the outer boundary, and the He I 10830 equivalent width read from the solution's own composition is $15.51 \% \text{ \AA}$ against 1.17 from the corrected one, a factor 13.3. Every line number in this memo is read from the `_adv` profiles; the solution's own composition is not usable above $4R_p$ and is not quoted. Carrying the three ionization fractions as transported variables is designed but not implemented (Sect. 2.).

Three systematics ride on the absolute numbers, and a convergence test catches none of them. They are stated here because they bound how far any single crossing should be read.

1. The outer region is inherited from the seed and is not resolved by the domain. A converged wind takes its $20\text{--}30R_p$ state from the solution it was started from, and $\|R\|$ cannot separate two such states. Sect. 13. bounds what that costs the line: moving the outer boundary to 45 and $60R_p$ moves the equivalent width by at most 1.7 per cent and the mass flux by 5×10^{-4} dex, while the profiles above $10R_p$ move by 22–29 per cent.

2. The closure tolerance is visible in the answer. It stops at $\text{tol} = 0.05$ on the undamped elemental residual, and the rungs of the ladder end at 0.045 (helium) and 0.018 (hydrogen). The driver refuses to tighten it further, because the radial spread of the flux over the measurement window, 3.7×10^{-3} in hydrogen and 4.9×10^{-3} in helium, would then approach the residual being decided on it.
3. Which chemistry produced the column. Every column here is Photochem 0.9.0 with the local corrections of docs/photochem_solver_modification_*.md.

Everything below uses inputs verified against the authors' own released material (Zenodo 20723095): their p-winds driver script, their Mega-MUSCLES v23 spectra, their reduced transmission spectrum and their likelihood grid. Products live in LHS1140b/; the notebook LHS1140b/LHS1140b_analysis.ipynb regenerates every number and figure here.

2. What changed since 2026-08-30

Every EXHALE number in this memo was re-solved between 2026-09-13 and 2026-09-16. This section states why, what was changed in the code and in the recipe, and how far the answers moved. The campaign itself, case by case, is LHS1140b/MODELS.md. The plan it executed is docs/PLAN_20260913_lhs_stationary.md, and its measurements are the items docs/lhs1140b_stationary_L*.md.

The catalog was solved twice. The campaign of 2026-09-14 was superseded three days later by the base-boundary correction of item L21 (Sect. 2.8), and every case was solved again. *The catalog of record* is LHS1140b/models/, solved on one binary whose md5 begins c2e9c9990b9f and whose source manifest sits beside the tree; Of the 86 atomic cases in the tree, 79 certify on it, each in one outer pass from its own previous state; the seven that do not are the XUV-grid gaps of Table 3. The results of 2026-09-14 are kept unchanged in models_20260914_preL21/, the intermediate generation in models_20260915_db87/, and the three are compared case by case in docs/lhs1140b_rerun_20260915.md. A number labelled *preL21* below is from 2026-09-14; everything else is the final pass.

2.1 Why the 2026-08-30 solutions were discarded

They were not stationary. Taken apart row by row on the current residual (item L1), the archived state carries an $O(1)$ residual everywhere between the base and about $1.1 R_p$. In every cell below $1.049 R_p$ the energy row is dominated by its own flux divergence rather than by its radiative terms ($|dF_3|/|\Gamma - \Lambda| = 7 \times 10^5$ at cell 1, 4.9×10^1 at cell 16), and the momentum row is out by a constant 2.8 per cent of the local weight over the whole column. The residual belongs to the state and not to the diagnostic. The base-layer velocity sawtooth contributes nothing to the energy row, and putting the archived heating back into the same hydrodynamic state lowers the worst cell only from 1.979 to 1.930. The old runs reported convergence because their flux gate began at $1.2 R_p$ and never looked below it.

Mapping such a state onto the current grid and re-solving does not repair it. All four routes tried stagnate: the JFNK solve at $\|R\| = 1.6\text{--}2.0$ for 56 iterations, the marching restart at a mass-flux spread du that never falls below the 10^{-2} hand-off in 5600 steps, the cold two-stage march at $du = 9.5$ after 954 steps, and the partitioned stationary route at $\text{info} = 1$ or 2 with the composition relaxation finding no admissible advance (MODELS.md Sect. 6).

Separately, the code has moved in ways that reach this wind directly. The changes are the photon-grid quadrature (a 9.5 per cent overcount of the H I photoionization rate removed), the cell width dr_j , the Jupiter radius (+2.26 per cent in R_0), the helium mass, the CODATA constants, the spectrum below 13.6 eV that the $\text{He}(2^3S)$ population depends on, and the ghost-row fix of the transit reader. The new Jupiter radius also changes the grid, so no archived state can be reloaded without interpolation.

2.2 The well-balanced reconstruction

No case in this campaign certifies without the well-balanced reconstruction. At the base of this planet the flow's own Mach number is 2.7×10^{-7} . The contact speed the HLLC flux puts through a base face is then set by the WENO3 reconstruction's third-order truncation error in the face pressure rather than by the velocity, which that error exceeds by one to three decades. The energy row of every base cell therefore measures its own largest term against itself, and no seed can lower it. A relative density jump d at a face enters the interface flux through its acoustic part as $d \rho c_s$, which is $d/\mathcal{M}$ times ρv . At $\mathcal{M} \sim 10^{-6}$ a jump of one per cent is a six-decade error in the flux.

Well balanced: True is the reconstruction of Käppeli & Mishra (2016). Every cell carries its own hydrostatic equilibrium, the reconstruction and the Riemann solver act on the *departure* from it, and the momentum source is the face-pressure difference of that same equilibrium. At a discrete hydrostatic equilibrium the flux difference and the source then cancel in the algebra instead of in floating point. The term-by-term statement, the mapping onto the code and the reason the key is off by default are in docs/well_balanced_and_hydrostatic_20260913. The table below shows what it does here, on the same seed, the same input and the same binary (item L5c).

Table 1: Hydrodynamic rows of the first outer pass (cellwise maximum of $|R|$ over the row's own largest term) and the outcome of the stationary solve on the reference case, $K_{zz} = 10^9$, $\text{He}/\text{H} = 1.6261$.

option	hydro	mass	momentum	energy	outcome
none	info= 2	1.08	3.2×10^{-2}	0.63	pass refused
Low-Mach damping: 2.0e-2	info= 2	1.32	1.4×10^{-2}	1.24	pass refused
Well balanced: True	info= 0	2.6×10^{-10}	1.8×10^{-14}	4.6×10^{-9}	certified at pass 8

Low-Mach damping does nothing to the energy row because it is a fourth difference of the *velocity* and does not touch the pressure truncation error. Three seeds whose base velocities differ by a factor 10^4 reach one solution (ρ to 3.4×10^{-10} , v to 2.5×10^{-8} , $\dot{M}$ to seven digits), so the campaign's answers do not depend on which solved state seeded them.

2.3 The secondary-ionization coupling, applied from the first evaluation

Secondary_ionization defaults to STAGED, which arms the Shull & van Steenberg (1985) channel only when a march first converges. The stationary route takes no time step, so a state solved under the default is written with the coupling off. The post-processing pass applies it unconditionally, and every number quoted here comes from that pass, so it reports the same wind with the coupling on. On the $K_{zz} = 10^9$, $\text{He}/\text{H} = 1.6261$ case we measure $\log_{10} \dot{M} = 8.29$ under

the default against 7.87 with `Secondary_ionization: Immediate`, and $n(\text{He } 2^3S)$ differs by 74 per cent beyond $1.5R_p$, where the line forms. Every case of the campaign carries the key.

2.4 The C, N and O line cooling at the local electron density

Six coolants the photochemical column brings into the base (C I, C II, N I, N II, O I and O II) were evaluated in the coronal limit, that is above the critical density of the levels they excite. Against the CHIANTI statistical equilibrium of the same ion, the coefficient the cooling assembly took stood at 0.02 to 5.4×10^4 times the right value over the wind-relevant domain. The replacement is a density-resolved table read at the cell's own electron density, and it stands at 0.969 to 1.009 at every point of a 7×6 grid spanning 1000 to 20000 K and 10^4 to 10^9 cm^{-3} (item L6b).

A second defect was found in the closed-form fits those six terms used, and is removed from the production path with them. The fits were built from the CHIANTI .scups *theoretical* transition energies, while the level populations that decide the emission use the *observed* level energies. The two differ by up to 14 per cent in ΔE for the O II $4S^{\circ}-2D^{\circ}$ excitation, which at 2000 K is a factor 14 in $\exp(-\Delta E/kT)$.

This is the change that reshaped the solved-column branch of this memo. The metal content of the base is no longer a thermostat. At $\text{He}/\text{H} = 2.13$ over the scalar base, adding the column's C, N and O reservoirs moves T_{max} from 5730 to 5566 K and the red-pair equivalent width from 1.3749 to 1.3721 % Å.

2.5 The outer boundary

The LHS 1140 b wind is subsonic at $30R_p$, so the free-outflow continuation of the outer ghost is not a passive extrapolation: one characteristic points inward, and the ghost rule is the external condition. Solving the same case on 45 and $60R_p$ domains settles what that costs (item L8). The wind is transonic, with its critical point at 40.06 and $40.08R_p$ in the two extended runs, and once the boundary is outside that point the two solutions agree with each other to better than 0.1 per cent everywhere.

The $30R_p$ answer holds for the quantities taken from it and not for the rest: $\log_{10} \dot{M} = 7.8737, 7.8740, 7.8735$ at 30, 45 and $60R_p$ and the He I 10830 equivalent width within 1.7 per cent, against 25 per cent in temperature, 29 in density and 22 in velocity at $28R_p$. No profile above about $10R_p$ is quoted from these runs. The exobase of the extended solutions sits at $29.0R_p$, below the critical point, so the continuum model fixes its own topology in gas it cannot describe. The insensitivity of the mass flux is a statement about the discrete problem and not a physical validation.

2.6 The composition above $4R_p$

The wind is solved with a photoionization-equilibrium composition, and above about $4R_p$ that composition is not the one the gas carries. Over a flow time r/v the hydrogen recombines 1.7 times at $2R_p$ and 5×10^{-3} times at $29R_p$, and is photoionized 7×10^{-2} and 10^{-2} times at the same radii (item L10). The ionization state is frozen into the parcel and carried outward. The equilibrium neutral fraction falls from 0.963 at $2R_p$ to 0.346 at $29R_p$ while the advection-corrected one goes from 0.974 to 0.941, and the photoheating that follows the neutral density stands a factor 5.9 apart at $44R_p$.

The consequence for the line is a choice of file rather than a correction. The He I 10830 equivalent width read from the solution's own composition is 15.5116 % Å against 1.1702 from the advection-corrected one, a factor 13.3. Feeding the transit tool the solution's *temperature* with the corrected composition instead moves it by 0.15 per cent (item L12a). The line depends on the composition and hardly at all on anything else in those two files, and every line number here is read from the `_adv` profiles. Carrying $x(\text{HII})$, $x(\text{HeII})$ and $x(\text{HeIII})$ as transported fractions per element nucleus is designed (item L12a) but not implemented.

One defect was found and fixed on the way. The advection correction discarded MINPACK roots returned with `info` = 4 even where the iterate satisfied the energy equation to 10^{-17} of the largest term the equation holds, and because the correction is an upwind recursion the discard restarted the temperature profile at every row above it. With the roots accepted by their residual the 45 and 60 R_p corrected temperatures agree to 0.19 per cent, which is the 0.28 per cent at which the two *solutions* agree.

2.7 The convergence procedure at low XUV

Five cases of the XUV grid did not certify on the first campaign, and the reason is in the outer loop rather than in the wind (item L14). The alternation is “solve the wind at fixed composition, relax the composition at fixed wind”. Its own error is the undamped distance the element relaxation returns, which the under-relaxation contracts by $1 - \omega$ per pass. The progress control reads a different quantity, the distance of the state from certification. Where that distance fails to fall it halves ω , taking the contraction from 2.00 to 1.14 per pass, and after three such passes it ends the loop. On these cases the two quantities disagree from the first pass: the composition distance falls every pass while the gated elemental He/H row rises. The certified case at 0.30 of the same spectrum is the control: it entered at the same composition distance, its gated row also failed to fall over its first pass, and it certified at outer pass 13 with ω at 0.500 throughout. The cell-1 hydrodynamic energy row was not what refused the five. At the last pass of the worst of them the three hydrodynamic rows read mass 2.30×10^{-8} , momentum 3.39×10^{-14} and energy 8.49×10^{-7} of 10^{-6} , and the gated elemental row read 2.12×10^{-2} of 10^{-5} .

On the corrected base boundary these cases enter further still from their fixed point, and the remedy is the one `run_case.sh` documents at its own EXHALE_PTC_DTAU0 default: start the pseudo-time at 10^8 instead of letting the ramp collapse from 1. MEASURED on five scalar cases, that turns hours of refusing into minutes of certifying: `x0.20/HeH2.13` went from four outer passes in 6 h 22, still refusing, to certified in 17 minutes. Nine of the thirteen cases treated this way certified within the hour. *Four did not*, and they are the gaps in Table 18: `atomic_scalar_gj1132x0.10_kzz1e9/HeH9.7` and the photochemical cases at 0.15, 0.20 and 0.25 of the fiducial spectrum keep the same behaviour at $d\tau_0 = 10^8$, the ramp falling back to $d\tau$ of order ten while the cell-1 mass row holds $\|R\|$ near 1.5. Raising the start is not what they need. Separately, the three cases at 0.01 of the fiducial have never solved on any boundary. Item L17 attributes that to the Newton model of a wind subsonic to Mach 10^{-8} over its whole lower atmosphere; it measured the finite-difference Jacobian and the certification anchor and cleared both.

2.8 The base boundary, which chose its branch on the wrong quantity

This is the change that forced the second campaign, and it is the largest single movement in the tree. The base face of a certified LHS 1140 b state is an *inflow* face carrying the wind’s own mass flux, and the boundary condition decided whether to treat it as inflow or as reversal on the cell-centred product $\rho_1 v_1 r_1^2$ of the first interior cell. At this base that product is not a flux: it is the collocated odd-even velocity mode of item P44. MEASURED on `atomic_scalar_gj1132_kzz1e9/HeH2.13`, the Riemann face mass flux is $+1.00000 F_{\text{wind}}$ at every one of the 500 faces including the base face (largest departure 1.6×10^{-9}), while the cell-centred product reads $-2.00 F_{\text{wind}}$ at cell 1 and -2.27 at cell 2. The boundary was therefore running 97 per cent in the reversal branch, where the face density is the interior’s isentrope rather than the reservoir’s, and the two differ by $\rho_{\text{rev}}/\rho_{\text{res}} = 0.531$ at the common face pressure. `base.inp`’s 226 K reservoir was entering the state with a weight of 0.026, and the first cell of a “226 K base” run sat at 417.6 K.

No choice of blend width repairs that, because the discriminant has the wrong sign: a narrower window takes the reversal branch more completely. Keying the branch on the mass flux the state carries is the fix. A second error sat beside it: the blend window was 10^{-6} in face Mach number, chosen against the hot-Uranus base, while LHS 1140 b’s own base operating Mach number is 4.6×10^{-7} , *below* the window, so even with the sign repaired a fifth of the reservoir’s entropy would be blended away on a clean inflow. The window is now 10^{-8} . On the same state the branch reads $M_{\text{branch}} = +4.02 \times 10^{-7}$ and a reversal weight of zero: the reservoir states the entropy of the gas that enters, which is what the boundary exists to do.

What it did to the catalog, over the 79 cases that certify both ways (`docs/lhs1140b_rerun_20260915.md`): the first physical cell falls by a median 46 per cent in temperature and rises by a median factor 1.82 in density, $\dot{M}$ falls by a median 2.0 per cent with no case rising, and the red-pair equivalent width falls by a median 2.0 per cent. On the fiducial case the first cell goes from 417.6 to 238.0 K. The profiles cross near cell 21 and agree to a fraction of a per cent in the outer wind. Four rungs of the GJ 699 ladder are bit-identical across the correction, because on that weak spectrum the interior isentrope at the base already *was* the reservoir, so the branch the old boundary took wrongly happened to give the right state: the correction changes what was wrong and nothing else.

2.9 What a written state means when it is read back

Two items tightened the contract between a state and the files that hold it, and neither changes physics.

A certified state did not reproduce its own certification when it was read back (item L18). The pair of files states the density twice, as the `rho` column of `Hydro_ioniz.txt` and as the species densities of `Ion_species.txt`, and the two agree only while the composition closes its own mass. It does not close exactly, and the loader took the species as the authority, so the state handed to the evaluation stood some 2000 units in the last place from the one the solve had certified. At this base the hydrodynamic rows cancel their largest term by 6.2×10^5 , and at that cancellation 2000 ulp of density is the whole row. The loader now takes the density from its own column and projects the loaded species onto it, one factor per cell, which leaves every element ratio and every ionization split where the file put them. One unit in the last place of the density still moves the cell-wise maximum of the energy row by about 11 per cent, so *bitwise* reproduction of a certification is not available and the contract says so.

The composition a sweep returns is now put on the density the sweep was given, and the mass closure $\max_j |\sum_i f_i A_i - 1|$ is reported by the certification (item L19). The companion question, whether the cool column of a written state is the cooling of that state, was posed as a defect and is negative (item L20): on every state the current code produces the column and the total of `Cooling_breakdown.txt` agree to between 3.4×10^{-15} and 3.0×10^{-10} . The $1.8\times$ discrepancy that raised the question was measured on a regression fixture pinned from an earlier code generation. What was missing was the assertion, which is now there.

2.10 The molecular rate network, and the pseudo-time ramp

Item L7f asked which chemistry is right about the base layer, where the lower-atmosphere handoff stands 7 to 21 times above the root of the wind network's own H_2 carrier row. It retired reaction R20, added the channel of Black (1978), and rewrote R17. Those repairs are in the binary every number here stands on. They move the molecular regression cases and leave an atomic wind alone, so nothing in Sects. 8. to 13. depends on them.

Item L4h took both ends of the pseudo-time ramp: a shrink on damped steps that could pin the pseudo-time at its floor, and a growth that could climb to 10^{14} while the residual stood at 1.9. The delivered rule clips the cut of an accepted step at ten and floors it a fixed span below the pseudo-time the solve has already earned at a whole step, with a cap on how many damped steps may run in a row. Turning the guard off reproduces the earlier arithmetic bitwise. What it does to the XUV grid of this planet is measured below.

2.11 How far the answers moved

Table 2: Three generations of the same quantities. The archived column is read from the record the 2026-08-30 runs left (LHS1140b/WORKPLAN.md); *preL21* is the campaign of 2026-09-14, read from LHS1140b/models_20260914_preL21/; *final* is the catalog of record, read from LHS1140b/models/. The intermediate generation is not tabulated: it agrees with the final one to seven decimal digits or better in every entry (docs/lhs1140b_rerun_20260915.md).

quantity	archived 2026-08-30	preL21	final
well-mixed crossing He/H, GJ 1132	0.4132	0.4012	0.4044
well-mixed crossing He/H, GJ 699	0.0483	0.0526	0.0526
crossing He/H, $K_{zz} \rightarrow 0$	3.81	2.72	3.10
crossing He/H, $K_{zz} = 10^9$	1.6108	1.4922	1.5161
crossing He/H, $K_{zz} = 10^{11}$	0.837	0.878	0.893
crossing He/H, flux-closed ladder	8.212	none	none
EW of the closed rung at He/H $\simeq 2.09$ [% Å]	0.545	1.5006	1.4765
T of the first cell, fiducial case [K]	–	417.6	238.0
$\log_{10} \dot{M}$, GJ 1132	7.7–7.8	7.84–7.88	7.84–7.87
$\log_{10} \dot{M}$, closed ladder	7.40–7.41	7.91–7.92	7.90–7.91
$\log_{10} \dot{M}$, GJ 699	8.58	8.43–8.57	8.42–8.57
matched Gaussian FWHM [km s ⁻¹]	21.95–22.23	21.97–22.22	21.97–22.22

Two things follow from the table. Against 2026-08-30 the prescribed-base branch moved by 2 to 19 per cent in composition and by about 0.1 dex in $\dot{M}$: its structure is unchanged and its numbers are revised. In the solved-column branch the sign of the effect reversed, since with the base metals no longer cooling the wind the photochemical column makes a *stronger* line than the scalar base at the same composition, and the ladder that used to straddle the measurement now stands entirely above it. Against preL21 the movement is smaller and it is all one thing: the base cell is the stated reservoir instead of the interior isentrope, so every case is a little cooler and denser at its first cell, a little smaller in $\dot{M}$, and a little weaker in the line. The crossings walk *up* by 1 to 14 per cent, most at $K_{zz} \leq 10^5$ where the ladders are shallowest and a 2 per cent shift in equivalent width buys a long step in composition.

2.12 The campaign, group by group

Table 3 collects the whole campaign on one page: twenty-one groups of the model tree, the reservoirs solved in each, where that group's own rungs cross the measured equivalent width, and the range of the escape rate over the group. It indexes the sections that follow, and it also records what is not solved: seven cases of the XUV grid, three of them at 0.01 of the fiducial and four between 0.10 and 0.25, and the molecular groups, which are still being worked and whose solved cases are recorded as they stand.

Two of the molecular cases do now certify, and they are the first LHS 1140 b winds solved with H_2 , H_2^+ , H_3^+ and HeH^+ carried through the wind rather than left at the handoff: $\text{He}/\text{H} = 2.13$ at $\log_{10} \dot{M} = 7.91$, red-pair $EW = 1.5758\%$ Å and depth 5.63 per cent, and 9.7 at 7.95, 2.4212 Å and 7.68 per cent, both with element diffusion at $K_{zz} = 10^9$. Both stand above the measured 1.108, as the flux-closed rungs do. The 0.55 rung of the same group is a marching stop rather than a certified state ($\log_{10} \dot{M} = 7.64$, $EW = 0.1591\%$ Å), so no crossing is read from this group here even though two of its rungs straddle the measurement. The remaining molecular cases are uncertified or in flight and are not quoted.

The catalog itself, with the input keys of every group and the provenance of every case, is LHS1140b/MODELS.md. Its Sects. 7 and 8 are written by models/status.py from the case directories and are the machine-readable form of this table.

3. Inputs, and how each was pinned

3.1 System

Planet and star from Cadieux et al. (2024) as quoted by C26: $R_p = 1.730 \pm 0.025 R_\oplus$, $M_p = 5.60 \pm 0.19 M_\oplus$, $T_{\text{eq}} = 226$ K, $a = 0.0946$ AU, $M_\star = 0.1844 M_\odot$, $R_\star = 0.2159 R_\odot$. One trap: EXHALE converts R_p with its own internal constants (mean radii, $R_J = 6.9911 \times 10^9$ cm, $R_\oplus = 6.3725 \times 10^8$ cm), so the input must be built with those rather than with IAU equatorial values. The wrong pair costs 2.2% in radius and 4.4% in transit depth.

3.2 Lower boundary condition

EXHALE anchors the lower boundary at the planetary radius R_p using the standard CETIMB pressure-anchored condition (Base BC: pressure 1.0), fixing the base pressure at $P_0 = 1.0 \mu\text{bar} = 1.0 \text{ dyn cm}^{-2} = 1.0 \text{ erg cm}^{-3}$ (0.1 Pa). The

Table 3: Every group of LHS1140b/models/ and what it returns, on the binary c2e9c9990b9f. The crossing is where the group’s own solved rungs reach the measured red-pair equivalent width, $1.108 \pm 0.030 \text{ \% } \text{\AA}$, and $\dot{M}$ is the range over the group. The cases at 0.01 of the fiducial XUV have never solved (item L17); the photochemical cases at 0.15, 0.20, 0.25 and the scalar He/H = 9.7 case at 0.10 stopped solving on the corrected base boundary (Sect. 12.9). In the molecular groups three cases are in flight at the time of writing and are marked so; the carrier balance they wait on is item L7e of docs/PLAN_20260913_lhs_stationary.md.

group	rungs solved	crossing He/H	$\log_{10} \dot{M}$
<i>prescribed scalar base, GJ 1132 proxy</i>			
well mixed	0.083–1000 (9)	0.4044	7.84–7.86
$K_{zz} = 0$	0.55–3.9 (6)	3.1010	7.84–7.87
$K_{zz} = 10^5$	2.6–3.93 (5)	3.0706	7.87
$K_{zz} = 10^6$	0.55–3.74 (6)	2.8930	7.84–7.87
$K_{zz} = 10^7$	0.55–3.18 (4)	2.5365	7.85–7.87
$K_{zz} = 10^8$	0.55–2.41 (4)	1.9729	7.85–7.87
$K_{zz} = 10^9$	0.55–9.7 (7)	1.5161	7.85–7.87
$K_{zz} = 10^{10}$	0.55–1.29 (4)	1.1579	7.86
$K_{zz} = 10^{11}$	0.55–0.93 (4)	0.8927	7.86
$K_{zz} = 10^9$, C/N/O base	2.13 (1)	-	7.87
<i>prescribed scalar base, GJ 699 proxy</i>			
well mixed	0.042–1000 (6)	0.0526	8.42–8.57
<i>photochemical column, flux closure</i>			
$K_{zz}(p)$ from the profile	2.09–12 (9)	none: every rung above	7.90–7.91
<i>XUV grid, lower atmosphere held fixed</i>			
scalar, $K_{zz} = 10^9$	0.10–0.33 $\times$ at He/H 2.13; 0.15–0.33 $\times$ at 9.7 0.01 $\times$, and 0.10 $\times$ at 9.7: not solved	depth limit at 0.28 $\times$	6.82–7.36
photochemical column	0.10, 0.30, 0.33 $\times$ at He/H 9.7 0.01, 0.15, 0.20, 0.25 $\times$: not solved	limit below 0.10 $\times$	6.84–7.39
<i>molecular chemistry in the wind</i>			
scalar base, well mixed	0.083, 0.55 uncertified; 2.13 in flight	-	-
scalar base, $K_{zz} = 10^9$	2.13, 9.7 certified; 0.55 a marching stop; 0.083 uncertified	-	7.91–7.95
photochemical column	2.09, 9 in flight	-	-

corresponding base total number density n_0 follows from the ideal gas law at the equilibrium temperature $T_0 = T_{\text{eq}} = 226.0 \text{ K}$:

$$n_0 = \frac{P_0}{k_B T_0 n_{\text{tot},\text{bc}}} = \frac{1.0 \text{ dyn cm}^{-2}}{(1.380649 \times 10^{-16} \text{ erg K}^{-1}) \times 226.0 \text{ K}} = 3.20485 \times 10^{13} \text{ cm}^{-3}, \quad (1)$$

where $n_{\text{tot},\text{bc}} = 1.0$ for neutral atomic gas. In logarithmic form, $\log_{10}(n_0/\text{cm}^{-3}) = 13.506$, which is entered in `input.inp` as `Log10_lower_boundary_number_density [cm^-3]: 13.506`. When Base BC: pressure is specified, EXHALE re-evaluates and enforces this relation internally (`input_read.f90`).

3.3 Stellar spectrum

The paper describes its normalization two ways that cannot both hold: the Supplement's prose says the proxy SED is scaled so its integrated X-ray flux matches that measured for LHS 1140, while the quoted constants ($A_{1132} = 0.59$, $A_{699} = 0.056$) are the catalog X-ray flux *ratios*, a factor 1.8 apart. The authors' script settles it:

```
Lx_ratio_1132 = 0.59
norm_constant_1132 = Lx_ratio_1132*(d_1140/(smax_1140*au2m))*2
```

applied to the plain `const-res v23` product. That combination also reproduces the fiducial XUV flux the main text quotes, 0.033 W m^{-2} (we measure $33.6 \text{ erg s}^{-1} \text{ cm}^{-2}$ over $10\text{--}1300 \text{ \AA}$). All results here use it. The proxy SED and the paper's own APEC fit disagree in the X-ray band (4.4 against $2.9 \text{ erg s}^{-1} \text{ cm}^{-2}$ at the orbit). That inconsistency comes from the paper's procedure and is not introduced here.

3.4 Measurement

The comparison target is the authors' released Fig. 3B spectrum (129 samples, vacuum, planetary rest frame, correlated-noise corrected), not a reading of a figure. Fitting it with our own three-Gaussian extractor (`he_line_metrics.py`, the same one applied to every model spectrum here) reproduces the paper's MCMC to within its published uncertainties:

	this extractor	paper
blended red depth	1.254%	$1.24^{+0.22}_{-0.23}\%$
blue depth	0.198%	$0.25^{+0.14}_{-0.12}\%$
red/blue	6.35	$6.7^{+12.7}_{-3.1}$
FWHM	$0.841 \text{ \AA}$	$0.86^{+0.15}_{-0.27} \text{ \AA}$

so models and measurement can be compared through one instrument.

3.5 The atomic input that sets the line: $\text{He}(2^3\text{S})$ destruction on atomic hydrogen

Every composition this memo quotes is read off the metastable population, and in this wind that population is drained almost entirely by one reaction, $\text{He}(2^3\text{S}) + \text{H}$. Its rate coefficient is therefore an input of the same standing as the spectrum, and it has been changed since the earlier draft. The change is recorded in `docs/Update_EXHALE.md`, Sect. 87. The grounds are given here because every number below moved with it.

The code carried the fit of Taylor et al. (2025), Table 2, transcribed correctly: $1.9 \times 10^{-9}(300/T)^{0.07}$ below 4000 K and $9.1 \times 10^{-9}(300/T)^{0.50}$ above it. Those two branches do not meet: at 4000 K they give 1.5849 and 2.4921×10^{-9} , a jump of 1.5724. A rate coefficient is a continuous function of temperature, and “it is the published value” is not a defense of an expression that is discontinuous in a state variable. In these solutions the seam falls at 1.11–1.18 R_p , inside the line-forming region. In the figures of the earlier draft it showed up as a one-cell step of 33 per cent in $n(2^3S)$.

The fit also disagrees with its own paper. Taylor et al.’s Fig. 19 (bottom right) plots the quantity Table 2 is meant to represent (their Maxwell–Boltzmann average of the Cohen & Lane (1971) and Morgner & Niehaus (1979) cross sections), and that curve has a maximum near 1400 K. A monotonically decreasing power law cannot produce a maximum, so the tabulated low branch cannot be the same quantity; digitized, the two differ by 1.13–1.62 over 500–4000 K. Above the seam the high branch is worse: 2.23×10^{-9} at 5000 K, above every cross-section determination collected in García Muñoz (2025), Fig. 5, which lie between about 1.3 and 1.7×10^{-9} there. Their appendix records how this happened: an equation already implemented in their KPP routine, fitted in two pieces with `curve_fit` and nothing tying the pieces together at the join.

What is used now. García Muñoz (2025), A&A 698, A199, publishes in the caption of its Fig. 5 a closed form for the Maxwell–Boltzmann average of the Movre & Meyer (1997) total ionization cross sections,

$$k = 10^{-9} \exp\left(\frac{c}{T} + d_1 \ln T + d_2 (\ln T)^2 + d_3 (\ln T)^3\right) \text{ cm}^3 \text{ s}^{-1}, \quad (2)$$

with $c = -8.64804 \times 10^1$, $d_1 = -2.86766 \times 10^{-1}$, $d_2 = +8.68445 \times 10^{-2}$ and $d_3 = -5.73001 \times 10^{-3}$, stated there as accurate to better than 2% over 200–10⁴ K. Measured here, it reproduces that paper’s own Table A.5 to 0.4%, and it is continuous at 4000 K to the eighth digit ($1.35630953 \times 10^{-9}$ at 3999.9 K against 1.35630905 at 4000.1).

The change is not one paper’s fit traded for another’s. Taylor’s own Fig. 19 and the García Muñoz curve agree to 10–20%; the outlier is the tabulated fit, and that is what the code was using.

The 0.9:0.1 branching. Equation (2) is the *total* ionization rate: Penning, $\text{He}(2^3S) + \text{H} \rightarrow \text{He}(1^1S) + \text{H}^+ + e^-$, and the associative channel that leaves HeH^+ , sharing one cross section. García Muñoz splits them “an average 0.9:0.1”, and the machine-readable network file deposited with that paper (references/garcia_munoz_2025_network/, Zenodo 15453858) applies that split through two amplitudes per channel. EXHALE now does the same. The split is more than a scale factor. The total rate removes the metastable, so every sink keeps the sum; using 0.9 there would make the observable, $n(2^3S)$, 10 per cent too high. The 0.9 scales only the terms that leave a lasting proton behind and deposit the Penning exothermicity, and the remaining 0.1 becomes an HeH^+ source in the molecular system.

Effect on the numbers. The new coefficient is lower than the old one everywhere in this wind (new/old 0.69 at 1000 K, 0.86 just below the seam, 0.54 just above, 0.81 at 10⁴ K), so the metastable level lives longer and the line is stronger at fixed composition. Re-solved, `heh0p55` goes from $EW = 1.109$ to 1.310 % Å and `heh2p13_diff_kzz1e9` from 1.129 to 1.338, with $\log_{10} \dot{M}$ unchanged in both. Every crossing therefore moves down, and the composition ladders were re-solved rather than scaled by a local slope, because a +18 per cent shift is well outside the range over which such a slope is valid. The step detector of LHS1140b/archive_20260830/exhale/metastable_step_diagnosis flags no cell at the seam in any of the runs quoted here.

4. Reproducing the paper’s p-winds model

We ran p-winds at the authors’ settings: their SED normalization, $R_p = 0.154 R_J$, $b = 0.23$, five-phase transit averaging, `turbulence_broadening=True`, and their fitted line-of-sight bulk velocity $v_{\text{wind}} = 2.26 \text{ km s}^{-1}$. The composition and outflow are the MCMC medians ($\dot{M} = 2.03 \times 10^8 \text{ g s}^{-1}$, $T = 5160 \text{ K}$, $\text{H}:\text{He} = 1.01 \times 10^{-3}$), the vector their Fig. 4 legend states for the plotted curve. Our reproduction gives a red depth of 1.376% and FWHM $0.575 \text{ \AA}$, against the 1.32% and $\sim 0.71 \text{ \AA}$ read off their Fig. 4. The depth matches to 3%, but the line is about 20% too narrow.

The width residual does not depend on which point in parameter space is drawn. Running the same reproduction at the grid maximum-likelihood point ($1.83 \times 10^8 \text{ g s}^{-1}$, 5131.6 K , 8.338×10^{-4}) gives red 1.366% and FWHM $0.581 \text{ \AA}$, i.e. the two vectors differ by less than $0.01 \text{ \AA}$ in width. What produces the extra width in the published curve has not been identified.

Two things here cost time to find.

First, the curves drawn over the spectrum in the paper’s Fig. 3 are the three-Gaussian empirical fits, not p-winds; the physical model is overlaid only in Fig. 4. The close agreement in Fig. 3 is a fit to the data, and says nothing about the model’s line shape.

Second, the paper’s plotted model is visibly broader than a plain p-winds run. We first read that as an undocumented instrumental kernel, and it is not one. Their script sets `turbulence_broadening=True` and averages over five transit phases. Both widen the line, and together they account for the difference.

5. Turbulent broadening: what the literature does

Since that switch turned out to matter, all 43 papers under references/ were checked: every PDF converted to text, searched for `turbulen`, `microturbulen` and `non-thermal broadening`, and each hit read in context. Fifteen matched; most use “turbulent” for eddy diffusion K_{zz} or for flow structure in 3D simulations, not for line broadening. Section 5.1 tabulates the ones that bear on broadening.

The convention, stated most explicitly by Yan et al. (2024), replaces the thermal Doppler width by

$$\Delta\nu_D = \frac{\nu_0}{c} \left(\frac{2kT}{m} + v_{\text{turb}}^2 \right)^{1/2}, \quad v_{\text{turb}} = \sqrt{\frac{5kT}{3m}} = c_s, \quad (3)$$

the monatomic sound speed, as assumed by Salz et al. (2018), Lampón et al. (2020) and Czesla et al. (2022). The p-winds implementation uses $\sqrt{5kT/6m}$ in a σ -convention, which is the same physics expressed against a Doppler width that lacks the factor 2; in EXHALE’s b -parameter convention the same term is $b \rightarrow \sqrt{11/6} v_{\text{th}}$, a factor 1.354. That is what `EXHALE_TRANSIT_TURB=1` applies. It is opt-in and off by default, since the field does not agree that it belongs.

5.1 Paper by paper

Adopt it as an assumed value. Salz et al. (2016), Eq. (5): “ b is the Doppler parameter, v_{th} the thermal Doppler velocity, and for the microturbulence v_{micro} we use the sound speed”, applied when synthesizing the transit profile. Their atmosphere solver separately carries a constant 1 km s^{-1} microturbulence at the lower boundary, a different quantity, which they note “has little impact on the overall atmospheric structure”. Lampón et al. (2020, 2021) is the reference

implementation, with turbulent broadening a free or assumed parameter of 1D isothermal Parker fits, and Czesla et al. (2022) follows it. Cherubim et al. (2026) inherits the choice through p-winds' `turbulence_broadening=True`, so the LHS 1140 b retrieval carries it. The paper text does not say so anywhere.

Test it, then leave it out. Yan et al. (2024) give the cleanest statement of the convention and of the doubt. They record that “these authors assumed that the turbulence velocity $v_{\text{turb}} = \sqrt{5kT/3m}$, which corresponds to the speed of sound of a monatomic gas and, thus, to the transonic case with a Mach number of 1”, then keep it out of their nominal models and test $v_{\text{turb}} = 0.5, 1.0$ and $1.5 c_s$ as a sensitivity. Their finding for He 10830 is that turbulence “decreases the absorption depth slightly at the line center, but doesn’t significantly alter the absorption at the line wings”. They argue Mach 1 is likely an overestimate, since in the diffuse ISM turbulent velocity anti-correlates with temperature and the Mach number is below unity at a few thousand K. They conclude that “the turbulence effect is likely less significant than the one explored in this paper”. Yan et al. (2019) computes the Doppler width explicitly “assuming no turbulence”.

Argue against it. Murray-Clay et al. (2009), on García Muñoz’s (2007) proposal to broaden $\text{Ly}\alpha$ by wind turbulence: “to generate the large velocities observed, the energy in such turbulence would need to exceed the thermal and bulk kinetic energies in the mean flow by a factor of ~ 100 . Such energy requirements seem insurmountable.” Koskinen et al. (2013b) find that “non-thermal broadening such as that proposed by Ben-Jaffel and Hosseini (2010) does not appear to be necessary to explain the current observations”, while allowing that stellar-wind interaction may produce broadening turbulence. Schulik & Owen (2025) offer a physical alternative, in which adiabatic cooling produces a metastable-helium population bump at high velocity and widens the line with no free parameter, and they are direct about the isothermal case: that model “can only explain the broadness of the data under physically inconsistent assumptions, i.e. those of even larger mass-loss rates, not supported by the energy input into this planet, and an empirical free parameter, turbulent broadening (Lampón et al. 2021)”. They add that $T = 10^4$ K already maximizes what thermal broadening alone can give.

Need it even so. Taylor et al. (2026) is the closest analogue to the present work: a self-consistent model (energy balance, multispecies diffusion, non-LTE $\text{H}(n=2)$) that reproduces the HD 189733 b line core but not its width. They write that “several prior analyses have matched the HD 189733 b width by introducing ad hoc (micro)turbulent or Gaussian broadening terms; here, we show that this step remains necessary even in our self-consistent framework”, and quantify it as $\sim 12 \text{ km s}^{-1}$ of additional non-thermal broadening. They suggest, calling the suggestion speculative, that it is plasma turbulence from ions trapped on low-latitude planetary field lines, strengthening with stellar activity. Rumenskikh et al. (2022), as reported there, instead imposed reduced hydrogen line cooling and prescribed zonal jets of $10\text{--}20 \text{ km s}^{-1}$ in a 3D multifluid model.

Matches that are not about broadening. Recorded so the sweep need not be repeated: Wogan (2025), Lavvas (2014, 2019), Robeling (2026) and the second half of Koskinen et al. (2013b) use “turbulent” for eddy diffusion K_{zz} ; 2607.18193v1 for flow–flow interaction in a 3D stellar-wind simulation; Jensen (2012) for photospheric turbulence driving stellar activity; Taylor et al. (2025) for gravity-wave breaking. Vidal_2003_Nature.pdf matches only on its

first page, which carries the tail of the preceding *Nature* letter (on Crab pulsar radio emission); Vidal-Madjar et al. themselves do not discuss turbulence.

5.2 Empirical baseline: turbulence in Solar System giant planets

Whether sonic or supersonic ($v_{\text{turb}} \sim c_s$) turbulence is physically plausible in a planetary atmosphere can be checked against the measured turbulence and non-thermal velocity fields in the stratified layers of Jupiter and Saturn:

- *Troposphere and cloud deck* ($\sim 0.1\text{--}1$ bar). Cloud-tracking measurements from *Voyager*, *Galileo*, *Cassini*, and *Juno* (e.g., Salyk et al. 2006; Del Genio et al. 2007; Ingersoll et al.) isolate the fluctuating eddy velocities $v' = \sqrt{u'^2 + v'^2}$ from the mean zonal jets ($100\text{--}500$ m s⁻¹). The resulting RMS turbulent eddy velocity is $v' \approx 3\text{--}10$ m s⁻¹ on Jupiter (locally $20\text{--}30$ m s⁻¹ in the Great Red Spot shear zone) and $5\text{--}15$ m s⁻¹ on Saturn. Compared to the tropospheric sound speed ($c_s \approx 800\text{--}900$ m s⁻¹), this is deeply subsonic turbulence, with Mach numbers $\mathcal{M} \sim 0.005\text{--}0.02$.
- *Stratosphere* ($\sim 10\text{ }\mu\text{bar--}10$ mbar). High-resolution submillimeter spectroscopy of impact-delivered HCN/CO molecules with ALMA (e.g., Cavalié et al. 2021) and Herschel reveals stratospheric jet velocities up to $300\text{--}400$ m s⁻¹ in the auroral zone, but the associated non-thermal velocity dispersion (turbulence and wave motions) remains limited to tens of m s⁻¹ up to $\sim 100\text{--}200$ m s⁻¹ ($\mathcal{M} \lesssim 0.2$).
- *Thermosphere and auroral ionosphere* ($\lesssim 1\text{ }\mu\text{bar}$). Ground-based high-resolution ($R \sim 10^5$) infrared spectroscopy of H₃⁺ line profiles (e.g., with VLT/CRIRES, Keck, and CFHT) disentangles thermal Doppler broadening from non-thermal velocity fields. In non-auroral mid-latitudes, the non-thermal velocity is $v_{\text{nth}} \approx 0.2\text{--}0.5$ km s⁻¹. In the auroral oval, where intense magnetosphere–ionosphere coupling, ion drag, and Joule heating inject substantial kinetic energy, non-thermal broadening reaches $v_{\text{nth}} \approx 0.8\text{--}2.0$ km s⁻¹.

These measurements show that pure hydrodynamic planetary turbulence is strictly subsonic ($\mathcal{M} \ll 1$), and that even strong magnetospheric auroral driving in upper thermospheres caps non-thermal broadening at $\lesssim 2$ km s⁻¹.

5.3 Turbulence and 3D hydrodynamic simulations of exoplanets

Beyond 1D parameterizations, 3D simulations show how turbulent mixing, non-thermal velocity structure and line broadening arise in several classes of exoplanet:

1. 3D hot Jupiter escape and circulation.

- *3D atmospheric escape and kinematic line broadening*. Hydrodynamic 3D escape simulations (e.g., McCann et al. 2019; Debrecht et al. 2020; Carolan et al. 2020, 2021; Wang & Dai 2021; MacLeod & Oklopčić 2022) show that apparent transmission line broadening comes mainly from large-scale 3D kinematic velocity gradients ($10\text{--}50$ km s⁻¹) across the day-to-night outflow, Coriolis-deflected tails, and tidal distortion, rather than from unresolved isotropic microscale turbulence. The collision between the expanding planetary wind and the incoming stellar wind creates a shear layer prone to Kelvin–Helmholtz instabilities and turbulent mixing at

the contact discontinuity. In 3D multifluid models of HD 189733 b, Rumenskikh et al. (2022, 2023) show that upper-atmospheric zonal jets of $10\text{--}20\text{ km s}^{-1}$ combined with 3D outflow geometry reproduce the observed broad transmission profiles without invoking ad hoc microturbulence.

- *3D general circulation models (GCMs) and macroturbulent mixing.* In the observable lower-to-middle atmosphere (1 bar to $\sim 1\text{ }\mu\text{bar}$), 3D GCMs (e.g., Showman & Polvani 2011; Showman et al. 2013, 2015; Parmentier et al. 2013; Charnay et al. 2015; Zhang & Showman 2018) show that equatorial superrotation ($1\text{--}5\text{ km s}^{-1}$) and vertical mixing ($K_{zz} \sim 10^7\text{--}10^{11}\text{ cm}^2\text{ s}^{-1}$) are driven by planetary-scale macroturbulence (non-linear interactions between planetary-scale Kelvin and Rossby waves and the mean flow) and overturning circulation. The sub-grid parameterization of small-scale turbulent dissipation (hyperdiffusion, Smagorinsky closures, or Shapiro filters) remains a known source of numerical sensitivity in GCM wind speeds and jet stability (Heng et al. 2011; Polichtchouk et al. 2014; Skinner & Cho 2021).
- *Magnetohydrodynamic (MHD) turbulence in ultra-hot atmospheres.* For planets with $T \gtrsim 1500\text{ K}$, thermal ionization of alkali metals (Na, K) couples the gas to the planetary magnetic field. 3D MHD simulations (e.g., Perna et al. 2010, 2012; Rauscher & Menou 2013; Rogers & Showman 2014; Rogers & Komacek 2014; Beltz et al. 2022) show that magnetic winding of the field by supersonic zonal winds generates Lorentz forces (magnetic drag) that brake the flow, producing 3D MHD turbulence, Ohmic dissipation, and time-dependent variability in wind patterns and hotspot offsets.

2. 3D simulations across other planetary classes. 3D fluid simulations reach well beyond hot Jupiters, into the other main classes of exoplanet and substellar object:

- *Warm Neptunes and sub-Neptunes.* For warm Neptunes such as GJ 436 b and GJ 3470 b, 3D particle-mesh and hydrodynamic escape simulations (Bourrier et al. 2015, 2016; Villarreal D’Angelo et al. 2021; Carolan et al. 2021) give large asymmetric cometary hydrogen tails spanning tens of planetary radii, which account for the 56% $\text{Ly}\alpha$ transit obscuration and the He 10830 Å absorption through 3D stellar radiation pressure and charge-exchange interactions. In metal- and water-rich sub-Neptune atmospheres (e.g., GJ 1214 b, K2-18 b), 3D GCMs model cloud/haze circulation, photochemical transport, and vertical eddy mixing ($K_{zz} \propto P^{-0.4}$; Charnay et al. 2015; Innes & Pierrehumbert 2022).
- *M-dwarf habitable zone terrestrial planets.* Tidally locked rocky worlds (such as TRAPPIST-1 e/f/g, Proxima b, and LHS 1140 b) have been modeled extensively with 3D climate GCMs (Turbet et al. 2016, 2018; Wolf et al. 2017; Sergeev et al. 2020, 2022; the THAI intercomparison, Fauchez et al. 2020, 2022). These simulations show how thick dayside convective cloud decks raise the planetary albedo and prevent runaway greenhouse states, while stationary Rossby–Kelvin wave patterns redistribute heat to the nightside and keep the atmosphere from freezing out. Complementary 3D multi-species MHD simulations (Dong et al. 2017, 2018; Cohen et al. 2014) assess O^+ and H^+ ion escape driven by extreme stellar coronal mass ejections and flaring winds.
- *Lava worlds and ultra-short-period rocky planets.* For scorched rocky planets with dayside magma oceans (e.g., 55 Cancri e, K2-141 b, CoRoT-7 b), 3D hydrodynamic models of mineral-vapor atmospheres (Castan & Menou

2011; Nguyen et al. 2020; Zilinskas et al. 2022; Kang et al. 2021) show supersonic dayside-to-nightside winds ($\mathcal{M} > 1-3$, $v \sim 1-3 \text{ km s}^{-1}$), standing hydraulic jumps/shocks, and mineral vapor condensation leading to rock rain on the nightside.

- *Brown dwarfs and directly imaged giant planets.* In isolated brown dwarfs and young, directly imaged giants (e.g., HR 8799 b/c/d/e, VHS 1256 b), 3D convective-radiative GCMs (Tan & Showman 2019, 2021a,b; Showman & Kaspi 2013; Zhang & Showman 2014) couple rapid planetary rotation (strong Coriolis forces) with deep geothermal convection and silicate/iron cloud radiative feedbacks. The resulting turbulent vortices, patchy cloud bands, and storm dynamics explain the observed rotational photometric variability.

5.4 Where that leaves this comparison

There is no settled treatment. Turbulence at $v_{\text{turb}} = c_s$ is a common convention in 1D isothermal Parker-wind fits, and hence in the paper being reproduced, while Solar System observations and self-consistent or 3D models show that isotropic microscale turbulence cannot reach the required supersonic levels. 3D models instead supply line width through kinematic flow gradients, zonal jets, and magnetospheric/plasma interactions. Enabling the $v_{\text{turb}} = c_s$ term in EXHALE is therefore right for a like-for-like comparison with C26 and wrong as a physical default, which is how it is implemented.

6. Searching for a match: the composition scan

Once the star, the planet and the spectrum are fixed, composition is the free parameter left to search over. This section searches it in the well-mixed limit: element diffusion off, so the reservoir ratio is also the ratio everywhere in the wind. That is the simpler of the two mixing assumptions available. Sect. 9.4 repeats the same scan with binary H/He diffusion active, where the answer depends on the eddy coefficient as well.

Eleven compositions were solved, from solar ($\text{He}/\text{H} = 0.083$) to $\text{He}/\text{H} = 10^4$, each a JFNK steady state ($\|R\| < 10^{-3}$, $\text{info} = 0$; one case, $\text{He}/\text{H} = 100$, stagnates at 1.18×10^{-3} and is recorded as such). Nothing is imposed: mass, momentum and energy are solved with photoionization heating and radiative cooling, so $\dot{M}$ and $T(r)$ are outputs.

He/H	$\log_{10} \dot{M}$ [g s ⁻¹]	red [%]	red (turb) [%]	FWHM [Å]	EW [% Å]
0.083	7.77	1.13	1.02	0.233	0.279
0.25	7.75	2.49	2.21	0.244	0.645
0.50	7.76	3.83	3.35	0.255	1.040
0.55	7.76	4.05	3.54	0.257	1.109
0.70	7.78	4.71	4.09	0.261	1.311
1	7.80	5.59	4.83	0.268	1.598
10	7.76	7.12	5.97	0.298	2.270
10 ³	7.74	5.46	4.55	0.315	1.844
10 ⁴	7.73	5.40	4.49	0.316	1.825
observed	-	1.254	-	0.841	1.108 ± 0.030

Nine of the eleven runs are listed; He/H = 0.6 falls between two tabulated entries and He/H = 100 is the case whose residual stagnates. The three helium-dominated rows are corrected values. Their depths and equivalent widths were an artifact of the old advection gate, which pinned the metastable population to its equilibrium value across the hydrogen ionization front. They fell by 89–93% when the gate was rebuilt on the slowest solved species (Update_EXHALE.md Sect. 72; at He/H = 10, $EW\ 21.53 \rightarrow 2.270\ \%$ Å). The crossing of Sect. 8. is interpolated inside He/H ≤ 1 and is untouched by that correction. The escape rate itself is nearly composition-independent, $\dot{M} \simeq 6 \times 10^7\ \text{g s}^{-1}$, about a third of the 2.03×10^8 the retrieval imposes.

This table is the 2026-08-30 scan, kept as the record of what was run then and not as the current value of any entry. It predates both the Penning coefficient replacement of Sect. 3.5 and every change of Sect. 2.. The current well-mixed ladder is the nine solved rungs of models/atomic_scalar_gj1132_wellmixed, He/H = 0.083 to 1000, and the crossing of Sect. 8. is read from it. For one point of comparison, He/H = 0.55 re-solved on the current code gives $EW = 1.3118\ \%$ Å and FWHM $0.2598\ \%$ Å with $\log_{10} \dot{M} = 7.85$, against the 1.109 and 0.2573 of this table.

7. Why the two models disagree

At the same composition the two models are not small perturbations of each other. The *total* metastable column is comparable (at H:He = 10⁻³ EXHALE’s radial column over 1–20 R_p is 1.04 of p-winds’), but its distribution is not, and transit depth is absorbing *area*:

	p-winds	EXHALE
50% of the 2 ³ S column inside	1.31 R_p	5.7 R_p
90% inside	1.58 R_p	11.7 R_p

Both radii are read from the local-equilibrium metastable profile, which is what Fig. 2 draws. The advection-corrected profile that the transit integral consumes is more concentrated, and for this helium-dominated composition it collapses onto the base once the advection gate is formed on the metastable’s own clock rather than on hydrogen’s

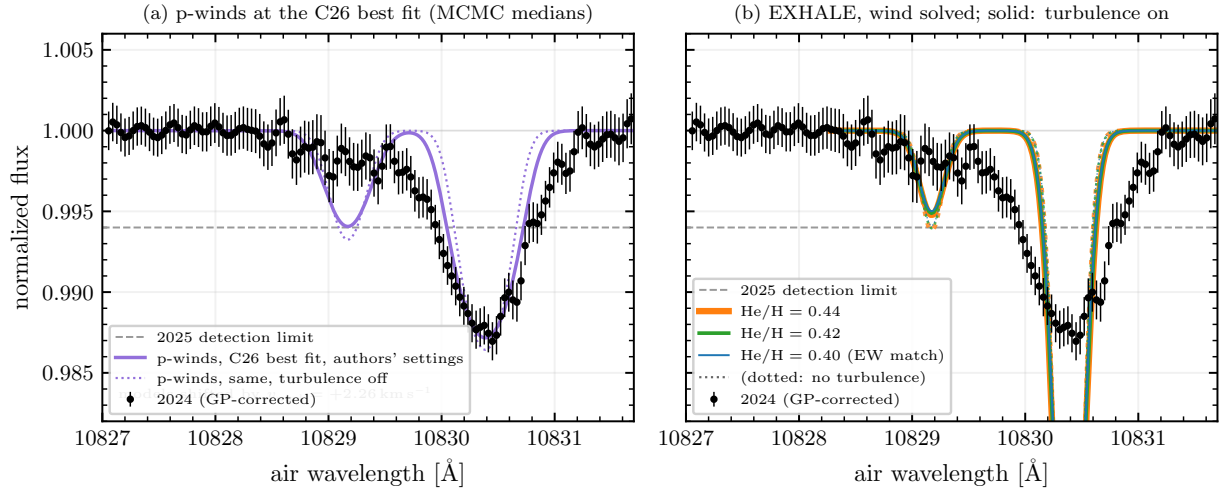


Figure 1: The measurement is in black. On the left is the paper's own p-winds model at its best fit, the MCMC medians its Fig. 4 legend states, computed with the authors' released settings; that model is the solid curve and the dotted one is the same run with turbulence off. On the right is EXHALE at the three rungs of the well-mixed ladder of record that bracket the equivalent-width crossing 0.4044: He/H = 0.40, the solved rung nearest the crossing, with 0.42 and 0.44. The turbulence twin of each rung was synthesized on the same solution, into its own `tpm_turb/`, so the solid EXHALE curves carry the same turbulence term p-winds uses and the dotted ones are without it. Over that range of composition the profile barely moves, so the three nearly coincide; the composition dependence of the absorbing power is Fig. 4. Neither code applies a line-of-sight bulk velocity, so both sets of model curves are shifted by the $v_{\text{wind}} = +2.26 \text{ km s}^{-1}$ that C26 fit, a rigid shift that leaves equivalent widths unchanged. The figure is drawn in the presentation of the paper's Fig. 4.

(Update_EXHALE.md Sect. 72). This section is therefore a statement about the two wind solutions; every transit number quoted elsewhere in the memo is measured on the advection-corrected profiles.

The star is $13.6R_p$ across, so metastable helium spread to $10R_p$ covers most of the disk while a column confined to $1.6R_p$ can darken only a narrow ring. The difference in distribution comes from the wind structure, which p-winds imposes and EXHALE solves. Take $r = 10R_p$ at the same composition and spectrum. p-winds is isothermal at 6100 K by construction, while EXHALE's energy balance gives 1499 K. p-winds flows at 4.3 km s^{-1} and EXHALE at 0.51 . The cold, slow, denser gas keeps helium neutral and the metastable level populated far out, $n(2^3S) = 48 \text{ cm}^{-3}$ against 2×10^{-4} (Fig. 2). That radius is at the edge of what these $30R_p$ solutions resolve (Sect. 13.). The same three comparisons at $5R_p$, well inside it, read 6100 against 3102 K, 1.9 against 0.20 km s^{-1} and 1.3×10^{-2} against 47 cm^{-3} .

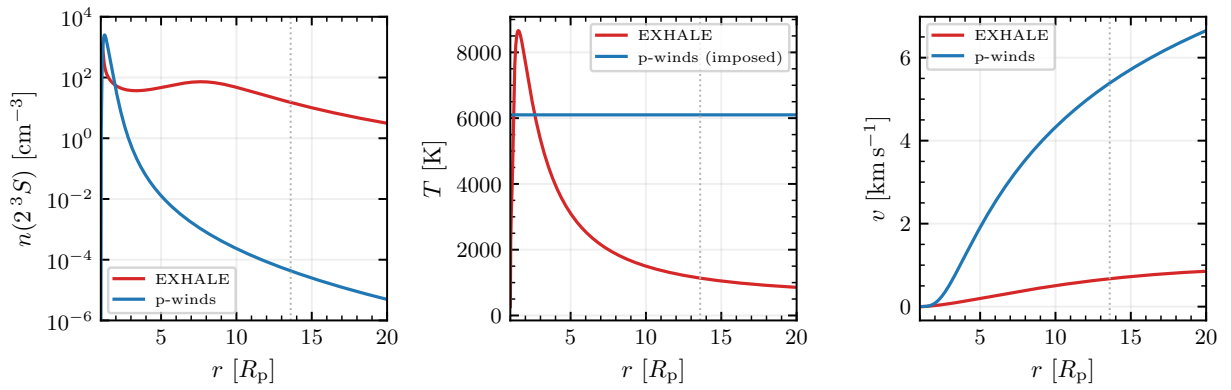


Figure 2: Same composition ($\text{H}:\text{He} = 10^{-3}$), same spectrum, two wind structures. Dotted vertical line: the stellar radius, $13.6R_p$.

7.1 Density and ionization, the same pair

Figure 2 draws three of the five profiles the comparison was set up to make; the other two, mass density and hydrogen ionization, are in Fig. 3, from the same pair of solutions.

The densities are the part that agrees. Outside the base the ratio EXHALE/p-winds runs from 0.15 at $1.2R_p$ through unity near $2R_p$ to 2.7 at $5R_p$, and stays inside a factor 2.3 from $3R_p$ out. At the base cell itself EXHALE sits $176\times$ higher, which is the two lower boundary conditions and not a profile difference: p-winds sets its base density from the imposed $\dot{M}$ through the Parker solution, EXHALE from its own base state.

The ionization does not agree at all. EXHALE has hydrogen completely ionized above $1.03R_p$ ($n(\text{HI})$ is an exact zero in the output), while p-winds keeps it 91–94% neutral throughout, $x(\text{HII})$ rising only from 0.056 at $1.2R_p$ to 0.086 at $20R_p$. Neither end of that is the other's error, and both need an explanation.

p-winds' hydrogen is frozen far below its own photoionization equilibrium. Integrating the same SED against the Verner et al. (1996) ground-state cross section gives an unattenuated rate $\phi = 5.17 \times 10^{-8} \text{ s}^{-1}$ (p-winds' internal value for this file agrees to 0.1%), an ionization time of 224 d, while the flow time r/v through the imposed $\dot{M} = 2.03 \times 10^8 \text{ g s}^{-1}$ falls to $2.6 \times 10^5 \text{ s}$ at $2R_p$ and $2.6 \times 10^4 \text{ s}$ at $10R_p$. Equilibrium on p-winds' own density and temperature would be $x(\text{HII}) = 0.76$ at $2R_p$ and 0.997 at $5R_p$ (dotted in Fig. 3b); the wind carries the gas out before it gets there. Shielding

is not what holds it down: hydrogen is a trace species here and the neutral-H column above $1.2R_p$ is optically thin at every energy.

EXHALE's hydrogen goes the other way: it is more ionized than any photoionization balance on its own structure allows. With the same ϕ , its own electron density at $3R_p$ ($2.0 \times 10^6 \text{ cm}^{-3}$, set by helium) and its own temperature there (5247 K), primary photoionization alone gives $x(\text{HII}) = 0.059$: close to what p-winds carries, and nowhere near 1. An exact zero can only come from a hydrogen ionization source that does not vanish with $n(\text{HI})$, and the code has one: the Shull & van Steenberg (1985) secondary channel, added in `util_ion_eq.f90` as $P_{\text{HI}} \rightarrow P_{\text{HI}} + R_{\text{sec,HI}}/n(\text{HI})$, where $R_{\text{sec,HI}}$ is accumulated over every absorber's photoelectrons. In a helium-dominated gas that numerator is helium's. Evaluating it at $3R_p$ on this solution's own radiation field (the He I photoelectrons above the code's 40 eV secondary threshold, with $f_{\text{ion,HI}} = 0.091$ at the local ionized fraction $x = 0.246$) gives $R_{\text{sec,HI}} = 4.9 \times 10^{-2} \text{ cm}^{-3} \text{ s}^{-1}$ against a hydrogen recombination rate of 6.6×10^{-3} , a factor 7.5. Complete ionization follows.

That is an approximation used outside the composition it was made at. The SvS85 fits are functions of the ionized fraction alone; the helium abundance of the gas their Monte Carlo followed is fixed inside them and is not the $\text{He}/\text{H} = 10^3$ of this run. Their two branches hand hydrogen $7 \times$ helium's share of the secondary ionizations ($f_{\text{ion,HI}}/f_{\text{ion,HeI}} = 0.3908/0.0554$ at $x \rightarrow 0$), where the abundances say the opposite by three orders of magnitude. It is recorded here rather than changed, because the reach was measured and is bounded. $n(\text{HI})$ reaches an exact zero only at $\text{He}/\text{H} \geq 10$ over a prescribed metal-free base (exhale/heh10 above $6.6R_p$, heh100 above 1.06, heh1000 above 1.03). It does not happen in any of the solutions this memo quotes science from: heh0p55, heh2p13_diff_kzz1e9, and every rung of the flux-closed ladder out to $\text{He}/\text{H} = 12$, where $x(\text{HII})$ at $2R_p$ runs 0.060 to 0.178 and no cell is fully ionized. The $\text{H}:\text{He} = 10^{-3}$ comparison of this section is the only place in the memo that sits inside it, and the quantity it distorts is hydrogen rather than the metastable helium the line is made of.

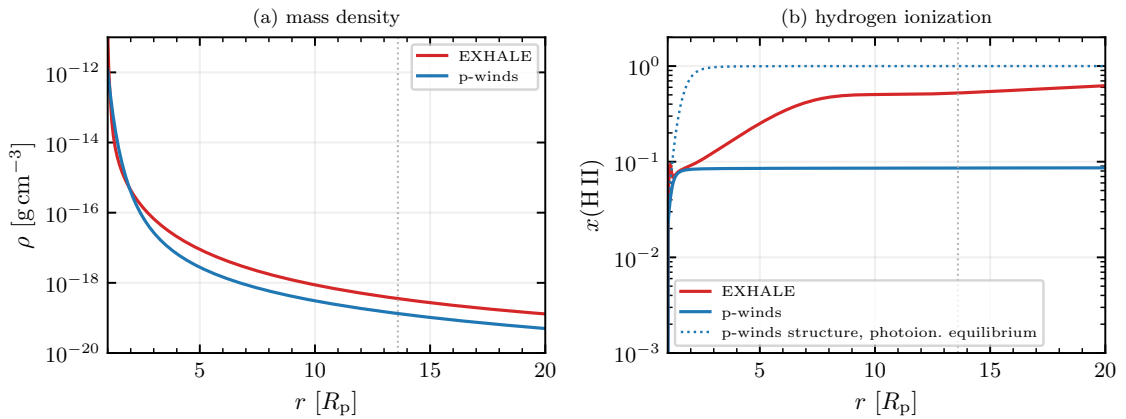


Figure 3: The other two profiles of the same comparison ($\text{H}:\text{He} = 10^{-3}$, same spectrum): (a) mass density, (b) hydrogen ionization. The dotted curve in (b) is photoionization equilibrium evaluated on p-winds' own density and temperature, so its distance from the solid p-winds curve is how far the imposed flow holds hydrogen out of equilibrium. Dotted vertical line: the stellar radius.

8. What can be matched: the equivalent width

Depth and width trade against each other under any broadening, but the equivalent width does not: it is invariant under the instrument profile, the turbulence term and the phase averaging. It is therefore the quantity on which a self-consistent model can be matched without first settling the broadening question.

Over the blended red pair (vacuum 10832.6–10834.2 Å), the released spectrum gives $EW = 1.108 \pm 0.030 \% \text{ Å}$. EXHALE crosses it at

$$\text{He}/\text{H} = 0.4044 \quad (0.3851\text{--}0.4201 \text{ on the } EW \text{ } 1\sigma), \quad \text{H}:\text{He} = 2.5, \quad (4)$$

that is $4.8\times$ solar, a helium mass fraction of 0.62. The crossing is read inside a bracket of two solved rungs, 0.40 at $EW = 1.1011$ and 0.42 at $1.1324 \% \text{ Å}$ against the observed 1.1084 ± 0.0295 , on the well-mixed ladder of models/atomic_scalar_gj1132_wellmixed; the rung at 0.40 is 0.07σ from the measurement. Enabling the turbulence term moves the crossing to 0.3985, 1.3 per cent away, so the estimate does not depend on the broadening treatment. The value sits three orders of magnitude away from the paper's $\text{H}:\text{He} \lesssim 10^{-3}$, and the reason is the structural difference of the previous section: a wind that spreads metastable helium over most of the stellar disk needs far less helium to absorb as much.

Two assumptions are carried inside that number and are taken up later: the gas is well mixed, and the spectrum is the GJ 1132 proxy. Sect. 9.4 rescans it with element diffusion active at every decade of the eddy coefficient, and Sect. 11. rescans it on the brighter proxy. Both move it, the spectrum by much the larger factor.

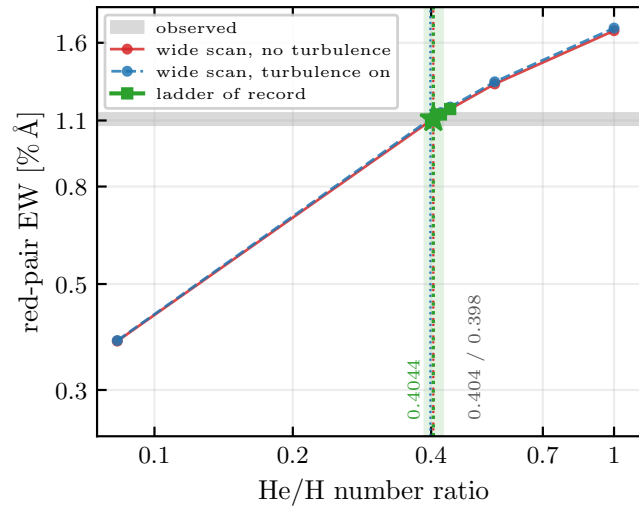


Figure 4: Red-pair equivalent width against composition, with and without the turbulence term, compared with the measured band. The crossing is insensitive to the broadening treatment: each rung of the well-mixed group up to $\text{He}/\text{H} = 1$ carries both syntheses of the same solution, so the two curves differ only in the broadening, and they cross the measured band at $\text{He}/\text{H} = 0.4044$ without the turbulence term and 0.3985 with it, 1.3 per cent apart. The three rungs 0.40, 0.42 and 0.44 drawn on top are where the spacing brackets the measurement directly.

9. What cannot be matched: the profile

Matching the equivalent width is not matching the line. At the crossing $\text{He}/\text{H} = 0.4044$ the profile is 3.3 times too deep (4.09% against 1.254%) and 3.3 times too narrow (0.253 against 0.841 Å). The equivalent width states how much absorbing power the model has. The profile states how that power is spread in velocity, and composition does not act on the second: every solved composition gives 0.23–0.32 Å, rising only to 0.26–0.39 Å with the turbulence term.

9.1 How much is missing, in velocity

The measured FWHM is 0.841 Å, that is 23.3 km s^{-1} at 10833 Å. Three ways to supply it, and what each would require:

- *Thermal alone.* A Doppler width of 23.3 km s^{-1} FWHM in helium needs $T = 4.7 \times 10^4 \text{ K}$. Helium is fully ionized far below that, so there would be no metastable population left to absorb. This is excluded.
- *Bulk outflow.* A line-of-sight velocity spread of order $\pm 12 \text{ km s}^{-1}$ across the line-forming region would do it. The EW-matched EXHALE solution reaches 1.1 km s^{-1} at $20R_p$; the paper's p-winds model, with its imposed temperature, reaches 4–7. Both are an order of magnitude short.
- *Non-thermal (turbulent) broadening.* Added in quadrature to what EXHALE already has, closing the gap takes 22.2 km s^{-1} FWHM: 9.4 km s^{-1} of turbulent velocity (σ), against the 2.9 km s^{-1} that the sound-speed convention of Sect. 5.1 supplies for helium at 5000 K. Of the three, this is the one that can be measured directly, and Sect. 9.2 measures it.

The model is short by an amount of velocity comparable to the escape speed, and no choice of composition supplies it.

9.2 The broadening the data demand, measured

The three routes above can be replaced by one measurement. Take the EW-matched EXHALE profile as solved, convolve it with a Gaussian of FWHM f in velocity, and ask which f reproduces the measured line. The convolution is applied to the excess absorption $(\max T - T)/\max T$ built from the instrument-convolved column of `tpm_He10830.txt`. Depth, width and the red/blue ratio are then read off with the same three-Gaussian extractor used on the data, and the equivalent width by the same integral over the vacuum window 10832.60–10834.20 Å. Only f is varied, and every number in Table 4 is measured the way the corresponding observed number was.

The width is reached at $f = 22.22 \text{ km s}^{-1}$ FWHM, that is $\sigma = 9.44 \text{ km s}^{-1}$. At that value the depth is 1.229% against the measured $1.24^{+0.22}_{-0.23}$, the blue component 0.161% against $0.25^{+0.14}_{-0.12}$, and the ratio 7.65 against $6.7^{+12.7}_{-3.1}$, so all three fall inside the observed 1σ interval at the value fixed by the width alone. None of the three was tuned. Figure 5 shows the profile and the two crossings.

The number does not depend on which solution is broadened. The neighboring rungs of the same ladder, $\text{He}/\text{H} = 0.42$ and 0.44, ask for 22.21 and 22.20 km s^{-1} . The $K_{zz} = 0$ run at $\text{He}/\text{H} = 3.0$ asks for 22.23 and the $K_{zz} = 10^9$ run at 1.50 for 22.22. Solar composition asks for 22.40, although that run carries a third of the observed absorption (1.533% red, EW 0.380) and so matches the shape while missing the amount. All six are solved cases of the model tree, and the spread over them is 0.9 per cent. The required broadening is therefore set by the velocity field rather than by composition or

Table 4: The EW-matched EXHALE profile (models/atomic_scalar_gj1132_wellmixed/HeH0.40, the solved rung nearest the well-mixed crossing $\text{He}/\text{H} = 0.4044$; no turbulence term) convolved with an added Gaussian of FWHM f . Observed values on the last row: red and blue depths from the C26 MCMC posterior, width and equivalent width measured here on their spectrum. Convolution, extractor and window are those of LHS1140b/make_memo_figures.py.

f [km s^{-1}]	red [%]	blue [%]	red/blue	FWHM [$\text{\AA}$]	EW [% $\text{\AA}$]
0	4.085	0.586	6.97	0.253	1.101
10	2.346	0.313	7.49	0.440	1.101
15	1.730	0.228	7.59	0.597	1.097
20	1.351	0.177	7.64	0.765	1.080
22.2	1.229	0.161	7.65	0.841	1.067
25	1.103	0.144	7.66	0.938	1.045
observed	$1.24^{+0.22}_{-0.23}$	$0.25^{+0.14}_{-0.12}$	$6.7^{+12.7}_{-3.1}$	0.841	1.108 ± 0.030

by the broadening treatment already in place, which is what Sect. 9. inferred. The equivalent width falls by 3% between $f = 0$ and $f = 22.2$ because the Gaussian wings carry absorption out of the red-pair integration window, not because absorbing power is lost.

This is a description, not an explanation. One free parameter applied to a solved profile reproduces the measured line, and nothing in that says what produces 22.2 km s^{-1} . The candidates are listed below, and the measurement sharpens them by fixing the number each has to deliver.

One conversion narrows that list. If the missing 22.2 km s^{-1} is called turbulence, it has to be turbulence in gas whose sound speed the solution itself fixes. Evaluating $c_s = (\gamma kT / \mu m_H)^{1/2}$ with $\gamma = 5/3$ on the advection-corrected profiles of the well-mixed $\text{He}/\text{H} = 0.55$ solution, taking μ from the solved H I/H II/He I/He II/He III and electron densities, gives $c_s = 5.3 \text{ km s}^{-1}$ at $2R_p$ ($T = 4108 \text{ K}$, $\mu = 2.00$), 5.6 at the metastable peak $1.53R_p$ (4607 K , 2.03) and 3.2 at $10R_p$ (1437 K , 1.90); weighted over the line-forming region by $n(2^3S)r$, $T = 2499 \text{ K}$, $\mu = 1.94$ and $c_s = 4.1 \text{ km s}^{-1}$. This is the bulk-gas sound speed, the quantity a Mach number is defined against, not the helium thermal speed that enters the line profile.

Against that, the kernel is supersonic on any convention. Read as a line-of-sight velocity dispersion, $\sigma = 9.44 \text{ km s}^{-1}$ is $\mathcal{M} = 2.3$. Read in the b -parameter convention of Lampón et al. and p-winds, where v_{turb} is the quantity added to $b^2 = 2kT/m + v_{\text{turb}}^2$, it is $v_{\text{turb}} = \sqrt{2}\sigma = 13.4 \text{ km s}^{-1}$, $\mathcal{M} = 3.3$; as an isotropic three-dimensional turbulent speed, $\sqrt{3}\sigma = 16.3 \text{ km s}^{-1}$, $\mathcal{M} = 4.0$. The $v_{\text{turb}} = c_s$ convention of Sect. 5.1 supplies only $3\text{--}5 \text{ km s}^{-1}$ in this gas, about a third of what the data demand. The literature reviewed there points the other way: Yan et al. (2024) argue $\mathcal{M} = 1$ is already an overestimate at a few thousand K, and Murray-Clay et al. (2009) call the energy requirement of supersonic turbulence “insurmountable”. Supplying the shortfall under the name of turbulence therefore looks physically hard to sustain, and the measurement points to the candidates of the next section (an ordered velocity field, three-dimensional flow, or heating the 1D energy equation does not carry) rather than to a missing thermal width.

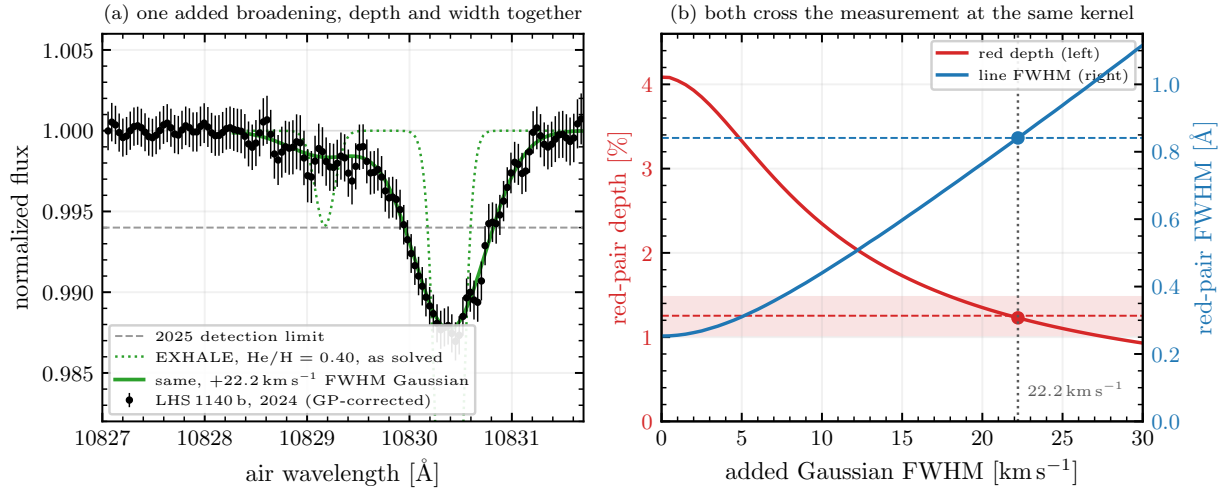


Figure 5: Left: the EW-matched EXHALE profile as solved (dotted, its core running below the plotted range) and convolved with a 22.22 km s^{-1} FWHM Gaussian (solid), against the measurement. Both are shifted by the same $v_{\text{wind}} = +2.26 \text{ km s}^{-1}$ used in the earlier figures. Right: red-pair depth (red, left axis) and red-pair FWHM (blue, right axis) as the added kernel is scanned. Dashed lines are the observed values, the shaded band the C26 1σ interval on the depth; the dotted vertical marks where the width is matched. Both curves meet the measurement at the same kernel.

9.3 Is the demand an artifact of the measurement?

The number above is a difference between two widths, so it is worth no more than the two definitions behind it. Before the missing velocity is treated as physics, the observing and fitting side was audited on its own. The record is docs/lhs1140b_width_measurement_audit.md, where every number was recomputed from the released spectrum, the released MCMC chain and the code in this repository.

Nothing on the measurement side supplies the width. Term by term, in the units the requirement is stated in:

- *The instrument profile.* The WINERED HIRES-Y kernel the Supplement states, and validates against nearby telluric features, is $10833 \text{ Å} / 68,000 = 0.1593 \text{ Å} = 4.41 \text{ km s}^{-1}$ FWHM. The observed line is 5.4 times wider, so it is resolved, and removing the kernel in quadrature moves the intrinsic width from 23.9 to 23.4 km s^{-1} .
- *The triplet splitting.* The 0.0897 Å separation of the red pair is present on both sides of the comparison. It is 0.7 per cent of the observed blend width but 8 per cent of the model's, because at $\sigma = 0.35 \text{ Å}$ the pair is a single hump; the asymmetry nearly cancels, moving the requirement by less than 0.1 km s^{-1} and in the direction that makes the gap larger, since the extractor already credits the model with width it does not get from gas motion.
- *Stellar rotation.* With $P_{\text{rot}} = 131 \text{ d}$ and $R_{\star} = 0.216 R_{\odot}$, $v_{\text{eq}} = 0.083 \text{ km s}^{-1}$ and the rotational broadening is 0.14 km s^{-1} . The Rossiter–McLaughlin distortion and the center-to-limb residual scale with the same $v \sin i$ and are correspondingly irrelevant. Neither is mentioned in the paper or its Supplement, and on this star there is nothing to mention.

- *Planetary rotation.* Tidally locked: 0.032 km s^{-1} at R_p and 0.097 at $3R_p$. It is already inside the model curves, which carry the solid-body spin.
- *Orbital motion.* 0.93 km s^{-1} across the full transit if left uncorrected, but the spectra are shifted into the planetary rest frame before averaging, which leaves the 0.022 km s^{-1} of the ephemeris uncertainty; a 300 s exposure smears 0.037 km s^{-1} .

Together these come to at most about 0.4 km s^{-1} , and most of that is removed as soon as each side is deconvolved from its own kernel. Done that way, component width against component width and both intrinsic, the requirement is 22.00 km s^{-1} FWHM ($\sigma = 9.34 \text{ km s}^{-1}$), against the 22.22 the convolution scan above returns without deconvolving either side. The audit moves the number by 0.2 km s^{-1} , not by a factor. (The audit itself was run on the earlier crossing; what is carried here is the differential it measured, -0.22 km s^{-1} , applied to the present scan.)

One defect was on our side, and it is corrected here. The transit synthesis convolves the He 10830 line with a Gaussian of FWHM λ/R , and until this revision every LHS 1140 b run in this memo took R from the built-in default of EXHALE_transit.py, which is the CARMENES 8×10^4 , whereas this observation was made at WINERED's 68,000. Every model curve quoted here has been re-synthesized at the WINERED value; the runs now read it from LHS1140b/winered_hires_y.sh, and the built-in default is left alone because the other planets in this repository were observed through other instruments. The correction widens the modeled blend by about $0.013 \text{ \AA}$ (at $\text{He}/\text{H} = 0.55$, $0.244 \rightarrow 0.257 \text{ \AA}$), lowers the modeled peak depth by about 5 per cent, and leaves every equivalent width in this memo unchanged to $10^{-4} \% \text{ \AA}$, because a convolution conserves absorbing power. So it moves no crossing and no composition, only the widths and the broadening they demand, the latter by 0.12 km s^{-1} . The p-winds curves were already computed at $R = 68,000$, so the comparison of Sect. 9.5 is now through one instrument on both sides.

The paper's own model does not reach the width either, and that changes what the shortfall means. Reproducing the authors' p-winds setting at their retrieved medians and measuring it with the same extractor gives FWHM $0.458 \text{ \AA}$ with turbulence off and $0.575 \text{ \AA}$ with it on, against the measured 0.841 . The retrieval that produced the published parameters therefore lands 1.5 times short of the width printed in the same paper. Taken with the accounting above, the reading this supports is that the observation is not described by a one-dimensional steady-state wind of either code, rather than that the wind solved here carries a defect the comparison model does not. That is an interpretation. What is measured is that both models are too narrow and that the measurement side does not account for the difference.

A second difference runs the same way. The p-winds curve reaches part of its width through saturation: its red-to-blue amplitude ratio is 2.36 with turbulence off and 2.57 with it on, against the optically thin limit of 8, whereas the measurement sits at 6.35 and the solutions here at 6.9–7.0. The premise that this line is optically thin, so that column density cannot be traded against width, holds for the data and for these solutions but not for the comparison model.

Two remarks on the published width, offered as observations rather than conclusions. First, the three widths the Supplement's stellar-variability test reports as FWHM ($0.22176 \text{ \AA}$ for the stellar helium line, 0.323 for the in-transit feature and 0.392 for the pre-ingress feature) appear to be Gaussian standard deviations rather than full widths. Fitting the same three-Gaussian model to the same pre-GP spectrum with our extractor returns $\sigma = 0.3241 \text{ \AA}$ against their 0.323 , a 0.3 per cent match, where the corresponding FWHM is $0.763 \text{ \AA}$ and matches nothing on that list. The $\sim 9\sigma$ separation the

paragraph concludes with is a difference between two widths and is unaffected; the absolute values are affected. Second, the asymmetric uncertainties on the headline FWHM appear to be transposed. The deposited MCMC chain (identified by its shift column reproducing the published $0.072^{+0.080}_{-0.073}$ Å exactly) gives $2.3548\sigma = 0.865^{+0.261}_{-0.155}$ Å ($23.93^{+7.24}_{-4.30}$ km s⁻¹), where the paper prints $0.86^{+0.15}_{-0.27}$ Å and $23.9^{+4.2}_{-7.5}$ km s⁻¹: the magnitudes agree once the two are swapped. Neither remark can be settled from outside the collaboration, since the printed bars may come from a chain other than the deposited one, and neither changes the comparison except in one direction: on the deposited posterior the 1σ lower bound on the measured width is 0.709 Å rather than 0.59 , so narrow models are excluded somewhat more firmly than the printed interval suggests.

What the audit could not exclude. The star's own metastable helium line is broad: taking the Supplement's 0.22176 Å as a standard deviation, it is 0.522 Å = 14.5 km s⁻¹ FWHM as observed and 13.8 km s⁻¹ with the instrument kernel removed. An imperfectly divided stellar line would therefore inject a residual much wider than the modeled planetary line, though still short of the observed feature on its own. The released data do not include the stellar template, so a mixture of a narrow planetary line with a displaced stellar residual could not be tested here and is not excluded. Whether the published σ is itself deconvolved from the instrument profile was also left unsettled; it is worth 1.7 per cent in width either way.

9.4 The same demand, under diffusion and the adopted K_{zz}

Everything above was solved in the well-mixed limit. With binary H/He element diffusion active, helium settles out of the base unless an eddy term lifts it, and how much of it reaches the line-forming region is set by the eddy coefficient K_{zz} , which is not measured for this system. The diffusion operator is described in `binary_diffusion_design.md`; the literature range at the $1\mu\text{bar}$ base of these runs, the ladder of solved cases, and the adoption of $K_{zz} = 1.0 \times 10^9$ cm² s⁻¹ are in `LHS1140b/kzz_decision.md`.

The composition, and how much of it K_{zz} carries

Turning the operator on moves the equivalent-width crossing of Fig. 4 away from $\text{He}/\text{H} = 0.4044$, and by how much depends on K_{zz} . The composition was therefore rescanned at every decade of the eddy coefficient from $K_{zz} = 0$ to 10^{11} cm² s⁻¹, by the same route in every case: the partitioned stationary route of `MODELS.md` Sect. 6, the same post-processing pass and transit synthesis, and the same red-pair equivalent width on the vacuum window 10832.60 – 10834.20 Å. Table 5 is the result; the groups are `models/atomic_scalar_gj1132_kzz*`, four to seven solved rungs each. Every entry is interpolated inside a bracket of two solved rungs, none extrapolated, but the brackets are not all equally tight: at 10^7 , 10^8 and 10^9 the rung below the crossing is the $\text{He}/\text{H} = 0.55$ case that group carries for the sensitivity scan, so the reading spans a factor 2.7–4.9 in composition rather than the 12–17 per cent the other decades give.

The full ladder is three regimes rather than one power law (Fig. 6). Above 10^7 the degeneracy is shallow and very nearly a power law, $d\log(\text{He}/\text{H})/d\log K_{zz} = -0.115$ (a least-squares fit over 10^8 – 10^{11} ; -0.109 , -0.114 , -0.117 and -0.113 decade by decade), so $\text{He}/\text{H} \propto K_{zz}^{-0.115}$. Below 10^7 the curve rolls over (-0.057 across 10^6 – 10^7 , -0.026 across 10^5 – 10^6) onto a plateau at $\text{He}/\text{H} = 3.10$, the value the line implies when the eddy term does nothing.

Table 5: Where the red-pair equivalent width crosses the measured $1.108 \pm 0.030 \text{ \% } \text{\AA}$, at every decade of the eddy coefficient, with binary H/He diffusion active throughout. The crossing is the log–log chord reading through the group’s solved rungs and the 1σ range propagates the measurement error alone, within one K_{zz} . At 10^{11} the upper edge of that range lies above the group’s highest solved rung ($EW = 1.1364$ at $\text{He}/\text{H} = 0.93$) and is not bracketed. The decades 10^1 – 10^4 were not scanned in composition on the current code: on the 2026-08-30 solutions their equivalent width at a fixed composition differed from the $K_{zz} = 0$ one by less than 10^{-13} of the measurement error, which is the entry they carry. The last column is the ratio to the 2026-08-30 value (Table 2).

$K_{zz} [\text{cm}^2 \text{ s}^{-1}]$	He/H at the crossing	1σ range	bracketing runs [EW in $\text{\% } \text{\AA}$]	new/old
0 (molecular only)	3.1010	2.9942–3.2093	3.0 [1.0798], 3.5 [1.2175]	0.814
10^1 – 10^4	not distinguishable from $K_{zz} = 0$ by this line (archived 2026-08-30)			
10^5	3.0706	2.9677–3.1772	3.0 [1.0880], 3.35 [1.1862]	0.809
10^6	2.8930	2.7928–2.9948	2.8 [1.0804], 3.19 [1.1949]	0.800
10^7	2.5365	2.4823–2.5903	0.55 [0.1585], 2.70 [1.1996]	0.826
10^8	1.9729	1.9195–2.0263	0.55 [0.3089], 2.05 [1.1513]	0.867
10^9 (adopted)	1.5161	1.4644–1.5744	0.55 [0.4793], 1.50 [1.0996]	0.941
10^{10}	1.1579	1.1104–1.2077	1.15 [1.1032], 1.29 [1.1867]	1.019
10^{11}	0.8927	0.8542–(not bracketed)	0.865 [1.0866], 0.93 [1.1364]	1.067
none (well mixed)	0.4044	0.3877–0.4237	Fig. 4	0.979

The shape has survived every revision; the level has not. Decade by decade the exponent was -0.021 , -0.071 , -0.130 , -0.150 , -0.151 and -0.133 on the 2026-08-30 solutions, against -0.026 , -0.057 , -0.109 , -0.114 , -0.117 and -0.113 here: the three regimes and their ordering are the same, the shallow branch is 24 per cent shallower and the roll-off below 10^7 is steeper. What moved is the level, and not by a uniform factor: the plateau falls to 0.81 of its archived value while the 10^{11} end rises to 1.07 (last column of Table 5). That is the shallow branch flattening, not a rescaling.

The plateau sits where it does because the operator adds K_{zz} to the molecular coefficient. The base coefficient of the $K_{zz} = 0$ run is $D_{\text{eff}} = 9.276 \times 10^5 \text{ cm}^2 \text{ s}^{-1}$ (measured at the $r = 1.00029 R_p$ face in `output/element_flux_profile.txt`, dominant pair He I–H I), so $K_{zz} = 10^6$ raises it by 108%, 10^5 by 11% and 10^4 by 1.1%. The crossings in the table say the same from the other side: 10^5 moves the inferred reservoir by 1.0 per cent against the 6.9 per cent the measurement error alone allows, and 10^6 is the first decade that moves it by more than that error. At the other end the operator never reaches the well-mixed limit either: even at 10^{11} the inferred reservoir, 0.8927, is 2.2 times the well-mixed 0.4044.

Over the span the literature leaves open at the base pressure of these runs the inferred reservoir ratio moves only from 1.16 to 1.97, a factor 1.7, against a factor 7 in equivalent width across the same two decades at fixed composition: the line responds to composition much faster than it responds to K_{zz} , so a large change in the eddy coefficient is bought back by a small change in the reservoir. The composition inferred from this line under diffusion is therefore only weakly conditional on K_{zz} , as $K_{zz}^{-0.115}$, and strongly conditional on the assumed spectrum. The SED moves the well-mixed crossing by a factor 7.7, from 0.4044 to 0.0526 (Sect. 11.), which is more than twice the entire K_{zz} lever.

What K_{zz} does act on strongly is the helium profile itself, and Fig. 7 separates that from the wind. At a fixed reservoir He/H = 0.55, raising K_{zz} from 0 to 10^{10} lifts the homopause from below the base to $1.154 R_p$ and the metastable peak by a factor 24; between 10^8 and 10^{10} the factor is 2.3, while the velocity at $10 R_p$ moves by 8 per cent, the density at $2 R_p$ by 28 per cent and $\log_{10} \dot{M}$ by 0.02 dex. Along the equivalent-width-matched sequence of Table 5, where each K_{zz} carries its own crossing composition, $\log_{10} \dot{M} = 7.86\text{--}7.87$ in every run: the escape rate does not carry the K_{zz} dependence at all.

The width the data demand, again

The remaining question is whether the width the data demand moves with the composition. It does not. Repeating the measurement of Sect. 9.2 on `models/atomic_scalar_gj1132_kzz1e9/HeH1.50`, the solved rung nearest the $K_{zz} = 10^9$ crossing He/H = 1.5161 (red-pair EW 1.100 % Å, red depth 4.091%, FWHM 0.252 Å), with the same convolution, the same three-Gaussian extractor and the same integration window, gives Table 6.

The width is reached at $f = 22.22 \text{ km s}^{-1}$ FWHM, the same value to four digits as the diffusion-off well-mixed solution of Sect. 9.2. At that kernel the depth is 1.227% against the measured $1.24^{+0.22}_{-0.23}$, the blue component 0.161% against $0.25^{+0.14}_{-0.12}$, the ratio 7.64 against $6.7^{+12.7}_{-3.1}$ and the equivalent width 1.067 against 1.108 ± 0.030 , all four inside the observed 1σ interval at the value fixed by the width alone, as in the well-mixed case. The velocity the profile demands is therefore unchanged by turning diffusion on and by the adopted eddy coefficient. Diffusion changes how much helium is there, not how fast it moves. Figure 8(a) shows the radial number density profiles of the hydrogen and helium species in this diffusive steady state, and Figure 8(b) the broadened transit spectrum against the measurement.

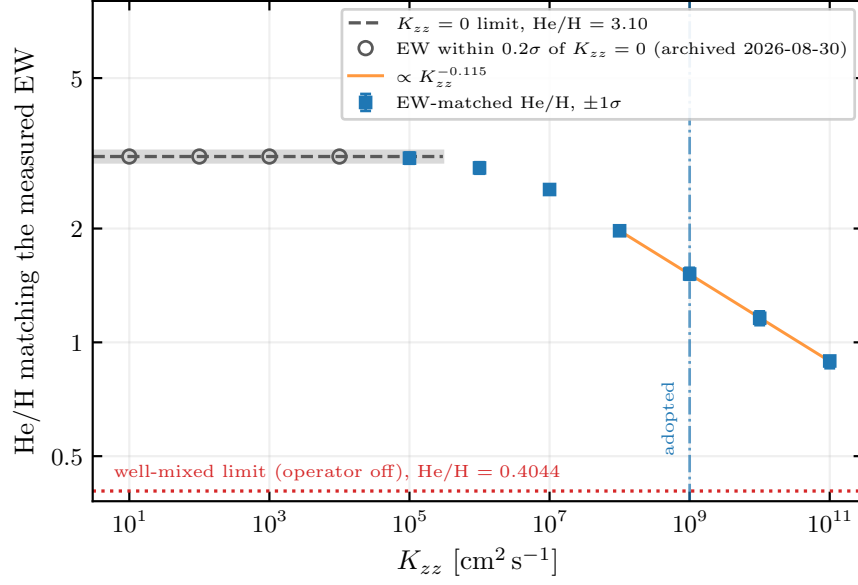


Figure 6: The composition–mixing degeneracy in full. Each square is the He/H reservoir ratio whose red-pair equivalent width equals the measured one, solved at one decade of K_{zz} with binary H/He diffusion active (Table 5); the error bars propagate the measurement error alone and are smaller than the symbols over most of the range. The landscape has three parts. Below $K_{zz} \simeq 10^5$ the eddy term is a small correction to the molecular coefficient the base already carries, the inference sits on a plateau at He/H = 3.10 (dashed, with its 1σ band), and the open circles mark the decades whose equivalent width, on the 2026-08-30 solutions, differed from the $K_{zz} = 0$ one by less than a fifth of the measurement error. Between 10^6 and 10^7 the curve rolls off. From 10^8 up it is the shallow power law $K_{zz}^{-0.115}$ (orange, fitted to those four points). Neither end of the axis reaches the well-mixed crossing of 0.4044 (dotted): with the operator active, helium settles at every eddy coefficient tried, and He/H inferred from this line is a statement about the assumed mixing as much as about the planet. The adopted $K_{zz} = 10^9 \text{ cm}^2 \text{ s}^{-1}$ is marked.

Table 6: The EW-matched diffusive solution (models/atomic_scalar_gj1132_kzz1e9/HeH1.50, the solved rung nearest the crossing He/H = 1.5161 at $K_{zz} = 10^9 \text{ cm}^2 \text{ s}^{-1}$) convolved with an added Gaussian of FWHM f , measured as in Table 4.

f [km s ⁻¹]	red [%]	blue [%]	red/blue	FWHM [Å]	EW [% Å]
0	4.091	0.588	6.96	0.252	1.100
10	2.344	0.313	7.48	0.440	1.100
15	1.728	0.228	7.58	0.597	1.096
20	1.349	0.177	7.63	0.765	1.080
22.2	1.227	0.161	7.64	0.841	1.067
25	1.101	0.144	7.65	0.938	1.047
observed	$1.24^{+0.22}_{-0.23}$	$0.25^{+0.14}_{-0.12}$	$6.7^{+12.7}_{-3.1}$	0.841	1.108 ± 0.030

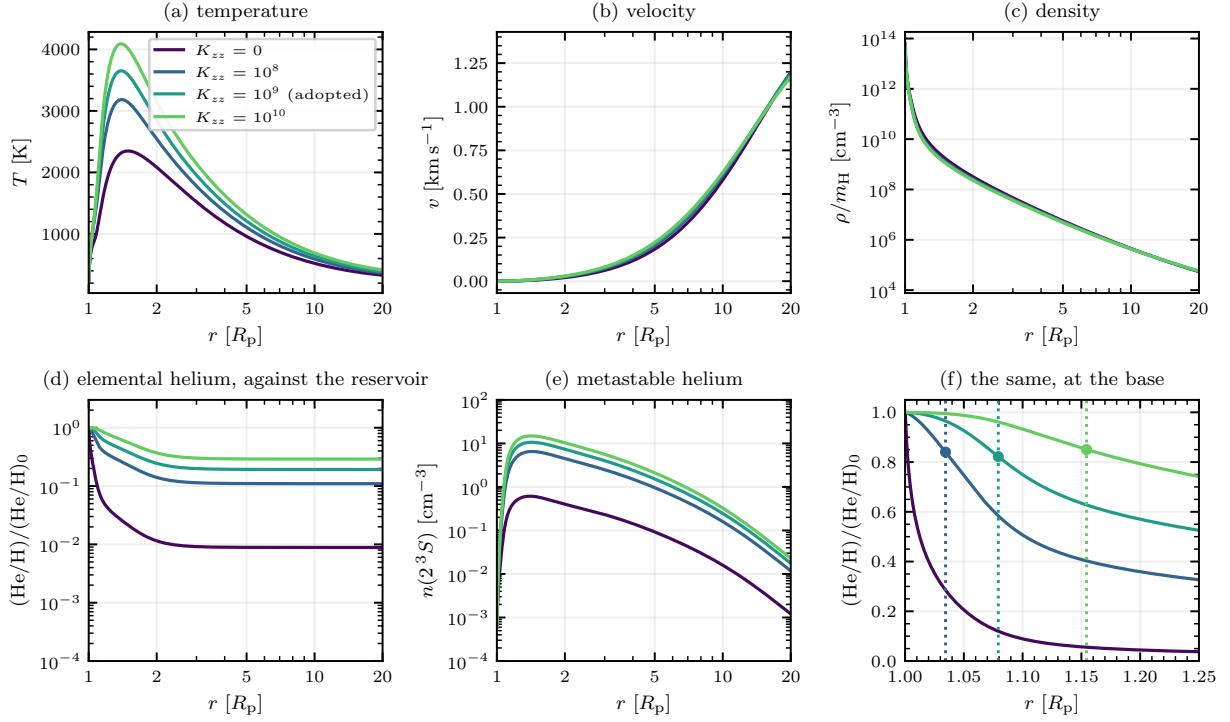


Figure 7: Four solutions differing only in the eddy coefficient, at a fixed reservoir $\text{He}/\text{H} = 0.55$ with binary H/He diffusion active: $K_{zz} = 0, 10^8, 10^9$ (adopted) and $10^{10} \text{ cm}^2 \text{ s}^{-1}$, dark to light. The eddy coefficient acts on the composition, not on the wind. Panels (d)–(f), the elemental helium against its base reservoir value and the metastable population: at $1.05 R_p$ the elemental ratio is 0.206, 0.745, 0.927 and 0.988 of the reservoir and at $5 R_p$ it is 0.009, 0.110, 0.193 and 0.291, while the peak $n(2^3S)$ is 0.61, 6.5, 10.6 and 14.9 cm^{-3} . In panels (b) and (c) the velocity at $10 R_p$ moves by 8% and the density at $2 R_p$ by 28%, and $\log_{10} \dot{M}$ by 0.02 dex. Panel (a) is the exception: T at $1.5 R_p$ rises from 2349 to 4006 K, because the helium the eddy term lifts is itself absorbing. Panel (f) repeats (d) on a linear axis over the base, where the dots and dotted lines mark each run’s homopause (the radius at which its own molecular coefficient equals its K_{zz}) at $1.034, 1.079$ and $1.154 R_p$, with the $K_{zz} = 0$ homopause below the base. The face-flux table the homopause is read from is written only under EXHALE_DIFFUSION_CHECK=1 and comes from models/diffusion_check/, the post-processing pass of these same four solutions repeated with that flag.

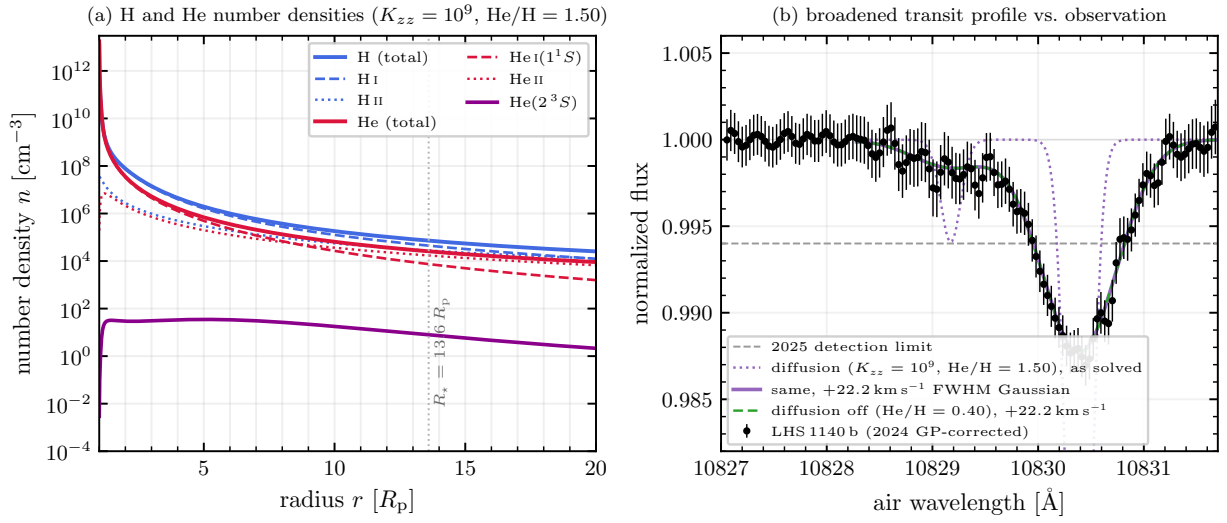


Figure 8: The EW-matched diffusive solution (models/atomic_scalar_gj1132_kzz1e9/HeH1.50; $K_{zz} = 10^9 \text{ cm}^2 \text{ s}^{-1}$). **(a)** Absolute number densities of hydrogen species (total H, H I, H II) and helium species (total He, He I(1^1S), He II, and metastable He(2^3S)) as a function of planetary radius r [R_p]. At the lower boundary ($r = 1.0 R_p$), $\text{He}/\text{H} = 1.50$; binary element diffusion causes helium to settle outward relative to hydrogen, with the local abundance falling to $\text{He}/\text{H} = 0.456$ at $2 R_p$ and 0.355 at $10 R_p$. Dotted vertical line marks the stellar disk radius ($13.6 R_p$). **(b)** The transmission spectrum as solved (dotted purple, its core running below the plotted range) and convolved with the 22.22 km s^{-1} FWHM Gaussian that reproduces the measured width (solid purple), against the measurement; the diffusion-off well-mixed curve at its own matched kernel (dashed green) is repeated from Fig. 5. The two broadened curves are indistinguishable: their kernels agree to four digits. All model curves are shifted by the fitted $v_{\text{wind}} = +2.26 \text{ km s}^{-1}$.

9.5 Broadening cannot repair a deficit in equivalent width

The same scan was applied to the p-winds curves, in their own vacuum frame and with the same three-Gaussian extractor and the same integration window (Table 7). The convolution conserves absorbing power: it redistributes the excess in velocity and leaves the equivalent width unchanged, apart from the few percent that the wings carry out of the integration window. Whether one kernel can reproduce depth and width together is therefore decided before the scan starts, by the equivalent width the curve already has at $f = 0$.

Table 7: The same added-Gaussian scan applied to the p-winds curves, with the EXHALE row of Table 4 repeated for comparison. The kernel is the one that reproduces the measured width, $0.841 \text{ \AA}$; the depths and the ratio are read at that kernel. Equivalent widths are for the unbroadened curve, over the vacuum window $10832.60\text{--}10834.20 \text{ \AA}$.

curve	EW($f=0$) [% $\text{\AA}$]	f [km s^{-1}]	red [%]	blue [%]	red/blue
p-winds best fit, turbulence on	0.837	17.8	0.926	0.339	2.73
p-winds best fit, turbulence off	0.701	20.1	0.777	0.301	2.58
p-winds depth-matched, 6100 K	0.480	20.2	0.535	0.140	3.82
EXHALE, He/H = 0.40	1.101	22.2	1.229	0.161	7.65
observed	1.108 ± 0.030	-	$1.24^{+0.22}_{-0.23}$	$0.25^{+0.14}_{-0.12}$	$6.7^{+12.7}_{-3.1}$

The p-winds best fit carries $0.837 \text{ \AA}$ against the measured 1.108 ± 0.030 , 24% short. At the kernel that reproduces the width, $f = 17.8 \text{ km s}^{-1}$, the red depth is 0.926% against $1.24^{+0.22}_{-0.23}$, about 1.4σ low. The broadening moves the disagreement rather than removing it. Unbroadened, the curve is too deep in the blue component (0.534% against $0.25^{+0.14}_{-0.12}$, some 2σ high) and too narrow. Broadened to the measured width, the blue component falls to 0.339% , inside 1σ , and the red component drops below the measurement. The blue side improves, the red side does not, and the total is fixed.

The intensity ratio barely moves, 2.57 at $f = 0$ against 2.73 at the matched kernel. It is the amplitude ratio of a saturated pair, set by how the two components share the optical depth, and a common convolution acts on both alike. It stays well below the measured $6.7^{+12.7}_{-3.1}$, though the observed interval is wide enough that this alone is not a discriminator.

Matching the depth as well would require 24% more absorbing power, and that is not something a kernel supplies: it is a change in $\dot{M}$, temperature or composition. Fitting the broadening as a free parameter alongside those (which is what the measurement asks for) would presumably shift the C26 parameter estimates, since the composition is constrained through the same absorbing power that would have to rise.

The asymmetry with EXHALE is not a statement about one code outperforming the other. The EXHALE curve entered the scan with its equivalent width already set to the observed value by the choice of He/H = 0.40 (Sect. 9.2), so only the shape remained free, and one kernel settled it. The p-winds curve entered at the composition C26 report, whose equivalent width falls short, so no kernel could settle both. The two scans test the same physical statement (broadening redistributes, it does not create) from two different starting points.

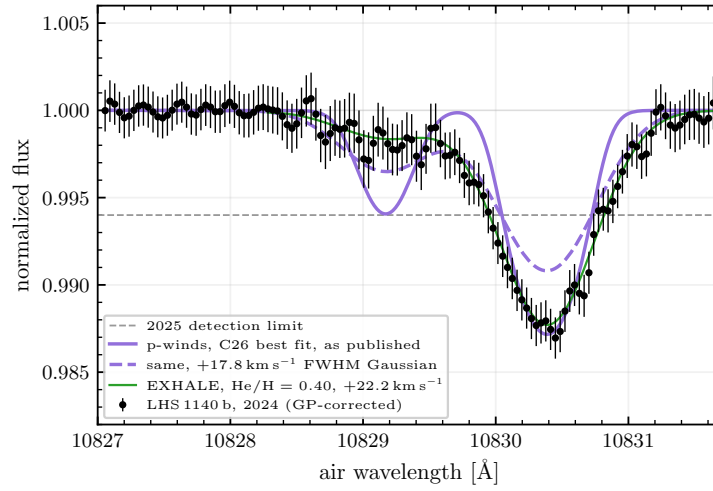


Figure 9: The C26 p-winds best fit as published (solid purple) and convolved with the 17.8 km s^{-1} FWHM Gaussian that reproduces the measured width (dashed purple), against the measurement. The EXHALE $\text{He}/\text{H} = 0.40$ curve at its own matched kernel (22.2 km s^{-1} , thin green) is repeated from Fig. 5. Broadened to the observed width, the p-winds curve passes above the red core, so the width is matched and the depth is not. All curves are shifted by the same $v_{\text{wind}} = +2.26 \text{ km s}^{-1}$.

9.6 The adiabatic-cooling population bump, looked for

Schulik & Owen (2025) propose that the missing width is already contained in the metastable population itself. Writing the steady state of their Eq. (1) as

$$n_{2^3S}(T_e) = \frac{\alpha(T_e)}{q(T_e) + t_{\text{ion}}/n_e} n_{\text{He}^+}, \quad (5)$$

they note that as adiabatic expansion drops the temperature “from $\simeq 10^4 \text{ K}$ to $\simeq 10^3 \text{ K}$ ”, “a change in slope in the population occurs as the dominant triplet loss rate $q(T_e)$ drops significantly at lower temperatures”: the metastable fraction climbs above the curve computed with the rates frozen at the peak temperature. The rise is “quenched” where the advection timescale overtakes the excitation to 2^1S , after which $\dot{m}_{2^3S} = 4\pi r^2 v n_{2^3S} m_{\text{He}}$ falls through ionization and the $2^3S \rightarrow 1^1S$ decay. They credit Yan et al. (2022) and Biassoni et al. (2024) with first identifying the effect, and their conclusion states the condition it must satisfy to be observable: “The helium triplet population bump must be situated at high velocities to help explain anomalous line widths.”

EXHALE carries the temperature dependence the mechanism needs, and it enters in the balance written above. The metastable row (`tr_triplet_row`, `ion_residual_core.f90:81`) is

$$n_e (n_{\text{He}^+} \alpha_{2^3S} + n_{\text{HeI}(1^1S)} q_{13}) = n_{2^3S} [\Gamma_{2^3S} + A_{31} + n_{\text{H}} Q_{31} + n_e (q_{31a} + q_{31b} + b_{2^3S})], \quad (6)$$

with every coefficient a function of T : $\alpha_{2^3S} = 2.10 \times 10^{-13} (T/10^4)^{-0.778}$ (`Cool_coeff.f90:985`), the collisional losses q_{31a} ($2^3S \rightarrow 2^1S$, 0.80 eV) and q_{31b} ($2^3S \rightarrow 2^1P$, 1.40 eV) as Bray (2000) collision strengths folded with $\exp(-\Delta E/kT)$ (`Cool_coeff.f90:920` and `943`), the destruction term Q_{31} on atomic hydrogen as the García Muñoz (2025) closed form of Eq. (2) (`ioniz_HeI23S_H` in `Cool_coeff.f90`), collisional ionization of the metastable (`Cool_coeff.f90:1091`),

and the radiative decay $A_{31} = 1.272 \times 10^{-4} \text{ s}^{-1}$ (util_ion_eq.f90:1235). Between 5000 and 1000 K the loss $q_{31a} + q_{31b}$ falls by a factor 1.3×10^3 while α_{2^3S} rises by a factor 3.5, so the ingredients of the bump are present and are not an imposed parameterization.

LHS1140b/bump_analysis.py measures the result. It reconstructs the balance above in Python from the solved profiles, recovering from the row itself the one term not written to file (the 2^3S photoionization rate Γ_{2^3S}), and then recomputes the reference curve with α , q_{13} , q_{31a} , q_{31b} and Q_{31} frozen at the peak temperature. Each transcribed coefficient was checked against the compiled Fortran over 300–10⁴ K: Q_{31} agrees to 8×10^{-16} and the other five to 1.2×10^{-5} or better, the residue being the single-precision literals the Fortran carries.

Table 8: The metastable bump in the solved winds. Columns 2–4 locate the n_{2^3S} maximum in the equilibrium profile and give the temperature and outflow speed there; the last three characterize the absorber-weighted ($w = n_{2^3S} r^2 dr$) distribution of $|v_r|$ over the advection-corrected 2^3S column that the transit tool integrates. The He/H = 1000 solutions have no outward density maximum (their n_{2^3S} peaks at the base), and are marked †. The six runs are the well-mixed groups of the model tree, models/atomic_scalar_gj1132_wellmixed and models/atomic_scalar_gj699_wellmixed, all certified (info=0, $\|R\| \leq 1.7 \times 10^{-7}$).

case	r_{peak} [R_p]	T [K]	v [km s ⁻¹]	med. $ v_r $ [km s ⁻¹]	> 2 [%]	> 5 [%]
<i>GJ 1132 proxy SED</i>						
He/H = 0.55	5.57	1616	0.28	0.41	0.0	0.0
solar He/H	4.17	1367	0.15	0.44	0.0	0.0
He/H = 1000	1.08 [†]	3258	0.00	0.33	0.0	0.0
<i>GJ 699 proxy SED</i>						
He/H = 0.050	5.82	917	1.56	2.49	63.8	0.0
solar He/H	5.74	982	1.50	2.41	62.1	0.0
He/H = 1000	1.05 [†]	3166	0.00	1.37	9.9	0.0

The bump is there. In every case the metastable fraction $f_3 = n_{2^3S}/n_{\text{He}}$ rises monotonically outward across the whole cooling leg, from the temperature maximum to a maximum of f_3 at 9–15 R_p (a rise of $1.2\text{--}4.4 \times 10^3$ in the equilibrium profiles and 20–140 in the advection-corrected ones), and the rise is attributable to the temperature dependence in the way Schulik & Owen describe. On the GJ 1132 proxy at the well-mixed He/H = 0.55 case the equilibrium maximum sits at 5.57 R_p where $T = 1616$ K, against $T_{\text{max}} = 4789$ K at the inner peak: the temperature has fallen to 0.34 of its peak, which is the regime the mechanism asks for.

The quench is visible too: in the advection-corrected profiles the density maximum has moved inward to 1.05–1.8 R_p and n_{2^3S} then falls monotonically, although $\dot{m}_{2^3S}$ itself does not drop past the peak on the GJ 1132 cases and falls by 33 per cent only on the helium-dominated He/H = 1000 run of the brighter spectrum.

The velocity condition, however, is not met, and the shortfall is large. At the bump the outflow is moving at 0.28 km s⁻¹ on the GJ 1132 proxy and 1.6 km s⁻¹ on the GJ 699 proxy. Weighting by absorbers over the whole 2^3S column, the median $|v_r|$ is at most 0.4 km s⁻¹ and 2.5 km s⁻¹ on the two spectra, and the largest $|v_r|$ anywhere in the column is 1.2 and

3.9 km s^{-1} . No part of the column on either spectrum reaches 5 km s^{-1} , against the $\sigma = 9.4 \text{ km s}^{-1}$ of extra broadening the measurement demands (Sect. 9.2). Radial velocity is the generous choice here: the line-of-sight projection of a radial field over the transit chord is smaller still. Figure 10 draws the three statements together.

So the bump exists in these solutions and contributes nothing measurable to the width. That is consistent with what Schulik & Owen themselves report for the coolest hosts: for their M2 case they find that “the transit signal probes the same velocity region for both treatments of thermodynamics, making the line insensitive to velocity”, and they conclude that K stars, not M stars, are the favorable testbeds: G-type hosts suppress the bump by photoionizing the triplet, and M-type hosts have weaker tidal fields and so rarely drive fast enough outflows. LHS 1140 is an M4.5 dwarf, cooler than the M2 they model, and its planet is a temperate super-Earth rather than a hot giant. The mechanism appears to be present and correctly captured; the host is the wrong one for it to help.

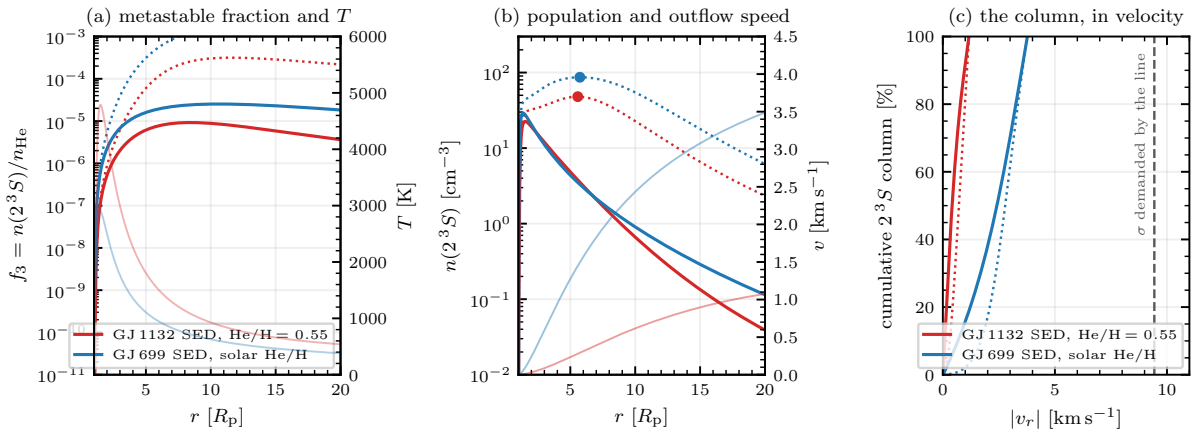


Figure 10: The adiabatic-cooling metastable bump in the EXHALE solutions (solid: advection-corrected, the profiles the transit integral uses; dotted: local equilibrium; faint curves on the right-hand axes). Two cases of Table 8 are drawn: He/H = 0.55 on the GJ 1132 proxy and solar He/H on the GJ 699 one. (a) The metastable fraction rises outward over the cooling leg, from the temperature maximum out to a maximum of f_3 at 11.6 and $11.8 R_p$ in equilibrium, which is the bump in the sense of Schulik & Owen (2025). (b) The equilibrium n_{2^3S} peaks at 5.6 – $5.7 R_p$ (markers); advection moves the maximum inward, to 1.4 – $1.5 R_p$, and quenches the outer rise. (c) The absorber-weighted distribution of $|v_r|$ over the 2^3S column: the whole column lies below 4 km s^{-1} , far short of the $\sigma = 9.4 \text{ km s}^{-1}$ the measured width demands.

9.7 The width in the wider He 10830 literature

The comparison so far has been between this model and this measurement. Against the published population, the measured width is ordinary and it is the model that is anomalous.

The measured width is typical. Few He 10830 analyses fit the width freely rather than predicting it; where they do, the answer is 19 – 33 km s^{-1} FWHM. Table 9 collects them. LHS 1140 b’s $23.9^{+7.2}_{-4.3} \text{ km s}^{-1}$ is an unremarkable member of that set. Guilluy et al. (2024) is the only survey that tabulates a free FWHM for several planets at once; their GJ 3470 b entry should not be taken at face value, since they write that “The larger FWHM for GJ 3470 b is believed

to be attributed to residuals from OH line contamination”. The entries are quoted as published (different instruments, different extractors, no common deconvolution of the instrumental profile), so the clustering is a statement about published numbers rather than a uniformly reduced sample.

Table 9: Published He 10830 transit widths where the width was fitted freely. Conversions use 10833 Å. The last two rows are velocity dispersions of a different definition, given for scale, not as FWHM.

system	instrument (R)	FWHM [km s^{-1}]	source
LHS 1140 b	WINERED (68,000)	$23.9^{+7.2}_{-4.3}$	this memo, on the released spectrum
WASP-107 b	GIANO-B ($\sim 50,000$)	$32.6^{+4.2}_{-3.6}$	Guilluy et al. (2024), Table A.3
HAT-P-11 b (3 nights)	GIANO-B	29.3/19.0/22.9	Guilluy et al. (2024), Table A.3
WASP-69 b	GIANO-B	26.6	Guilluy et al. (2024), Table A.3
GJ 3470 b	GIANO-B	60.3	Guilluy et al. (2024); OH residuals
4 young sub-Neptunes	NIRSPEC (32,000)	27–33	Zhang et al. (2023)
SPIRou detections	SPIRou	~ 22	Masson et al. (2024)
HD 189733 b	CARMENES	$b = 15.1 \pm 0.6$	Salz et al. (2018), Table 3
HAT-P-32 b	CARMENES	$v_{t,a} = 15.2 \pm 3.1$	Czesla et al. (2022)

Zhang et al. (2023) treat that width as *evidence for photoevaporation*: “The width is the most consistent metric and is within 10% of 1.1 Å for all planets”, because “For a Parker outflow, the sound, bulk outflow, and thermal speeds are all close to $\sqrt{2k_B T/\mu m_H} = 9.4 \text{ km s}^{-1}$ for $T = 7000 \text{ K}$ and $\mu = 1.3$, leading to an FWHM of $2.355 \text{ sigma} \sim 22 \text{ km s}^{-1}$.” That argument needs a 7000 K hydrogen-dominated transonic outflow. This solution has 5200 K at $2R_p$, a mean mass of 2.6–3.5 amu because of the helium enrichment, and an outflow of 0.024 km s^{-1} there; the same construction returns the model’s 2.4 km s^{-1} , not 9.4.

The turbulence term the field offers. There is a single turbulence formula in this literature, $v_{\text{turb}} = \sqrt{5kT/3m}$ with m the mass of a helium atom (Lampón et al. 2020, Eq. 16), adopted unchanged by p-winds as its Eq. (21) (Dos Santos et al. 2022). It is the adiabatic sound speed of a monatomic gas, that is, a Mach number of one, and it is a fixed function of T , not a free parameter. Evaluated on this solution it gives 4.2, 4.2, 3.6, 3.0 and 2.7 km s^{-1} at 1.2, 2, 3, 5 and $8R_p$, with a metastable-weighted mean of $3.8\text{--}4.0 \text{ km s}^{-1}$ over $1\text{--}8R_p$. Against the required $\sigma = 9.34 \text{ km s}^{-1}$ that is short by a factor 2.1–3.7 across the absorbing region and 2.3–2.5 on the absorber-weighted average. The measurement asks for Mach 2–4 and the term on offer is Mach 1. It cannot absorb the requirement, whatever one concludes about its physical justification. Three published analyses report the same failure in their own systems: Pallé et al. (2020) on GJ 3470 b (“The inclusion of the broadening of the lines due to turbulence ($v_{\text{turb}} = \sqrt{5kT/3m}$, where m is the mass of a helium atom), in addition to the standard Doppler broadening, was not enough to explain the measured broadening in the observations”), Lampón et al. (2021) on HD 189733 b (“Including a turbulence broadening component (see Sect. 3.3), the profile broadens, but it is still narrower than the measured absorption. . . . Hence, our hydrodynamic model alone is not able to explain the width of the absorption profile.”), and Czesla et al. (2022) on HAT-P-32 b (“Thermal broadening alone remains insufficient to explain the observed line width, even if the turbulence term is included.”).

Where a width is instead fitted freely, it comes back two to three times larger: Salz et al. (2018) obtain $b = 15.1 \pm 0.6 \text{ km s}^{-1}$ (extended model, $f = 20\%$) and 11.3 ± 0.5 (compact) at mid-transit for HD 189733 b, and Czesla et al. (2022) obtain $v_{t,a} = 15.2 \pm 3.1$, 11.2 ± 3.1 and $11.7 \pm 1.0 \text{ km s}^{-1}$ for HAT-P-32 b. Our own 9.34 km s^{-1} sits at the low end of that set.

An independently written code on another planet. Taylor et al. (2026) apply a 1D self-consistent model of the same class (energy balance, lower-atmosphere coupling, multispecies diffusion, non-LTE $\text{H}(n=2)$) to HD 189733 b and report the same shortfall: “several prior analyses have matched the HD 189733 b width by introducing ad hoc (micro)turbulent or Gaussian broadening terms; here, we show that this step remains necessary even in our self-consistent framework”, at “a factor of $\sim 2\text{--}3$ ” in width. The axes they varied are the axes varied here: “We tested variations in the free parameters (heating efficiency, K_{zz} , stellar activity level/XUV input, base composition, and lower-boundary placement): configurations that increase the width either deepen the core transit depth too much or violate the $\text{H}\alpha$ constraints, and none simultaneously reproduce both core depth and width.” Their proposed mechanism is stated by them as speculative, “Given that excited He is mostly produced by recombination of helium ions, this broadening can be driven by plasma turbulence . . . While these ideas are speculative, we encourage further exploration of plasma effects and magnetic field interactions”, and its first clause is the same physics Sect. 9.8 derives independently below. Rumeniskikh et al. (2022), as reported there, reached the observed width in 3D only by prescribing zonal jets of $10\text{--}20 \text{ km s}^{-1}$ together with reduced hydrogen line cooling; Schulik & Owen (2025) offer the adiabatic-cooling population bump as the one route with no free parameter, which Sect. 9.6 shows is present here and worth $0.25\text{--}1.5 \text{ km s}^{-1}$.

What follows. The width shortfall is not specific to EXHALE and not specific to LHS 1140 b. It is an unresolved problem of the 1D steady-state model class. The evidence for that statement is threefold: the measured width is a typical member of the published population (Table 9); the deficit is reproduced at the same factor of 2–3 by an independently written code, on a different planet, around a star at the opposite end of the activity range, after testing the same axes (Taylor et al. 2026); and the field’s own remedy is a term that is Mach 1 by construction and that its users report in print to be insufficient. Every published route to the observed width adds velocity by hand (a sonic turbulence term, a free Gaussian, or prescribed jets), and none of them follows from the 1D energy equation.

One citation needs a note here, as an observation rather than a correction to anyone’s result. Lampón et al. (2023) and Yan et al. (2024) both cite Salz et al. (2018) as a source of the $v_{\text{turb}} = \sqrt{5kT/3m}$ convention, but “turbulen” does not occur anywhere in the published Salz et al. (2018), which fits a free Doppler parameter instead. The convention itself is stated in Salz et al. (2016), Eq. (5); the 2018 paper appears to be intended as the source of the measured *width*. As written, the attribution suggests that a Mach-1 turbulence term reproduced the HD 189733 b width, when in that paper it was a free 15.1 km s^{-1} Doppler parameter.

9.8 The size of the non-thermal candidates

If the missing velocity is not thermal and not in the 1D energy equation, the next question is which non-thermal mechanism could supply $\sigma = 9.34 \text{ km s}^{-1}$. The estimates below are order-of-magnitude requirements computed on the reference solution (flux_closure/heh11p1/k01); the full set, including the stellar-wind standoff, the turbulent and kinetic

power budgets, charge exchange and radiation pressure, is in docs/lhs1140b_width_nonthermal_candidates.md.

Magnetic. Setting the Alfvén speed equal to the required dispersion, $v_A = B/\sqrt{4\pi\rho} = 9.34 \text{ km s}^{-1}$, and reading ρ off the advection-corrected profile, the field needed in the line-forming region is $B = 0.423, 0.118, 0.043$ and 0.016 G at $1.2, 1.5, 2$ and $3 R_p$. Continued inward as a dipole, $B_s = B(r/R_p)^3$, those are surface fields of $0.73, 0.40, 0.34$ and 0.43 G : a shallow minimum near $2 R_p$, so a dipole of a few tenths of a gauss would put v_A near the required value over the whole region at once. As a scale comparison that is unremarkable: it lies between Earth’s surface dipole ($\approx 0.3 \text{ G}$) and Jupiter’s ($\approx 4 \text{ G}$). The difficulty is what such a field would do rather than whether a planet can have it: the plasma β at that strength is $0.28\text{--}0.38$ over $1.2\text{--}3 R_p$, so the field would be dynamically dominant and would restructure the outflow rather than decorate it, and this solution does not model that. The wave energy flux required to sustain $\sigma = 9.34 \text{ km s}^{-1}$, $\rho(3\sigma^2)v_A$ through a sphere, is $12\text{--}870$ times the solution’s total XUV heating of $1.22 \times 10^{20} \text{ erg s}^{-1}$ over the same radii, which only a star–planet interaction driven by a wind of at least solar strength at this orbit could feed. The field strength does not exclude this route. The power the field would have to be given does constrain it, as does the fact that the comparison is being made for a 131 d rotator with $L_X/L_{\text{bol}} = 6.5 \times 10^{-6}$.

Where a metastable atom can be made. This is the constraint that bears hardest on any mechanism that manufactures fast helium somewhere else (a stellar-wind interaction region, a charge-exchange or energetic-neutral population), and brings it into the beam. $\text{He } 2^3\text{S}$ is fed by He^+ recombination and destroyed by Penning ionization with atomic hydrogen (tr_triplet_row in ion_residual_core.f90). The collisional channel from the singlet ground state supplies a fraction 1×10^{-13} of the source at $1.2 R_p$ and 2×10^{-14} at $2 R_p$, falling to 10^{-33} by $5 R_p$: the metastable population is a recombination product without qualification. Table 10 evaluates the two timescales on the solution, with $\alpha_{2^3\text{S}} = 2.10 \times 10^{-13} (T/10^4)^{-0.778}$ and the García Muñoz (2025) destruction rate of Eq. (2). Both columns are recomputed here from the solved profiles. The coefficient replacement leaves alone the wind the table stands on: between the last solution of this case that predates the replacement (flux_closure/heh11p1/k00) and the re-solved one T agrees to 1.2 per cent and t_{rec} to 1.8 at every radius, because the metastable is a trace population and not an energy-budget term. The last three columns are not preserved that way and should not be: they *are* the destruction rate, and it is the quantity Sect. 3.5 replaced, so they move by factors of $1.0\text{--}1.8$ between the two fits. A reconstruction that reproduced them would be evidence that the replacement had not been applied.

Two requirements follow, and they are severe. First, recombination is slower than destruction by $7.5 \times 10^3\text{--}4.8 \times 10^4$ everywhere, so only a tiny standing fraction of the helium is ever metastable and the population is set locally and continuously. Second, and more restrictive, within its own lifetime an atom moving at the required speed travels $0.05 R_p$ at $2 R_p$ and $0.30 R_p$ at $3 R_p$ (Table 10, last column). The inner metastable helium, which carries roughly a third of the absorption, cannot have been manufactured anywhere but where it is seen. A stellar-wind or energetic-neutral population would therefore have to be re-ionized and recombine *inside* the absorbing region: the recombination time there is $10^6\text{--}10^7 \text{ s}$ against a crossing time of $5 R_p$ at 9.34 km s^{-1} of $5.9 \times 10^3 \text{ s}$, short by three to four orders of magnitude. The candidate is not excluded as a source of *some* broad outer emission, but it cannot be the source of the bulk of the line.

Table 10: Metastable helium source and sink timescales in the reference solution, and the distance an atom moving at the required 9.34 km s^{-1} covers within its total lifetime. The reference solution is re-solved on the current binary (LHS1140b/archive_20260830/exhale/refresh_j96/misc/fc_heh11p1). The composition-renormalized split of *Update_EXHALE* Sect. 96 moves T here by at most 0.3 per cent and t_{rec} by at most 1.4: this is a trace-population diagnostic on a wind that does not move.

$r [R_p]$	$T [\text{K}]$	$t_{\text{rec}} [\text{s}]$	$t_{\text{Penning}} [\text{s}]$	$t_{\text{life}} [\text{s}]$	$L = 9.34 \text{ km s}^{-1} t_{\text{life}} [R_p]$
1.2	5111	8.4×10^4	1.72	1.72	0.0014
2.0	5283	8.9×10^5	59.6	59.1	0.049
3.0	3866	3.5×10^6	3.8×10^2	3.6×10^2	0.30
5.0	2662	1.6×10^7	2.8×10^3	2.1×10^3	1.7
8.0	2088	6.0×10^7	1.5×10^4	5.1×10^3	4.2
20	538	2.6×10^8	2.8×10^5	7.7×10^3	6.3

Three-dimensional flow. With $P = 24.737 \text{ d}$ and tidal locking, $\Omega = 2.94 \times 10^{-6} \text{ s}^{-1}$ and the corotation speed Ωr is 0.032 km s^{-1} at the surface, 0.098 at $3 R_p$ and 0.94 at the $29 R_p$ domain edge: already measured in Sect. 9.3 and two orders below the requirement. A pressure-driven day–night flow is bounded above by the thermal energy available, i.e. by a few times the $4\text{--}5 \text{ km s}^{-1}$ sound speed in the complete-conversion limit, and independently by the measured line shift, whose posterior caps any coherent line-of-sight flow of the whole absorbing column at 4.7 km s^{-1} . Both are small here for the same reason: the driver of 3D circulation is the day–night temperature contrast, and LHS 1140 b is a temperate planet at $T_{\text{eq}} = 226 \text{ K}$ around an M4.5 dwarf, with an irradiation and a rotation rate one to two orders of magnitude below the hot Jupiters for which $10\text{--}20 \text{ km s}^{-1}$ jets are prescribed. Taylor et al. (2026) reach the same conclusion for HD 189733 b, where the case is far more favorable: “typical GCM wind speeds (a few kilometers per second) are unlikely to supply the additional $\sim 12 \text{ km s}^{-1}$ required to match the observed width.”

What this does not settle. These are requirement estimates evaluated on *this* solution’s density, temperature and ionization structure. They say what each mechanism would have to be given in order to produce 9.34 km s^{-1} here, and on that basis none of them is comfortable: the magnetic route needs a power it has no obvious source for around this star, the fast-neutral route cannot make metastables where the absorption is, and the 3D route is short by two orders of magnitude. They do not identify which mechanism is in fact responsible, and they do not exclude the possibility that the absorbing gas is not this solution’s gas at all: the measured pre-ingress signal, at least as broad as the in-transit one at a phase when the planet occults nothing, is direct evidence that some of the signal comes from a structure no 1D radial model contains. Which mechanism supplies the width is not determined by this work, and that is also the state of the literature.

9.9 What would have to change

The measured line wants gas that is fast enough to be broad but not so extended that it absorbs too much. EXHALE at this XUV flux delivers neither. The candidates, in the order they should be tested:

1. *More energy*: tested (Sect. 11.), and it does not do it. The GJ 699 proxy carries five times the XUV of the GJ 1132 proxy used here, and the paper itself runs both. Solving the scan again on it raises $\dot{M}$ sevenfold and moves the composition estimate by an order of magnitude, but leaves the required extra broadening at 21.95 km s^{-1} against 22.18: the width gap is not an energy-input problem.
2. *A different helium distribution*: tested (Sect. 9.4), and it does not do it either. Letting helium settle against binary diffusion and lifting it back with an eddy term redistributes the absorber by orders of magnitude and moves the inferred composition with it, but the required broadening stays at 22.19 km s^{-1} : it changes how much helium is there, not how fast it moves.
3. *Heating the code does not carry*. Taylor et al.'s trapped-ion turbulence, or any magnetically mediated dissipation, is outside a 1D hydrodynamic energy equation by construction. Sect. 9.8 sizes it: the field needed to make $v_A = 9.34 \text{ km s}^{-1}$ is a few tenths of a gauss, which is ordinary, but the wave power that field must carry is 12–870 times the XUV heating.
4. *Geometry*. A line-of-sight velocity field from day–night or zonal flow is a 3D effect; Rumenskikh et al. reach the observed width with $10\text{--}20 \text{ km s}^{-1}$ jets, which is the right order. For this planet, though, rotation gives $0.03\text{--}0.10 \text{ km s}^{-1}$ and the measured line shift caps any coherent line-of-sight flow at 4.7 km s^{-1} (Sect. 9.8).
5. *Adiabatic-cooling population bump*. Looked for (Sect. 9.6): the bump is present in these solutions, but it sits at $0.28\text{--}1.6 \text{ km s}^{-1}$, so it does not supply width.

The paper's composition constraint should be read in the same light: it inherits the isothermal assumption. Impose a 6100 K structure and only helium-rich gas absorbs enough; solve the energy balance on this spectrum and, in the well-mixed limit, the same absorption is reached near $\text{He}/\text{H} = 0.42$, and near 0.048 on the other proxy spectrum, or near 1.6 once helium is allowed to settle against the adopted eddy coefficient. No answer on that list is secure while the width is unexplained, because the width is what tells us whether the line-forming gas is the gas either model thinks it is.

10. The best each model can do

Figure 11 puts only the two best models on the measurement: the paper's p-winds model at its own best fit (the MCMC medians), and the EXHALE solution whose equivalent width matches. They fail in different ways, and neither reaches the observed width. p-winds is the broader of the two, because its temperature is imposed where the line forms, but at $0.575 \text{ \AA}$ it is still 1.5 times narrower than the measurement, and it overshoots the blue component by a factor of about three. EXHALE carries the right total absorption but concentrates it into a line a third as wide as observed, so its core is too deep. Only the second failure can be undone by broadening alone (Sects. 9.2 and 9.5).

Figure 12 shows what that EXHALE solution actually is. It is not a fast, hot outflow. The temperature peaks at 4282 K near $1.51 R_p$, comparable to the isothermal value p-winds imposes, and then falls by adiabatic expansion to 855 K by $10 R_p$ over the region that forms the line. The velocity reaches only 1.1 km s^{-1} at $20 R_p$. Both elements stay mostly neutral through the line-forming region, and helium is the *more* ionized of the two: $x(\text{He II}) = 0.03, 0.07$ and 0.16 at

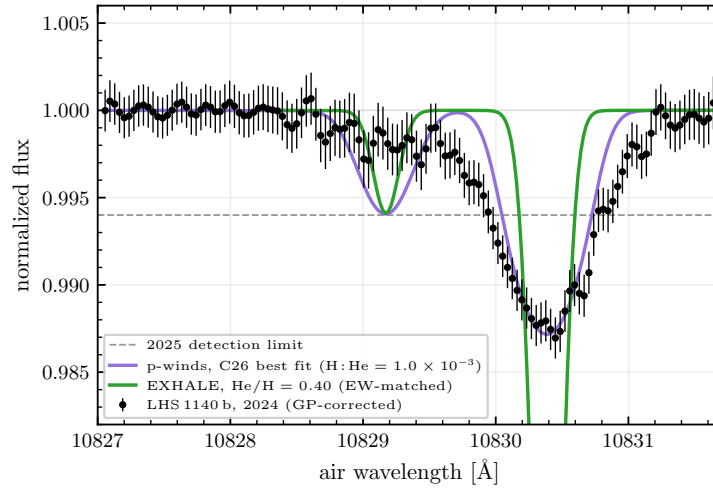


Figure 11: The two best models over the 2024 spectrum, in the presentation of the paper’s Fig. 4: p-winds at the C26 best fit (the MCMC medians) with the authors’ settings, and EXHALE at the equivalent-width-matched composition $\text{He}/\text{H} = 0.40$. Both are shifted by the fitted $v_{\text{wind}} = +2.26 \text{ km s}^{-1}$.

2, 3 and $5R_p$ against $x(\text{He II} + \text{He III}) = 0.06, 0.14$ and 0.29 . Helium carries the electrons that make the gas, and its ionization threshold is where this SED puts most of its photons. The metastable population peaks well out, at $5.2R_p$ with $n(2^3S) = 38 \text{ cm}^{-3}$ in local equilibrium. A large, cold, slow, metastable-rich envelope of this kind gives the right equivalent width with the wrong profile: plenty of absorbers spread over most of the stellar disk, but almost no velocity spread to distribute the absorption in wavelength.

11. The other proxy: five times the XUV

The first item on the list above (more energy) can be tested without adding any physics, because the paper itself runs two proxy spectra. Both stand in for LHS 1140’s unobserved XUV: GJ 1132 (used everywhere above) and GJ 699, scaled to the orbit by the catalog X-ray flux ratios $A_{1132} = 0.59$ and $A_{699} = 0.056$ following the authors’ script, the same convention adopted in Sect. 3.3. Integrated over $10\text{--}1300 \text{ \AA}$ at the orbit of the planet, the GJ 699 file carries $169.9 \text{ erg s}^{-1} \text{ cm}^{-2}$ against 33.5 for the GJ 1132 one: a factor 5.1 in XUV input.

On the EXHALE side only the spectrum changes. Each GJ 699 case is the matching GJ 1132 input with one key replaced, `Spectrum file:`. The luminosity keys in the file are irrelevant, because `sed_read` overwrites them with the integral of the spectrum it reads. The group is `models/atomic_scalar_gj699_wellmixed`, six solved reservoirs, all of them certified.

11.1 Escape rate: the retrieval value, on this spectrum

Unlike a change of composition, the brighter spectrum moves $\dot{M}$:

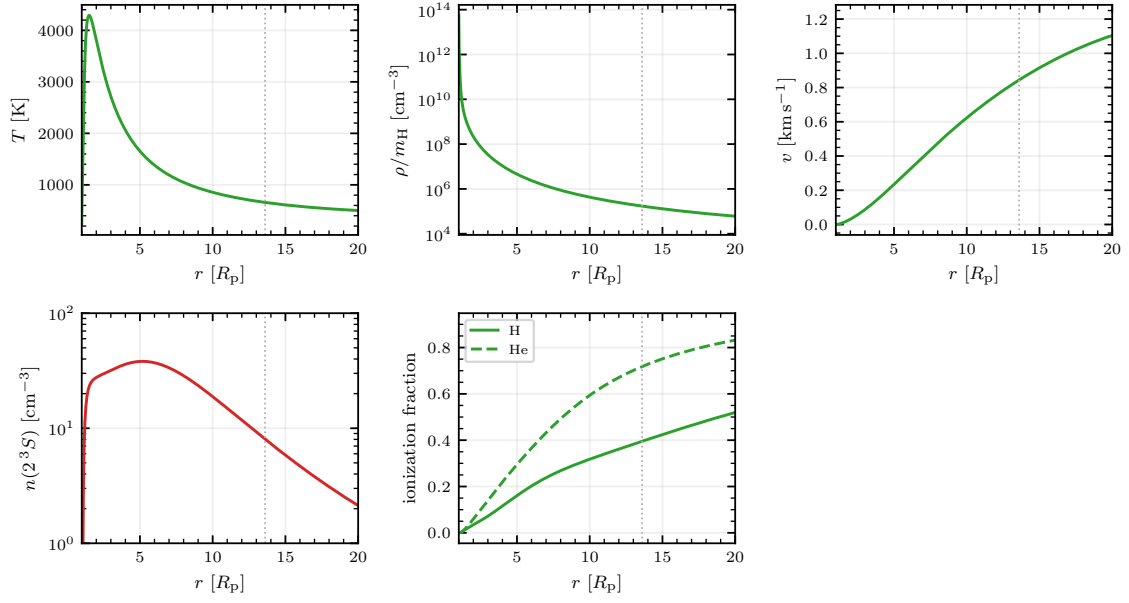


Figure 12: The EW-matched EXHALE solution (models/atomic_scalar_gj1132_wellmixed/HeH0.40, the solved rung nearest the well-mixed crossing 0.4044): temperature, density, velocity, metastable helium density, and ionization fractions. Dotted vertical line: the stellar radius.

	GJ 1132 proxy	GJ 699 proxy
$F_{\text{XUV}} (10\text{--}1300 \text{ \AA}) [\text{erg s}^{-1} \text{ cm}^{-2}]$	33.5	169.9
$\log_{10} \dot{M} [\text{g s}^{-1}], \text{He/H} \leq 1$	7.84–7.86	8.55–8.57
$\dot{M} [\text{g s}^{-1}]$	7.1×10^7	3.7×10^8
C26 retrieval on the same proxy	2.03×10^8	$4.22^{+1.14}_{-1.00} \times 10^8$

A factor 5.1 in XUV input gives a factor 5.1 in $\dot{M}$ here. On this spectrum EXHALE’s escape rate agrees with the paper’s retrieved value, $\log_{10} \dot{M} = 8.57$ at the crossing against 8.63, inside 0.06 dex and inside the retrieval’s own ± 0.11 dex. It is also 1.3 times the energy-limited estimate the paper quotes for this proxy, $2.90 \times 10^8 \text{ g s}^{-1}$. On the GJ 1132 spectrum the same code falls a factor 2.8 short of the retrieval. Whether a solved wind and an imposed one agree on $\dot{M}$ therefore depends on which proxy is assumed, and the disagreement reported in Sect. 9, is not a fixed property of the code.

11.2 Structure and line

Six compositions were solved on this spectrum, three of them ($\text{He/H} = 0.042, 0.046$ and 0.050) placed to bracket the equivalent-width crossing:

He/H	$\log_{10} \dot{M}$ [g s ⁻¹]	$T_{\max}$ [K] (at r/R_p)	$v(20R_p)$ [km s ⁻¹]	$n(2^3S)_{\max}$ [cm ⁻³] (at r/R_p)	red [%]	FWHM [Å]	EW [% Å]
0.042	8.57	2614 (1.47)	3.62	44.7 (5.8)	3.04	0.282	0.909
0.046	8.57	2645 (1.47)	3.60	48.9 (5.8)	3.29	0.282	0.986
0.050	8.57	2676 (1.47)	3.59	53.0 (5.8)	3.54	0.282	1.061
0.083	8.57	2919 (1.45)	3.49	86.7 (5.7)	5.40	0.284	1.627
1	8.55	6240 (1.34)	2.59	431 (6.8)	13.60	0.313	4.525
10 ³	8.42	9518 (1.36)	2.03	2.6×10^4 (1.0)	7.51	0.359	2.873
observed	-	-	-	-	1.254	0.841	1.108 ± 0.030

The wind is hotter and faster than on the fainter spectrum: at He/H = 0.050 it reaches 3.6 km s⁻¹ at $20R_p$ against 1.1 for the equivalent-width-matched GJ 1132 solution, and the metastable population still peaks far out, near $6R_p$.

11.3 The composition estimate moves by an order of magnitude

Interpolating the equivalent width in log–log across the ladder and solving for the measured value gives

$$\text{He/H} = 0.0526 \quad (0.0510\text{--}0.0543 \text{ on the } EW \text{ } 1\sigma), \quad \text{H:He} = 19.0, \quad (7)$$

read inside the bracket of the solved rungs at 0.050 ($EW = 1.0609$) and 0.083 (1.6271 % Å), the first of which is 1.6σ below the measurement and the second far above it, so the crossing is measured rather than extrapolated from the solar case.

The same code gives He/H = 0.4044 on the GJ 1132 proxy, an order of magnitude away, and the crossing passes the solar value on the way: the brighter spectrum wants a helium-*poor* atmosphere, $0.63\times$ solar, where the fainter one wanted a helium-rich one, $4.8\times$ solar. The measurement cannot arbitrate, because the two proxies are two guesses at the same unobserved spectrum and both reproduce the observed equivalent width once composition is free. The conclusion is therefore about the inference rather than about the planet: what the helium line constrains is the product of composition and XUV input, and any composition quoted from it (the paper’s H:He $\lesssim 10^{-3}$ as much as the values here) is conditional on the assumed SED. On the present evidence the composition inference is limited by the assumed SED.

11.4 The width does not move

On this spectrum too the width does not move. Across the whole GJ 699 ladder the red-pair FWHM is 0.282–0.359 Å against the measured 0.841 Å. Measuring the deficit the way Sect. 9.2 does (convolving with a Gaussian and asking which width reproduces the observation), the rung at the crossing needs 21.97 km s⁻¹ FWHM of added velocity spread, against 22.22 km s⁻¹ for the equivalent-width-matched GJ 1132 solution and the same 22.22 for the diffusive one of Sect. 9.4. Raising the XUV input by 5.1, $\dot{M}$ by 5 and the outflow velocity at $20R_p$ by 3 changes the required broadening by 1 per cent.

The reason is visible in the table: the outflow velocity that grows is radial and small compared with the missing spread, while the line-forming region stays cold. More energy is therefore tested and rejected as an explanation of the width. It moves $\dot{M}$ and the composition estimate, but not the profile. The remaining candidates (heating outside a 1D hydrodynamic energy equation, 3D velocity fields, and an adiabatic-cooling metastable bump) are unaffected by this test, and the last of them has since been looked for on both spectra (Sect. 9.6) and does not help either.

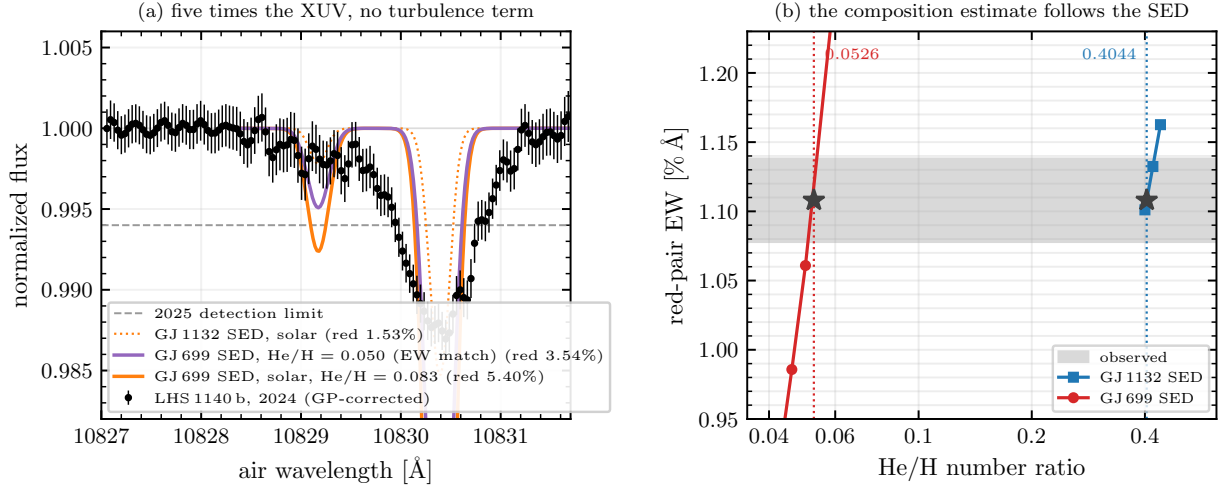


Figure 13: The GJ 699 proxy, five times the XUV of the GJ 1132 proxy used elsewhere in this memo. Left: the measurement with the EXHALE solution nearest the equivalent-width crossing 0.0526, the $\text{He}/\text{H} = 0.050$ rung, and the solar case on this spectrum, with the GJ 1132 solar case dotted for contrast; all curves without the turbulence term and shifted by the fitted $v_{\text{wind}} = +2.26 \text{ km s}^{-1}$. The lines are deeper but no wider, the GJ 699 solar curve reaching $3.5\times$ the depth of the GJ 1132 one (5.40% against 1.53%) while the added broadening the measurement demands of the two differs by 0.45 km s^{-1} , or 2 per cent (21.95 against 22.40 km s^{-1}). Right: red-pair equivalent width against composition for the two spectra, with the measured band and the two crossings; the composition the line implies moves from 0.4044 to 0.0526 when the assumed SED is changed.

12. Closing the loop: the composition as an output

Every number above took the helium abundance as an input. The scans of Sects. 6. and 9.4 chose it, solved the wind, and asked what the line did. The reservoir at the base was a parameter set by hand, and the lower atmosphere below the $1 \mu\text{bar}$ matching level was a handful of scalars. This section removes that freedom on one axis: the lower atmosphere is solved as a photochemical column, handed over as a profile, and the elemental fluxes are iterated until they are continuous across the matching level. The composition at the match is then a solution of the coupled problem rather than a number chosen for it.

The three runs of this section have since been re-run end to end (chemistry, wind, elemental-flux measurement and damped update) on Photochem 0.9.0 with the re-pinned adapter of *Update_EXHALE* Sect. 99 and on the current binary, which is what Table 14 reports. What they establish is that the closure is single-valued and that at

$K_{zz} = 10^9$ it hands back the reservoir it was given, which is a statement about the iteration rather than about the line. The ladder of Sect. 12.7, which is what the composition is read from, was re-solved on both. The implementation and its tests are recorded in docs/phase_e_flux_closure_design.md and docs/Update_EXHALE.md, Sects. 76–79; the design runs are in LHS1140b/archive_20260830/exhale/flux_closure/ and the re-run ones in LHS1140b/archive_20260830/exhale/refresh_j96/closure/.

12.1 The setup

The column below the match is solved with Photochem (Wogan 2025) on the Zahnle reaction set restricted to H/He/N/O/C. The temperature is not prescribed: clima’s AdiabaticClimate is run with solve_for_T_trop on, so the stratospheric temperature is the skin temperature of the solution. The background gas is chosen by abundance rather than by name, and on this atmosphere that is helium rather than H_2 . The climate solve is taken from a deep boundary at 20 bar, the shallowest one at which the photochemical model bottom fits inside the climate column. Chemistry and climate are run on the same elemental vector, He/H starting at 2.09 (the $K_{zz} = 10^9$ crossing of Table 5) with Lodders C, N and O, on the GJ 1132 proxy spectrum at the planet’s orbit and at the adopted constant $K_{zz} = 10^9 \text{ cm}^2 \text{ s}^{-1}$.

The result is handed to EXHALE as a profile of 102 levels from 16.73 bar to 1.10×10^{-8} bar, carrying p , r , T , n , ρ , K_{zz} , the molecular fractions and the elemental reservoirs He/H, C/H, N/H and O/H, matched to the wind at $1 \mu\text{bar}$. Sect. 12.2 reads that solution column by column, and says which of the columns the wind is actually built from. The loop is then closed in the other direction. The elemental escape fluxes EXHALE computes are imposed back on the photochemical model at its top, $\phi_{\text{El}} = \Phi_{\text{El}} / (m_{\text{El}} 4\pi r_{\text{top}}^2)$, helium on He and hydrogen on H_2 at $\phi_{\text{H}}/2$, which are the dominant carriers; the atomic-H share at that level is 10^{-4} . The pair is iterated by damped Picard ($\omega = 0.5$) on the residual $|F_{\text{measured}} - \Phi_{\text{trial}}| / |F_{\text{measured}}|$ to a tolerance of 5%.

Two things about that measurement decide whether it means anything. The elemental flux is read in the *steady* window $r \geq r_{\text{esc}}$ rather than in the interval the two models share: at a steady state $4\pi r^2(\rho X v + J)$ does not depend on radius, so a flux measured there is the flux through the matching level. The overlap window cannot carry it. From the match to the profile top the column spans 1.96 decades of pressure in $0.0098 R_p$, entirely inside the region where the wind’s mass flux is not yet its own, and the flux across it has a radial spread of 13 in F_{H} and 35 in F_{He} . In the steady window the same solution holds 191 faces at spreads of 0.44% and 0.46%.

That flatness depends on how far the wind was converged, and we measured it: at $\|R\|_{\text{tol}} = 10^{-3}$ the same solution spreads by 7.4%, which would fail the closure’s own validity test, and continuing it to 10^{-4} brings it to 0.50% and moves $\log_{10} \dot{M}$ from 7.47 to 7.50. The closure runs the wind at 10^{-4} .

The cross-check with a second chemistry code is not verified. The same handoff was set up with VULCAN (Tsai et al. 2017) on the production profile’s own $T(p)$ and K_{zz} , and VULCAN did not converge on this cold helium-rich column: its convergence measure rose from 2.65 to 2.75 against a criterion of 0.01 with the timestep pinned, the limiting cell being H_2O at the cold trap, and the run exited on its step limit. No VULCAN composition at the match was produced and no wind was run on one, so the cross-code spread is not measured here. The record is LHS1140b/lower_profile_vulcan/vulcan_work/STOPPED.txt.

12.2 The column the chemistry returns

The settings above produce one file, and everything EXHALE does on the profile branch rests on it, so the solution itself is read here column by column. Figs. 14 and 15 draw every column of the schema except the two elemental-flux columns, which are constants carrying the trial fluxes the header states. The file measured here is LHS1140b/lower_profile/lower_atmosphere_profile.dat, the one-way handoff at iteration 0; the closed rungs of Sect. 12.7 carry their own copies and are compared at the end of this section.

Grid and thermal structure. 102 levels from 16.73 bar to 1.098×10^{-8} bar: the photochemical model's own domain, sitting inside the 20 bar column the climate step was taken from. Temperature runs from 520.3 K at the deep boundary to a floor of 185.05 K, which it reaches at 0.94 bar and holds at every level above. The stratosphere is isothermal at the skin temperature, and the tropopause the climate solve reports, 1.031 bar, is one level below the first isothermal level. K_{zz} is $10^9 \text{ cm}^2 \text{ s}^{-1}$ at all 102 levels, so the eddy coefficient the wind runs at and the one the column was mixed at are the same constant. The column spans 553.6 km in radius, 0.15608 to 0.16400 R_J , with the 1 μbar match at 0.16240 $R_J = 1.782 R_\oplus$, which fixes the base radius of the wind. Density falls from $n = 2.33 \times 10^{20} \text{ cm}^{-3}$ ($\rho = 1.40 \times 10^{-3} \text{ g cm}^{-3}$) to $4.30 \times 10^{11} \text{ cm}^{-3}$ (2.58×10^{-12}).

The column is chemically almost frozen. Between the deep boundary and the match, 17 decades of pressure, q_{H_2} moves from 0.19258 to 0.19264, q_{CH_4} from 1.072×10^{-4} to 1.025×10^{-4} and q_{NH_3} from 3.158×10^{-5} to 2.919×10^{-5} . What changes at all is water, which condenses, and a set of trace photochemical products at the very top. In an eddy-mixed column at $K_{zz} = 10^9$ that is what one expects, and Sect. 12. makes the same point from the other side: the eddy term homogenizes the column all the way to the match.

Hydrogen stays molecular to the top. q_{H} rises from 5.9×10^{-21} at the deep boundary to 6.5×10^{-6} at the match and 4.0×10^{-5} at the model top, four to five decades below q_{H_2} throughout: there is no level in this column at which atomic hydrogen overtakes H_2 . The base EXHALE is handed is an H_2 base in that sense. It does not enter the wind that way in these runs: EXHALE's molecular chemistry is off, and q_{H_2} is read but inert (see below).

The composition is cold and reducing. At 520 K and 17 bar CO is already negligible, $q_{\text{CO}} = 8.1 \times 10^{-18}$, and CO_2 smaller still. Carbon is CH_4 and nitrogen is NH_3 , the opposite of the CO/ N_2 -dominated deep chemistry of a hot Jupiter. Counting carriers at the match against the hydrogen-nucleus fraction of the gas, $2q_{\text{H}_2} + q_{\text{H}} = 0.38532$:

- carbon: CH_4 gives 2.661×10^{-4} per H, 99.7% of the carbon the profile's carrier columns hold, with C_2H_2 and HCN supplying the rest;
- nitrogen: NH_3 gives 7.576×10^{-5} per H, 96.3%, with N_2 at 3.4%;
- oxygen: the water that survived the cold trap gives 4.831×10^{-7} per H, 99.9%.

Those carrier sums come to 96.0, 96.0 and 97.6 per cent of the elemental columns $\text{C}/\text{H} = 2.778 \times 10^{-4}$, $\text{N}/\text{H} = 8.187 \times 10^{-5}$ and $\text{O}/\text{H} = 4.957 \times 10^{-7}$. The remainder sits in species the profile does not carry as columns, so the carrier accounting here is a decomposition of the elemental ratios rather than an independent measurement of them.

This is the direct link to Sect. 12.6. The C and N that cool the wind and take its line down by $2.6\times$ are held in the column as CH_4 and NH_3 , both of which the solution keeps in the gas phase across the trap, so they cross the tropopause intact and arrive at the match at essentially their deep values. Oxygen, whose carrier is water, does not.

Photochemistry acts only in the top decades. From the tropopause to the match, HCN goes $1.3 \times 10^{-13} \rightarrow 6.5 \times 10^{-8}$, N_2 $9.2 \times 10^{-9} \rightarrow 5.2 \times 10^{-7}$, C_2H_2 $2.5 \times 10^{-13} \rightarrow 1.1 \times 10^{-7}$, CO $5.2 \times 10^{-16} \rightarrow 1.8 \times 10^{-10}$ and OH $4.5 \times 10^{-27} \rightarrow 7.1 \times 10^{-11}$: several decades each, and all of them still trace. They do not move the elemental budget: the carrier accounting above puts every one of them below 4% of its element.

What EXHALE consumes from this file was read in the code rather than inferred. The reader in `files_IO/lower_atmosphere_profile.f90` requires the columns `p`, `r`, `T`, `n_tot`, `rho`, `Kzz`, `q_H2`, `q_H` and `X_He` to be present, and `apply_lower_atmosphere_profile` (`input_read.f90`) then takes, *at the matching level only*, `T` into T_0 , `r` into R_0 , `q_H2` into `q_h2_base`, `X_He` into `HeH`, and `X_C`, `X_N`, `X_O` (and any other `X_<El>` the file carries) into the metal abundances, through the same door `metals.inp` uses. `p_match_bar` comes from the header. The `Kzz` column is the only one read over the whole profile, interpolated onto the radial grid through the file's own `r` column (`eddy_diffusion_on_grid`); a scalar `He_Kzz`: in `input.inp` is then announced as inert.

Everything else goes unread. `n_tot` and `rho` are required by the schema but never consumed (the base density is n_0 of `input.inp` and the base particle count follows from T_0 , `He/H` and the elemental ratios), `q_H` is required and never consumed, the two flux columns `F_H` and `F_He` are not read by the Fortran at all (the closure driver reads them on the Python side), and none of the molecular columns beyond `q_H2` is read. `q_H2` itself only acts when a molecular base is switched on: it reaches the base EOS through `h2_mixing_ratio_base`, which `comp_ntot_bc` calls only under `molecular_base`. In the runs of this memo that flag is off, and the resolved report confirms it: `mass_per_H` = 9.37393 and `ntot_bc` = 1.000116 are $1 + 4\text{He}/H + \sum X_{El}A_{El}$ and $(1 + \text{He}/H + \sum X_{El})/(1 + \text{He}/H)$ with no H_2 term. So of the 23 columns, six numbers at one level plus one full column are what the wind is actually built from.

The closure rungs each carry their own solved column, and raising the reservoir changes that column in three measurable ways (Fig. 15d–f). The deep boundary cools, $520.3 \rightarrow 439.4 \rightarrow 387.9$ K at 16.73 bar for `He/H` = 2.09, 5.01 and 10.31, while the stratosphere stays at 185 K. H_2 is diluted by number, $q_{H_2} = 0.1926 \rightarrow 0.0907 \rightarrow 0.0462$ at the match, with `C/H` and `N/H` unchanged to four digits because they are stated per hydrogen nucleus. The cold trap loosens: `O/H` at the match goes $4.96 \times 10^{-7} \rightarrow 6.43 \times 10^{-7} \rightarrow 9.09 \times 10^{-7}$, the trend the `O/H` column of Table 16 continues over 8.45–11.0. Oxygen is the only element the reservoir moves here, and it is the one whose carrier condenses.

12.3 The pressure form of K_{zz}

Every number in this subsection is the 2026-08-30 measurement, read from `LHS1140b/archive_20260830/`. The mechanism it establishes does not depend on the physics that has changed since (Sect. 2.), but its magnitudes have not been re-measured on the current code.

The eddy coefficient of the last section is a stated number rather than a solved one, and it is stated in a particular way: constant, $1.0 \times 10^9 \text{ cm}^2 \text{ s}^{-1}$ at every one of the 102 levels, which is what the adapter's `--kzz-const 1.0e9` writes into the `Kzz` column. The form is an assumption laid on top of the value. What the *value* carries is already measured: the equivalent width moves by a factor $\simeq 9$ between $K_{zz} = 10^8$ and 10^{10} (`LHS1140b/kzz_decision.md`, Sect. 3), so the open question is whether the *shape* carries anything as well. We tested that against the constant, and the shape does not carry the conclusions of this memo. The reason is short: of the profile, EXHALE consumes elemental ratios and thermodynamics, and both are the same at the match to better than 1.2 per cent under either form.

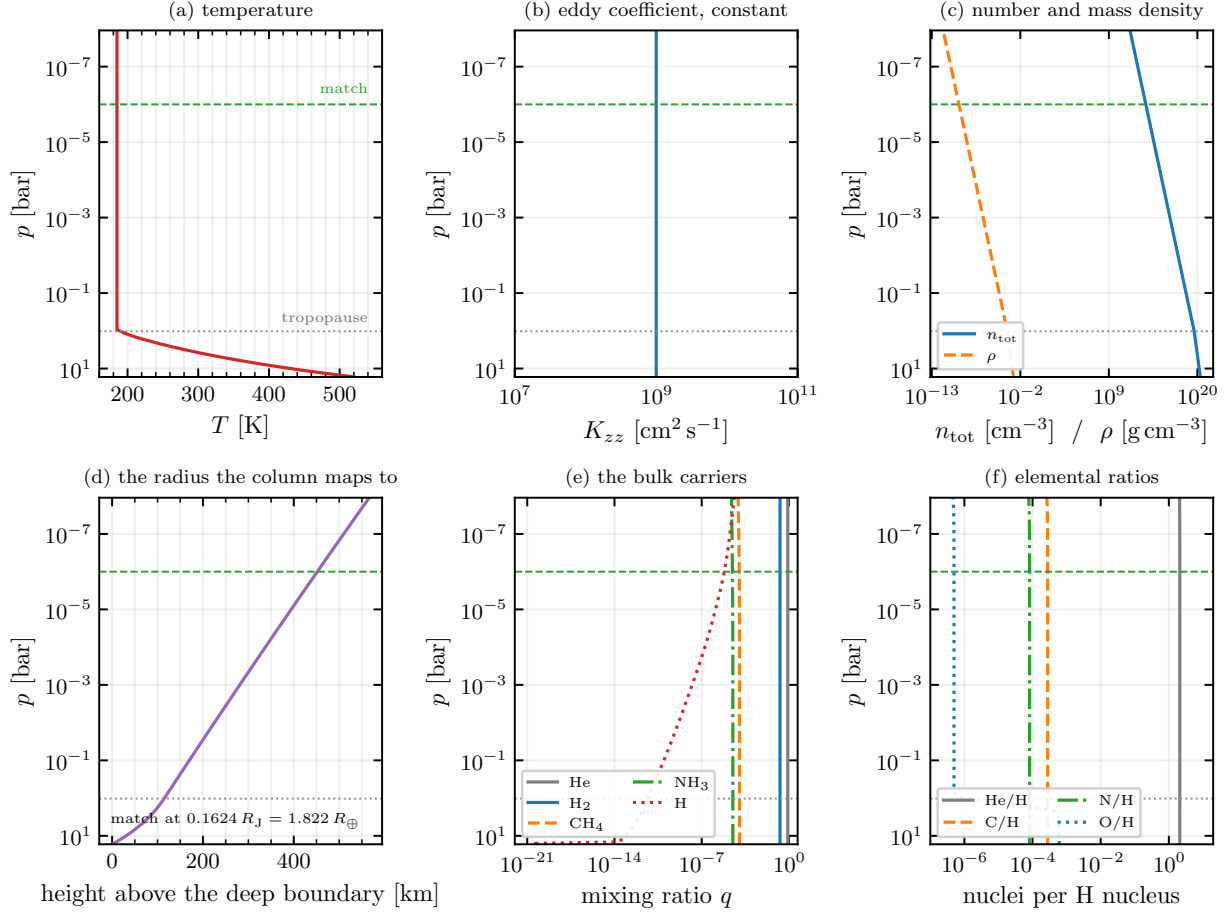


Figure 14: The photochemical column, its structure and its bulk carriers. (a) the solved temperature; (b) the eddy coefficient, constant at $10^9 \text{ cm}^2 \text{ s}^{-1}$ over all 102 levels; (c) number and mass density; (d) the radius the pressure grid maps to, which is what fixes the base radius of the wind; (e) the carriers that hold the bulk: helium reconstructed as $\text{He}/\text{H}(2q_{\text{H}_2} + q_{\text{H}})$, and atomic H four to five decades below H_2 everywhere; (f) the elemental ratios the profile hands over. The tropopause (1.031 bar, dotted) and the $1 \mu\text{bar}$ matching level (dashed) are marked on every panel. Fig. 19b,c shows the same climate and elemental ratios in the context of the closure, while this figure shows the file itself, column by column.

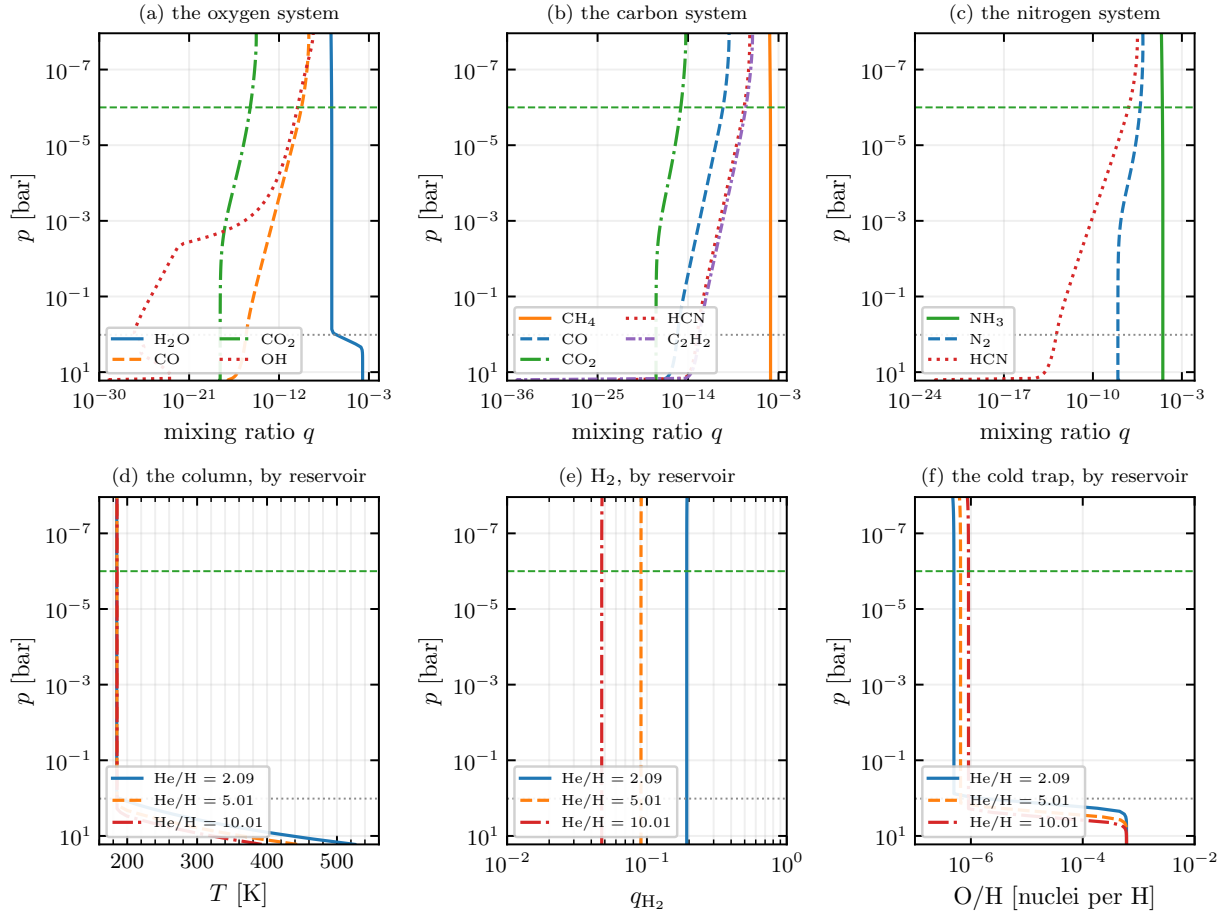


Figure 15: The molecular systems, and what the reservoir does to them. (a)–(c) the oxygen, carbon and nitrogen carriers of the same file: water condenses out at the tropopause while CH_4 and NH_3 cross it unchanged, and the photochemical products (HCN , C_2H_2 , N_2 , CO , OH) climb several decades in the top of the column while staying trace. (d)–(f) the same column solved at three reservoirs of the earlier ladder: the deep boundary cools as helium is added, H_2 is diluted by number, and the cold trap loosens, so oxygen is the one element the reservoir moves. Levels are marked as in Fig. 14.

What was tested. The saturated gravity-wave form,

$$K_{zz}(p) = 1.0 \times 10^9 \text{ cm}^2 \text{ s}^{-1} \left(\frac{p}{1.1 \times 10^{-8} \text{ bar}} \right)^{-1/2}, \quad (8)$$

anchored at the top of the profile and not at 1 bar. The anchor is the point of the test rather than a detail of it: EXHALE gives every cell above the shallowest tabulated level that level's value (`eddy_diffusion_on_grid`, Sect. 12.2), so anchoring at the top leaves the number the wind inherits above the file unchanged and confines the change to the interval the file describes. Equation (8) then implies 1.05×10^8 at the $1 \mu\text{bar}$ match, 1.05×10^5 at 1 bar and 2.56×10^4 at the 16.73 bar deep boundary: a factor 9.5 below the constant at the match, and four decades below it at the bottom. Two cases separate the chemistry from the wind, because the profile hands over both:

- *A, chemistry only.* The column is solved on Eq. (8) and then its K_{zz} column alone is written back to the constant 10^9 before the handoff, so the wind sees the eddy coefficient it saw before and only the composition it is handed changes.
- *B, chemistry and wind.* The profile as produced; the wind reads $K_{zz}(p)$ as well, but only where the file speaks. The column reaches $1.0091 R_p$ above the match, and every cell above that takes the top level's value, so the wind-side change is a shell nine thousandths of a radius deep in which the coefficient is 9.5 times lower at the base and equal at the top. The homopause under the constant sits at $1.056 R_p$ (`kzz_decision.md`, Sect. 3), outside that shell. Case B is therefore a bounded test of the wind side rather than a test of lowering K_{zz} where helium settles; the unbounded version of that is the magnitude scan, which is already measured.

The reference is the constant- K_{zz} profile re-solved with the same seed and the same binary, so the comparison is not against a number recorded on an earlier occasion; the recorded number is carried in the tables anyway, because the difference between it and the reference is the size of the procedure's own path spread. Two reservoirs of the earlier closed ladder were run, one at each end of it: the 11.11 rung (`flux_closure/heh11p1`, $k = 1$) and the 2.09 one on its high-trial-flux closure branch (`flux_closure/hi`, $k = 6$). The test has since been repeated run by run on the current binary, as the two “recorded, other seed” rows in L11p1 and L2p09, and the reasoning it was first reported under is confirmed: the coefficient is common to both sides of every entry, so a ratio between two chemistries at fixed everything else does not turn on it, and every ratio the test reports moves by less than the procedure's own path spread. The table below is the repeated version. The runs and the measurement script are in `LHS1140b/archive_20260830/exhale/kzz_power/`, the repeat in `LHS1140b/archive_20260830/exhale/refresh_j96/xuvkzz/`.

What did not move. The climate is identical: deep boundary 411.4 K, tropopause 2.636 bar at 185.5 K, cold-trap $f_{\text{H}_2\text{O}} = 5.590 \times 10^{-8}$ and match temperature 185.478 K, the same digits under either form. That is structural rather than coincidental. The adapter's `--climate` path solves the radiative-convective column with no eddy coefficient entering it, and hands the photochemistry a (P, T) pair with the K_{zz} slot empty, so the thermal structure, the tropopause and the trap the profile carries cannot depend on the eddy form.

The elemental ratios at the match, which with the temperature are what EXHALE consumes, move by at most 1.2 per cent:

What did move is the carriers, by up to four decades. The lower part of Table 12 is the real effect of the eddy form, and it is large. Nitrogen changes carrier outright: on the 11.11 rung q_{NH_3} at the match falls from 6.32×10^{-6} to 2.43×10^{-8}

Table 11: The pressure form of K_{zz} against the constant, at two reservoirs. *A* solves the chemistry on Eq. (8) and hands the wind the constant; *B* hands it Eq. (8) as well. The reference row is the constant- K_{zz} profile re-solved with the same seed, and the row above it is the same profile solved on an earlier occasion from a different seed: the path spread of the procedure, against which the effect of the eddy form is to be read. All eight runs are on the binary of 2026-08-30. *EW* is on this memo's red-pair window, comparable with the observed $1.108 \pm 0.030 \text{ \% } \text{\AA}$.

run	red [%]	FWHM [$\text{\AA}$]	<i>EW</i> [% $\text{\AA}$]	<i>EW</i> /obs	He/H at $2R_p$	$\log_{10} \dot{M}$
<i>reservoir</i> He/H = 11.11						
recorded, other seed	3.8857	0.2845	1.1798	1.065	1.5498	7.400
reference, $K_{zz} = 10^9$	3.9267	0.2845	1.1927	1.076	1.5664	7.410
A, chemistry only	3.9422	0.2845	1.1976	1.081	1.5743	7.410
B, chemistry and wind	3.9416	0.2845	1.1965	1.080	1.5582	7.410
<i>reservoir</i> He/H = 2.09						
recorded, other seed	2.0683	0.2472	0.5445	0.491	0.2351	7.500
reference, $K_{zz} = 10^9$	2.0760	0.2472	0.5466	0.493	0.2359	7.500
A, chemistry only	2.0929	0.2472	0.5511	0.497	0.2377	7.510
B, chemistry and wind	2.0846	0.2467	0.5486	0.495	0.2355	7.510

Table 12: At the $1 \mu\text{bar}$ match, the ratio of the $K_{zz}(p)$ column to the constant- K_{zz} one. The top block is what EXHALE reads; the bottom block is what it does not.

quantity at the match	11.11 rung	2.09 rung
<i>consumed by the wind</i>		
X_{He}	0.9985	0.9985
X_{C}	0.9899	0.9903
X_{N}	0.9878	0.9875
X_{O}	0.9894	0.9821
T	1.0000	1.0000
<i>carried, not consumed</i>		
q_{CH_4}	0.443	0.665
q_{CO}	54.7	47.6
q_{CO_2}	9.3×10^3	4.3×10^3
q_{NH_3}	0.0038	0.0496
q_{N_2}	28.7	28.1
q_{HCN}	0.566	2.95
$q_{\text{H}_2\text{O}}$	0.714	0.804
q_{H}	4.99	5.55

while q_{N_2} rises from 1.21×10^{-7} to 3.46×10^{-6} , so a column that was NH_3 -dominated becomes N_2 -dominated. Carbon partly follows, CH_4 down by 2.3 and CO up by 55; CO_2 rises by nearly four decades from a negligible base. Deeper the redistribution is more extreme still (at 1 bar the same ratios are 1.3×10^5 for CO, 3.0×10^7 for CO_2 and 8.8×10^3 for N_2) because that is where the two forms differ most. *Elemental* nitrogen at the match nevertheless moves by -1.2 per cent, because a carrier swap conserves nuclei, and it is elemental nuclei that the handoff transmits.

The crossing. Converted with the local slope $d \ln EW / d \ln(He/H) = 0.382$ measured over the 11.11–12.01 interval of the ladder these rungs belong to, the four equivalent widths move that ladder’s crossing from 11.73 to 11.61 and 11.63 on the 11.11 rung and to 11.48 and 11.62 on the 2.09 one: -0.25 to -0.10 in He/H, under a tenth of the 10.92–12.58 interval the measurement’s own 1σ band allowed there. All four now move it the same way, downward. In relative terms that is -2.1 to -0.8 per cent, and it is the relative figure that carries over to the present crossing. It is also no larger than the path spread of the procedure itself: re-solving the *same* profile from a different seed moves the equivalent width by 1.08 per cent on the 11.11 rung and 0.38 per cent on the 2.09 one, against the 0.32–0.83 per cent the eddy form produces. The signal is buried in the spread of the procedure that would have to measure it, by a narrower margin than before: 1.08 against 0.83 per cent where the previous branching gave 1.29 against 0.79.

One objection to the test itself has to be stated. The deep half of Eq. (8) is not the right law. The $P^{-1/2}$ growth is the saturated gravity-wave scaling of a *stratified* region (Lindzen 1981), and the climate solve puts this planet’s tropopause at 2.636 bar: below it the column is convective, where the mixing-length form of Gierasch & Conrath applies and K_{zz} should flatten or grow with depth, not collapse. Nor is the amplitude defensible there. At 1 bar Eq. (8) gives $1.05 \times 10^5 \text{ cm}^2 \text{ s}^{-1}$, which is at or below the floor of the modeling literature, one and a half to two and a half decades under the 3×10^6 – 3×10^7 Charnay et al. (2015) use for GJ 1214 b, 3.7 decades under the 5×10^8 of Parmentier et al. (2013) for HD 209458 b, and equal to the 10^5 floor Ackerman & Marley (2001) impose by hand. The physically correct construction is two-branch, a wave law above the radiative–convective boundary joined to a mixing-length law below it; this test ran one branch all the way down to 16.73 bar. Read as interpretation, that is very likely why the carriers moved as far as they did, since the 0.1–10 bar interval the single branch over-quenches is the one that fixes where CH_4 and NH_3 quench. The objection does not reach the verdict, though: a form this wrong in the deep column still failed to move an elemental ratio at the match by more than 1.2 per cent.

At the other end the same equation is on firmer ground than the adopted constant. Its value at the match, $1.05 \times 10^8 \text{ cm}^2 \text{ s}^{-1}$, falls inside the 2.8×10^7 – 1.4×10^8 that the Arfaux & Lavvas homopause scaling gives when it is evaluated with *this* planet’s gravity and temperature: the one estimate in the survey derived from LHS 1140 b itself, and the direction in which docs/eddy_diffusion_kzz.tex already records the adopted 10^9 as having no planet-specific basis.

Verdict. The constant assumption was carrying a great deal of the lower atmosphere’s carrier chemistry, up to four decades of it, and none of this memo’s conclusions. What EXHALE consumes from the profile is the elemental ratios and the thermodynamics, and those agree at the match to 1.2 per cent under a form that differs from the constant by a factor 9.5 there and four decades at the bottom. The match is still well mixed under either form (1.05×10^8 is 200 times the molecular coefficient of $\simeq 4.6 \times 10^5 \text{ cm}^2 \text{ s}^{-1}$ at the base, kzz_decision.md Sect. 1). What the results carry is therefore the magnitude of K_{zz} , the 10^8 – 10^{10} bracket across which the equivalent width moves by a factor 9, rather than its pressure form. One limit: the verdict is tied to this anchor. Anchoring Eq. (8) at 1 μ bar instead would raise by

9.5 the value the wind inherits over its whole domain, which is a change of magnitude and therefore a different test; it was not run.

12.4 The cold trap the climate solve puts in

Solving the temperature rather than prescribing it puts a cold trap into the handoff, at the level the climate solution places it rather than at the level a closed form assumes.

Table 13: The climate solve over its one free parameter, the deep boundary. $f_{\text{H}_2\text{O}}$ is the water mixing ratio above the tropopause, i.e. what crosses into the photochemical column. The stratospheric temperature is robust and the tropopause pressure is not. The trapped water follows the pressure, since at fixed T_{trop} the trapped ratio is $p_{\text{sat}}(T_{\text{trop}})/P_{\text{trop}}$.

deep boundary [bar]	T_{deep} [K]	P_{trop} [bar]	T_{trop} [K]	$f_{\text{H}_2\text{O}}$
1	242.4	0.488	186.0	3.33×10^{-7}
3	308.5	0.765	185.8	2.06×10^{-7}
10	439.9	0.974	185.4	1.49×10^{-7}
20 (adopted)	555.9	1.031	185.0	1.32×10^{-7}
30, 50	refused: the pseudoadiabats leaves the thermodynamic data			

T_{trop} moves by 1 K over a factor of 20 in the deep boundary, against the skin temperature $T_{\text{eq}}/2^{1/4} = 190$ K. P_{trop} moves by a factor of two, and the water moves with it. The scan itself gives the sensitivity, since $f_{\text{H}_2\text{O}}P_{\text{trop}}$ is $p_{\text{sat}}(T_{\text{trop}})$: across the rows it comes out at 19% per K at 185 K. The trapped water is therefore conditional on the tropopause temperature at about a fifth per kelvin, and linearly on the tropopause pressure.

Against the published estimate. Cherubim et al. (2026) quote $f_{\text{H}_2\text{O}} \sim 7$ ppm from a closed-form cold trap at a 0.1 bar tropopause with $T_{\text{skin}} = 194$ K. The climate solution gives 0.13–0.33 ppm at a tropopause of 0.5–1.0 bar: a factor 20–50 less water at a tropopause 5–10 $\times$ deeper. The saturation physics agrees: $p_{\text{sat}}(185 \text{ K})/1 \text{ bar}$ is 1.3×10^{-7} and $p_{\text{sat}}(194 \text{ K})/0.1 \text{ bar}$ about 5×10^{-6} , so each number follows from its own tropopause, and what differs is where the tropopause is put. In our case it is a solved level in a helium-dominated column ($\mu = 3.6$ at $\text{He}/\text{H} = 2.09$); in theirs it is assumed. We report this as a result of the climate model, and it was not tuned toward the published number.

In the handoff the trap shows up as a depletion with height rather than as a condensate sitting at the base. Gas-phase oxygen falls from $\text{O}/\text{H} = 6.062 \times 10^{-4}$ at the deep boundary to 4.957×10^{-7} at the match, a factor 1223, while C/H (2.778×10^{-4}), N/H (8.187×10^{-5}) and He/H cross the tropopause unchanged. The carriers of carbon and nitrogen in this column are CH_4 and NH_3 (Sect. 12.2), and this solution keeps both of them in the gas phase across the trap (Figs. 19c and 15a–c).

12.5 The closure, and what it returns

Every number in this subsection is the 2026-08-30 measurement, read from LHS1140b/archive_20260830/. The mechanism it establishes does not depend on the physics that has changed since (Sect. 2.), but its magnitudes have not been re-measured on the current code.

Three runs were made, from the reference trial pair $(\Phi_{\text{H}}, \Phi_{\text{He}}) = (1.8, 1.3) \times 10^7 \text{ g s}^{-1}$ and from $0.3\times$ and $3\times$ it, all seeded from the same converged wind and all starting from the same $\text{He}/\text{H} = 2.09$ reservoir.

Table 14: The elemental-flux closure on LHS 1140 b. k is the iteration at which the 5% residual criterion was met. The equivalent width is re-measured here on this memo's own red-pair window (10832.60–10834.20 Å vacuum), so it is directly comparable with the observed $1.108 \pm 0.030 \text{ \% Å}$; integrated instead over the whole synthesized band, as `Update_EXHALE.md` Sect. 79 records them, the same curves give 5.99–6.03 mÅ. Every column comes from one closure re-run end to end on the current binary and on Photochem 0.9.0 with the re-pinned adapter of *Update_EXHALE* Sect. 99 (LHS1140b/archive_20260830/exhale/refresh_j96/closure/). As on the ladder of Sect. 12.7, each converged iterate's wind is then re-solved on its own output with the profile held fixed, so that runs which stopped at different k are read at the same place. Both line columns are at $R = 68,000$, the resolving power of the measurement and of the rest of this memo. An earlier version of this table printed its red column from the $R = 80,000$ synthesis while its *EW* column and the observed row were at $R = 68,000$, and that mixture is corrected here.

start	k	$F_{\text{H}} [\text{g s}^{-1}]$	$F_{\text{He}} [\text{g s}^{-1}]$	He/H at match	$\log_{10} \dot{M}$	red [%]	<i>EW</i> [% Å]
1.0×	0	1.8017×10^7	1.2969×10^7	2.09235	7.500	2.022	0.532
0.3×	4	1.8161×10^7	1.3436×10^7	2.09234	7.500	2.030	0.534
3.0×	6	1.8147×10^7	1.3364×10^7	2.09236	7.500	2.019	0.531
observed	-	-	-	-	-	$1.254^{+0.22}_{-0.23}$	1.108 ± 0.030

The three runs agree on He/H to 7.8×10^{-6} , on $\log_{10} \dot{M}$ to better than the 0.005 the log prints, and on the fluxes to 0.8% and 3.5%, over a factor of ten in where they started, so the closure is single-valued on this planet. The damping was never exercised (the residual fell on every iteration), but the $3\times$ run used six of its eight allowed steps, so a start further than an order of magnitude out would need a larger iteration budget.

One systematic on the profile route is larger than anything the closure resolves, and it is stated here rather than corrected. A four-pass JFNK re-solve of the same run (identical input.inp, identical handoff column, identical binary, each pass seeded from the previous one and each reaching `info=0` at $\|R\| \simeq 10^{-4}$) lands on different fixed points depending on which converged wind it started from: on the 1.0×

The composition is an output for the first time, and at this eddy coefficient it is the well-mixed value. He/H at the match comes back at 2.09235 against the 2.09 the reservoir was started from, a change of 1.1×10^{-3} , of which only 2×10^{-4} comes from the escape flux itself. The reason is measured: at $K_{zz} = 10^9 \text{ cm}^2 \text{ s}^{-1}$ eddy mixing homogenizes the column

all the way to $1 \mu\text{bar}$, so a $1.8 \times 10^7 \text{ g s}^{-1}$ hydrogen escape imposed at the model top moves He/H by 1.8×10^{-2} at that top and by 2×10^{-4} at the match. At the matching level the composition is set by the eddy coefficient rather than by the escape flux. Closing the loop therefore does not retire the conditionality of Sect. 9.4, but moves it onto the same adopted K_{zz} .

Above the match the wind separates the two elements as strongly as before. The elemental fluxes carry $F_{\text{He}}/F_{\text{H}} = 0.73$ by mass against the base reservoir's 8.31, and He/H falls from 2.09 at the base to 0.183 at $30R_p$. The closure does not weaken that; it places the fractionation in the wind rather than below the match. One caveat on the budget, stated rather than corrected: the diffusion operator's mass fraction is that of the $\text{H} + \text{He}$ mixture and the trace metals ride inside ρ , so the measured F_{H} is hydrogen plus metals and overstates elemental hydrogen by 4.8×10^{-4} , two decades below the tolerance.

12.6 What the base metals do, and what the closed solution does not reproduce

The profile brings something the scalar base never had: the carbon, nitrogen and oxygen of the photochemical column. Isolating it takes a ladder of controls in which the scalar-base run is moved one key at a time onto the profile's own base state, every rung a converged solution (Update_EXHALE.md Sect. 78). That ladder was run on the 2026-08-30 code and its magnitudes are superseded. The paragraph after it gives the current measurement, and Sect. 2.4 says what moved. Setting the base helium ratio to the profile's 2.0921 moves $\log_{10} \dot{M}$ by less than 0.01. Setting the base temperature to the solved 185.05 K, from the 226 K the scalar runs used, moves it by less than 0.01 more, so the base temperature is not the lever. Adding the base radius moves it to 7.85 (the molecular fraction does not enter: $q_{\text{H}_2\text{base}}$ is inert unless `Molecular base` is set, which neither route sets). Adding the profile's C/H , N/H and O/H takes it to **7.49** and brings the red-pair depth down by a factor **2.9** on that rung ($5.59 \rightarrow 1.93$ per cent), that is -0.36 dex in $\dot{M}$ and a correspondingly shallower helium line, from metal-line cooling of a 185 K base at $\text{C}/\text{H} = 2.8 \times 10^{-4}$.

Over the ladder end to end the same pair is -0.32 dex and a factor 2.7. The profile route reproduces that rung to 0.00 dex in $\dot{M}$ and 0.5 per cent in the line, inside the workflow's own path spread, which is the consistency check the two routes exist to give. The six rungs are re-solved in `LHS1140b/archive_20260830/exhale/refresh_j96/basemetals/`, at $\|R\|_{\text{tol}} = 1.0 \times 10^{-4}$ on the four scalar rungs and 4.0×10^{-4} and 5.0×10^{-4} on the two metal ones. The scalar rung and the scalar column of Table 15 are now the same configuration at the same tolerance and give 1.420 and 1.424 % Å, 0.3 per cent apart, which is the seed-path spread of this workflow.

Every number in the two paragraphs above is the 2026-08-30 measurement, and the C, N and O cooling that carries it has since been replaced (Sect. 2.4). The decomposition of the handoff into its five terms (reservoir, base temperature, base radius, molecular fraction, elemental vector) stands as an account of which key does what; the size of the last term does not. Re-measured on the current code at $K_{zz} = 10^9$ and $\text{He}/\text{H} = 2.13$, adding the column's C, N and O to the scalar base moves $\log_{10} \dot{M}$ from 7.87 to 7.87 and the red-pair equivalent width from 1.3749 to 1.3721 % Å, 0.12 per cent, where the archived pair moved by -0.36 dex and a factor 2.9 in depth. The base metals are no longer the lever that separates the two lower boundaries. The flux-closed solution at $\text{He}/\text{H} = 2.09$ now gives $EW = 1.4765$ % Å against the metal-free base's 1.3749 at 2.13, above the measurement rather than a factor 2.0 below it.

Closing the elemental fluxes and matching the equivalent width are still two different constraints, but they no longer

pull in opposite directions. Fixing the reservoir by the line, as Sects. 8. and 9.4 do, was done above a base carrying no metals; with the photochemical column beneath it the same reservoir now absorbs *more*, so the composition that satisfies both conditions at once is smaller than the metal-free crossing, and smaller than every rung the closure ladder was run at (Sect. 12.7). The lower boundary is still an axis of its own, of the same standing as the assumed spectrum, the assumed mixing and composition itself. What changed is its sign and its size.

The mechanism: helium as a thermostat, and what is left of it

Why the same reservoir absorbs as it does over the two bases can be read off the solved structures. Table 15 measures the four solutions that bracket the question: the metal-free scalar base at $\text{He}/\text{H} = 2.13$; the same scalar base carrying the column's C, N and O reservoirs at the same composition, which isolates the metals from everything else the profile changes; the flux-closed solution on the photochemical column at 2.0924; and the closed solution at 10.01. All four run with H/He element diffusion active at $K_{zz} = 10^9 \text{ cm}^2 \text{ s}^{-1}$ (constant through the profile in the two profile runs, the scalar value in the other two), and with EXHALE's molecular chemistry off, so no molecular carrier appears in the wind. Mean mass per particle is $(n_{\text{H}} + 4n_{\text{He}})/(n_{\text{H}} + n_{\text{He}} + n_{\text{met}} + n_e)$; nuclei are counted as H I + H II and He I + He II + He III. The values are from the advection-corrected profiles.

Table 15: The two directions the thermostat runs, measured on the solved structures of the 2026-09-14 campaign. *EW* is on this memo's red-pair window against the observed $1.108 \pm 0.030 \% \text{ \AA}$. The C, N and O line cooling of all four is the density-resolved CHIANTI treatment of Sect. 2.4, which is what makes columns one and two nearly identical where the 2026-08-30 solutions put a factor 2.6 between them.

	scalar base no metals	scalar base C, N, O	photochem base C, N, O	photochem base C, N, O
reservoir He/H	2.13	2.13	2.0924	10.011
T_{max} [K]	5730	5566	5415	7682
$T(2R_p)$ [K]	4488	4423	4324	6780
mean mass per particle at $2R_p$ [m_{H}]	2.14	2.14	2.15	3.17
$n(\text{H I})$ at $2R_p$ [cm^{-3}]	5.69×10^7	5.71×10^7	5.45×10^7	1.47×10^7
He/H by nuclei at $2R_p$	0.682	0.681	0.696	3.587
$\max n(2^3S)$ [cm^{-3}]	34.6	35.5	35.8	88.8
$\int n(2^3S) dr, 1-10R_p$ [$\text{cm}^{-3}R_p$]	68.7	69.4	70.2	112.0
of which $1-2R_p$	25.3	25.9	26.2	52.7
of which $2-10R_p$	43.0	43.1	43.6	58.8
<i>EW</i> [% $\text{\AA}$]	1.3749	1.3721	1.4765	2.2224

The base metals are no longer a thermostat setting. With the C, N and O line cooling evaluated at the cell's own electron density rather than in the coronal limit (Sect. 2.4), adding the column's elemental reservoirs to the scalar base at fixed composition costs the wind 164 K in T_{max} and the line 0.20 per cent of its equivalent width. The comparison the 2026-08-30 solutions made between columns one and three ($T_{\text{max}} 5447 \rightarrow 3955 \text{ K}$ and $EW 1.424 \rightarrow 0.545 \% \text{ \AA}$, a factor 2.6) was a consequence of the coronal coefficient, which stood up to four decades above the statistical equilibrium of the

same ion at these densities. What is left between the scalar base and the photochemical column at the same reservoir is 315 K in $T_{\max}$ and +7.4 per cent in the line, and it is no longer dominated by the metals. Columns two and three still differ by the matching-level temperature, the base radius and the elemental profile the column carries.

The cost to the wind. With no metals, radiative cooling is 4.1–6.9 per cent of the heating over $1.2\text{--}2R_p$ and the wind is limited by expansion rather than by radiation. With the column's C, N and O beneath it, cooling is 5.1–7.9 per cent, of which the metal lines are 23 per cent at $1.2R_p$ and 3 per cent at $2R_p$, and $T_{\max}$ falls from 5730 to 5415 K. Both the metastable peak and the metastable column rise slightly ($34.6 \rightarrow 35.8 \text{ cm}^{-3}$ and $68.7 \rightarrow 70.2 \text{ cm}^{-3} R_p$ over $1\text{--}10R_p$), and the line rises with them. The 2026-08-30 draft stated here that the metals take the outer column, because the cooler wind has the smaller scale height and does not carry the metastable population out to where it adds absorbing area. That is a real mechanism at a cooling rate that large, but it is not what these solutions do at the cooling rate the density-resolved coefficient gives.

Raising the reservoir reheats the wind, by two terms that are both measured. From 2.09 to 10.01 on the same base, $T_{\max}$ goes $5415 \rightarrow 7682$ K. Two things change together. First, there are fewer particles to share the deposited energy. The mean mass per particle at $2R_p$ rises from 2.15 to $3.17 m_H$, a factor 1.47; the heating rate at $2R_p$ divided by the particle density rises by 1.50; and $T(2R_p)$ rises by 1.57. Here n_{tot} follows the code's `calc_ntot` convention, one particle per species with electrons excluded, as elsewhere in this section. The three factors agree to within 6 per cent, which is a consistency check on that reading rather than a proof of it, since the cooling term is not held fixed between the two runs. Second, the coolant is diluted along with the hydrogen it is referred to. The profile states C, N and O per hydrogen nucleus (schema column `X_*[El/H nuclei]`, and in the code $n_X = \text{melem_ab}(e)n_H$), and the two profiles carry C/H within 0.1 per cent of each other, so when $n(\text{H I})$ at $2R_p$ falls by 3.7 the carbon density falls with it by the same factor. The volumetric metal cooling nonetheless *rises*, $2.36 \rightarrow 4.65 \times 10^{-11} \text{ erg cm}^{-3} \text{ s}^{-1}$ at $1.5R_p$, because the cooling function is steeply temperature dependent. What the dilution does is let the gas settle at the higher temperature at which that happens.

The metastable balance. The 2^3S population in these solutions is fed almost entirely by recombination and drained by Penning ionization on neutral hydrogen and by electron collisional de-excitation. Reconstructing the residual EXHALE solves apports the budget: the source is $n_e n_{\text{He II } \alpha_{2^3S}}$ plus collisional excitation from the singlet ground state, against radiative decay A_{31} , photoionization, destruction on H I, electron collisional de-excitation to 2^1S and 2^1P , and electron collisional ionization (`ion_residual_core.f90`, `tr_triplet_row`). At the metastable peak of the advection-corrected profiles, collisional excitation supplies a negligible part of the source and radiative decay plus collisional ionization take 0.02–0.03 per cent of the sink. The rest of the loss is Penning ionization and electron de-excitation: 72% and 28% at $\text{He}/\text{H} = 2.09$, 54% and 46% at 10.01. Between the two the source rate rises by 2.76 (more He II and more electrons in the hotter gas) while the metastable lifetime is essentially unchanged, 1.09 against 0.98 s. The neutral hydrogen the Penning term feeds on is thinned by only 1.20 at the peak, and the faster electron de-excitation of the hotter gas takes back what that gains. The product, 2.48, is the peak ratio $35.8 \rightarrow 88.8 \text{ cm}^{-3}$. There is no charge-exchange term on the metastable in this code (the He/H charge exchange acts on ground-state He and He^+), so it is not part of this balance.

Read as an interpretation, the thermal structure of the 10.01 solution ($T_{\max} = 7682$ K, column 112.0 over $1\text{--}10R_p$) now stands well above that of the metal-free 2.13 solution (5730 K, 68.7), and its equivalent width is correspondingly above

the measured value rather than back at it. The ladder therefore still looks like a thermal trade between the metal content of the base and the helium content above it, but the metal side of the trade has become small (Sect. 2.4) and what is left is the helium. Helium in these solutions is not only the absorber; it is also the thermostat that sets how far the absorber is carried.

Composition and ionization. Fig. 16 carries the elemental fractions and the ionization state of the same three solutions, on the same advection-corrected profiles as Table 15. Two things follow from it that the thermostat argument rests on. Diffusion keeps the wind less helium-rich than the reservoir. In both solutions the helium fraction falls through the line-forming region and then flattens: $\text{He}/\text{H} = 2.09$ at the base gives 0.696 by nuclei at $2R_p$ and 0.554 at $5R_p$; 10.01 gives 3.587 and 2.747. That is a factor 3.0 and 2.8 between the reservoir and the wind at $2R_p$. In the composition's own terms, the flux-closed solution at 2.09 (a reservoir that is 68 per cent helium by number) has a *hydrogen-dominated* line-forming region, $n_{\text{H}}/n_{\text{tot}} = 0.59$ against $n_{\text{He}}/n_{\text{tot}} = 0.41$ at $2R_p$. At 10.01 helium leads there, 0.78 against 0.22, but by a factor 2.8 less than the reservoir states. A composition quoted for this planet therefore has to say at which level it is quoted; the reservoir and the absorbing layer are not the same number, and the memo's crossings are reservoir values.

The hotter solution is also the more ionized one. At $2R_p$, $x(\text{H II}) = 0.031$ on the photochemical base at 2.09 and 0.057 at 10.01: a factor 1.9, in the run whose neutral hydrogen density is 3.7 times lower. Both layers are optically thin at 13.6 eV, so the reading that thinner hydrogen shields itself less does not account for it. What grows is the ionization of hydrogen *by helium*, through Penning ionization $\text{He}(2^3\text{S}) + \text{H}$, charge exchange $\text{He}^+ + \text{H}$ and the He-recombination continuum above 24.6 eV. The split between those channels cannot be settled from the outputs: the individual rates (direct $\Gamma_{\text{H I}}$, the He-recombination-driven addition to it, and the secondary-ionization yield) are not written to any output file. One channel can be excluded, $n = 2$ photoionization, since `heat_Hpe[excitedH]` is zero in all cells of all three runs. Helium ionization itself tracks the temperature more simply: $x(\text{He II})$ is 0.046, 0.046 and 0.055 at $2R_p$ across the three, and $x(\text{He III})$ is at most 1.3×10^{-4} at $2R_p$ (9.1×10^{-5} , 9.0×10^{-5} and 1.3×10^{-4}).

Fig. 12 shows the same kind of structure for the single well-mixed solution that reproduces the line with no diffusion and no base metals, while Fig. 16 is the set the lower-boundary axis is built on. The two are complementary rather than alternative views.

12.7 The composition the line needs above a photochemical base

Sect. 12.6 left the composition that closes the fluxes *and* reproduces the line unsolved. This section solves the ladder, at the adopted eddy coefficient, and finds that on the current code there is nothing to solve for: every closed rung stands above the measured equivalent width. The closure driver is run at nine starting reservoirs between 2.09 and 12, each iterating Photochem and EXHALE to elemental-flux continuity across the $1\mu\text{bar}$ match, with the `clima` step solved, $K_{zz} = 10^9$, the GJ 1132 proxy spectrum, a 5% residual tolerance and the partitioned stationary route of MODELS.md Sect. 6 inside every iteration. Each rung is read at its last iterate. The runs are `models/atomic_photochem_gj1132_kzzprofile`.

Every rung returns a matching-level He/H equal to the reservoir it was started from to four digits, for the reason Sect. 12. measured at 2.09: at $K_{zz} = 10^9$ the eddy-mixed column reaches the microbar match, so the escape flux does

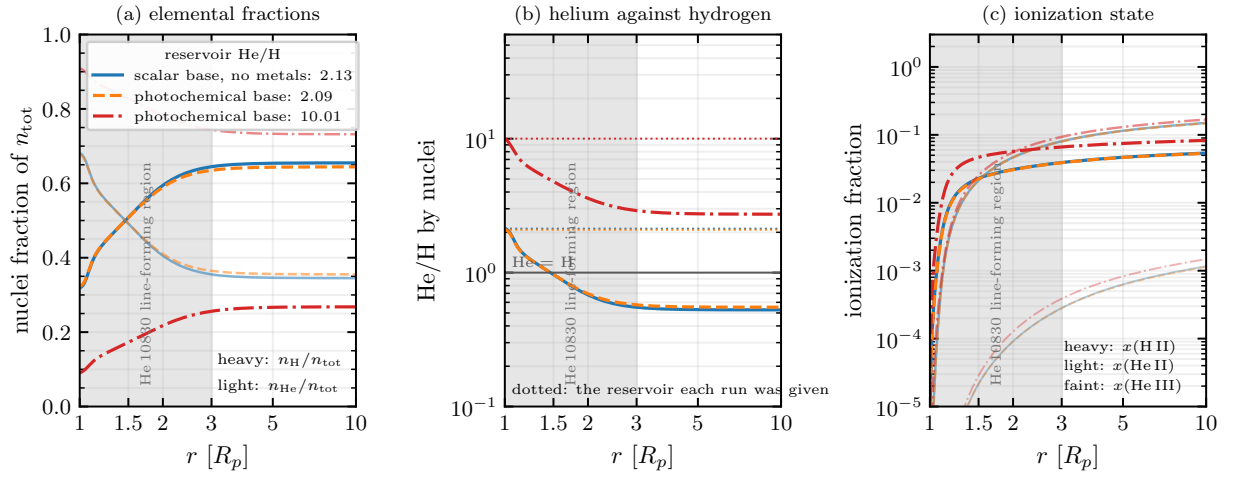


Figure 16: Composition and ionization against radius, for the three solutions of Table 15, from the same advection-corrected `Ion_species_adv.txt` profiles. **(a)** the hydrogen and helium nuclei fractions of n_{tot} ; **(b)** He/H by nuclei, with the reservoir each run was given as a dotted line, so the vertical drop from dotted line to curve is what element diffusion does; **(c)** the ionization fractions. n_{tot} follows the code's `calc_ntot` convention: one particle per species, H I, H II, He I, He II, He III and every metal ion stage, electrons excluded, and He 2^3S not added because it is an excited level inside He I, so (a) does not sum to one on the photochemical base. The metals it misses are negligible: their largest share of n_{tot} anywhere in $1\text{--}30R_p$ is 2.3×10^{-4} at He/H = 2.09 and 9.7×10^{-5} at 10.01, and zero on the metal-free base. Nuclei are counted as H I + H II and He I + He II + He III. All three solutions are cases of the model tree. The closure ladder does not cross the measurement anywhere (Sect. 12.7), so no crossing is marked between the two profile runs drawn here. The He 10830 line-forming region $1\text{--}3R_p$ is shaded.

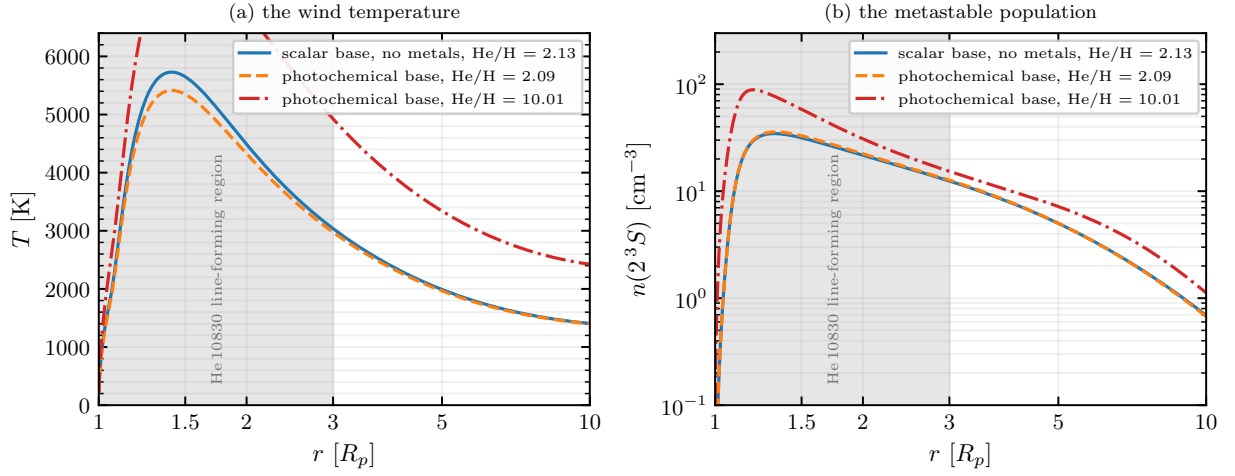


Figure 17: The thermostat, on the three solutions of Table 15. **(a)** temperature and **(b)** the metastable density, against radius, with the He 10830 line-forming region 1–3 R_p shaded. Adding the photochemical column’s C, N and O at fixed reservoir (solid to dashed) now costs the wind 315 K in $T_{\max}$ and leaves the metastable population where it was, to within 2 per cent in the peak and in the column. Raising the reservoir on that same base (dashed to dot-dashed) reheats it by 2267 K and lifts the metastable peak by 2.5. All three solutions are cases of the model tree.

Table 16: The reservoir ladder under flux closure, 2026-09-16. One row per rung, at the iteration k where the 5% residual criterion was met; fluxes in 10^6 g s^{-1} , the cold-trap oxygen in 10^{-7} nuclei per H interpolated to the matching level. The equivalent width is on this memo’s red-pair window, directly comparable with the observed $1.108 \pm 0.030 \text{ \AA}$, and no rung reaches it from below. ε is the larger of the two elemental closure residuals. The first column is the reservoir the rung was started from, the sixth the one the closure returned.

start	k	ε	F_{H}	F_{He}	He/H at match	$\log_{10} \dot{M}$	red [%]	FWHM [$\text{\AA}$]	EW [% $\text{\AA}$]	O/H trap
2.09	5	0.026	26.34	54.42	2.0924	7.910	5.321	0.2603	1.4765	4.97
3	4	0.048	19.89	60.52	3.0034	7.900	6.156	0.2684	1.7640	5.52
5	4	0.043	13.37	66.32	5.0057	7.900	6.933	0.2805	2.0746	6.53
7	4	0.044	9.17	70.06	7.0080	7.900	7.102	0.2871	2.1768	7.77
8	4	0.045	8.11	70.98	8.0091	7.900	7.115	0.2896	2.2004	7.99
9	4	0.045	7.37	71.66	9.0103	7.900	7.111	0.2916	2.2137	8.86
9.7	4	0.045	6.88	72.10	9.7111	7.900	7.101	0.2931	2.2206	9.03
10	4	0.045	6.70	72.28	10.0114	7.900	7.094	0.2936	2.2224	9.06
12	4	0.045	5.56	73.36	12.0137	7.900	7.037	0.2966	2.2289	10.51
observed	-	-	-	-	-	-	$1.254^{+0.22}_{-0.23}$	0.841	1.108 ± 0.030	-

not set the composition there. What the ladder adds is the line.

There is no crossing on the ladder. The lowest rung solved, $\text{He}/\text{H} = 2.0924$, already makes $EW = 1.4765 \text{ \% } \text{\AA}$ against the measured 1.108 ± 0.030 : 33 per cent, that is 12σ , above it. The ladder rises from there and flattens near $2.22 \text{ \% } \text{\AA}$ above $\text{He}/\text{H} \simeq 8$, so the measurement is not reached anywhere on it. Read against the ladder's own local slope $d\ln EW/d\ln(\text{He}/\text{H}) = 0.49$ between the two lowest rungs, the measured equivalent width would be reached near $\text{He}/\text{H} \approx 1.1$; that is an extrapolation below every solved rung and is not offered as the answer. What the ladder does say is that the photochemical lower boundary no longer makes the line harder to reach than the metal-free one does; it makes it easier. The comparison at one composition is Table 15, and its cause is the density-resolved C, N and O cooling of Sect. 2.4.

The ladder is shallow and becomes shallower: the local slope runs 0.49, 0.32, 0.14, 0.08, 0.05, 0.04, 0.03 and 0.02 over the successive intervals, so above $\text{He}/\text{H} \simeq 7$ the line stops responding to composition. The saturation is a result in its own right: on this base the equivalent width is a weak lever on the reservoir at the helium-rich end, and a composition read there would carry a large error bar even if the ladder did cross.

At the lowest rung the escape rate is $\log_{10} \dot{M} = 7.91$, i.e. $8.1 \times 10^7 \text{ g s}^{-1}$, a factor 2.5 below the $2.03^{+0.58}_{-0.67} \times 10^8 \text{ g s}^{-1}$ the p-winds retrieval imposes, and the elemental fluxes are $F_{\text{H}} = 2.6 \times 10^7$ and $F_{\text{He}} = 5.4 \times 10^7 \text{ g s}^{-1}$.

Two separate statements follow.

- The closure by itself does not select a reservoir on this planet. Every rung of Table 16 satisfies it, at elemental residuals of 2.6–4.8% against the 5% tolerance. What used to select one was the line, as in Sects. 8. and 9.4; on the current code the line selects none of them, and the composition this base would need lies below the ladder.
- The escape rate is flat along the whole ladder, 7.90 to 7.91 in $\log_{10} \dot{M}$ over a factor 5.7 in reservoir, so here too the wind does not fix the composition. It agrees with the 7.84–7.87 of the prescribed-base branch to within 0.05 dex, so what is placed at the lower boundary is no longer visible in the escape rate either.

The elemental residual ε of Table 16 is the test the closure applies; the radial spread of the flux over the window it is measured on is 3.2×10^{-3} in hydrogen and 6.5×10^{-4} in helium.

The ladder does not fix the width. The line is still too narrow: FWHM 0.26–0.30 $\text{\AA}$ against the measured 0.841, unchanged from Sect. 9.4 and from the diffusion-off solutions. The winds these numbers come from also carry the caveats of Sect. 13. in full: their domain stops inside the critical point, and nothing in their profiles above about $10R_p$ is usable. The He 10830 layer is not affected: it sits at $\text{Kn} \leq 0.011$.

12.8 The outer region a converged wind inherits

Every number in this subsection is the 2026-08-30 measurement, read from LHS1140b/archive_20260830/. The mechanism it establishes does not depend on the physics that has changed since (Sect. 2.), but its magnitudes have not been re-measured on the current code.

The effect was found on the earlier ladder, at its 12.01 rung, and it is why the ladder of Table 16 is climbed from one common parent. That rung was first reached in one step from the 8.0 rung, a $1.50\times$ jump in reservoir; a control

run reached the same reservoir from 11.11, a $1.08\times$ step. The two solutions agree on everything the closure and the chemistry set, and disagree on the line:

- Same reservoir: He/H at the match 12.013522 against 12.013520, seven digits.
- Same lower atmosphere: cold-trap O/H 1.0656×10^{-6} against 1.0660×10^{-6} nuclei per H.
- Both converged: info = 0 at $\|R\| = 1.866 \times 10^{-4}$ (jump-seeded) and 1.287×10^{-4} (control), both inside the 4.0×10^{-4} target the rung was run to, and both inside the closure's own flux-spread tolerance.
- Different line: $EW = 1.4293$ against 1.2094% Å and a red depth of 4.789 against 3.956 per cent, an 18 per cent difference in equivalent width at fixed composition.

The difference sits entirely in the outer wind. Two diagnostics that do not depend on where a spectral window is placed separate the two solutions, and both are monotonic along the ladder except at the discarded point:

Table 17: Two sampling-free measures of the outer wind along the earlier closed ladder, from the advection-corrected profiles. r_{drop} is the first radius outside $1.5R_p$ at which T falls below half of $T(12R_p)$, and the last column is the share of the He 2^3S radial column that lies outside $10R_p$; every entry was recomputed for this table from the run directories, which hold these rungs re-solved one rung at a time on the current binary (LHS1140b/archive_20260830/exhale/refresh_j96/xuvkzz/). On the single-parent ladder of Table 16 the same two diagnostics give $r_{\text{drop}} = 20.46$ falling to $19.47R_p$ and an outer column share of 0.0091–0.0095, flat across every rung, with no rung standing out.

case	$r_{\text{drop}} [R_p]$	$T(12R_p)$ [K]	He 2^3S column outside $10R_p$
2.09	30.00	899	0.0117
3.0	27.63	1026	0.0096
5.0	23.76	1299	0.0084
8.0	20.80	1630	0.0083
9.7	19.79	1794	0.0090
10.3	19.79	1839	0.0091
10.7	19.47	1870	0.0092
11.1	19.47	1898	0.0092
12.0	19.47	1959	0.0095
12.0 discarded	25.83	1846	0.0195

r_{drop} moves inward monotonically as the reservoir rises, from 30.0 to $19.5R_p$, and the outer share of the metastable column stays in a narrow band, 0.0083–0.0117, across the whole ladder. The discarded solution departs from both: $25.8R_p$ where its own reservoir gives 19.5, and about twice the outer column share of any other rung (2.05 times its control's, 1.67 times the largest any other rung reaches). Its $T(12R_p)$ does *not* match its control's (1846 K against 1959, 5.8 per cent apart), so that quantity no longer shows that the two agree on the base. The agreement rests on the

matching-level composition, identical to seven digits, and on the profile ratio below. The discarded solution is a hotter, more extended outer atmosphere sitting on an identical base.

The mechanism appears to be that the outer state is inherited from the seed and the JFNK finish does not re-solve it. The density there is low enough that it contributes almost nothing to the residual norm, so $\|R\|$ cannot tell the two states apart. Both report $\text{info} = 0$ at residuals inside the target, and which of the two reports the smaller one says nothing about which is right: on the current branching, as on the one before it, the discarded solution reports the larger residual, and only on the retired Penning coefficient did it report the smaller. He 10830 is optically thick at these columns, so a hot, extended outer region fills the line wings and raises the equivalent width even though it carries about one per cent of the metastable column.

Measured on the metastable profiles, the two solutions agree to 2.3 per cent out to $11.9R_p$ and then separate abruptly (the departure passes 10 per cent at $11.95R_p$ and 50 per cent at 12.14, reaching a factor 5.1), so the region in which they differ is $12\text{--}30R_p$ rather than the $20\text{--}30R_p$ an earlier draft of this paragraph gave. The exobase of these two runs is also not the $18.8\text{--}25.9R_p$ of Sect. 13.: that is the three-case range of Table 19, whose cases are two non-closure runs and the 2.09 closure run. For this pair it is $11.81R_p$ (jump-seeded) and 14.73 (control).

Putting the corrected radii together: for the solution being discarded the whole differing region lies outside its own exobase, where $\text{Kn} > 1$; for its control the innermost $2.6R_p$ of that region is nominally still collisional. The fluid solution is therefore not valid in most of the region that makes the difference, and in the discarded solution it is not valid anywhere in it. The ground for discarding that solution is continuity of the trend rather than a physical criterion, and by the same token the absolute equivalent widths of the whole ladder carry a systematic error that the closure tolerance (a 5% flux spread) does not catch. Its size is small, because the outer region is about one per cent of the metastable column, but it is not zero. The exobase itself does not move with either correction: across the ladder it is $11.8\text{--}18.9R_p$ on the current branching against $11.8\text{--}18.8$ on the one before it.

Path independence was checked directly at $\text{He}/\text{H} = 3.0$, from a seed $3.7\times$ away in composition and in the downward direction the ladder never takes: it returns the same r_{drop} to the digit, with everything else inside the closure tolerance. The 5.0, 8.0 and 2.09 rungs were not checked this way. Separately, the 10.31 rung was re-run from the same seed in its own directory and reproduced bitwise, which fixes determinism but says nothing about the seed dependence. The rule the episode leaves is that a ladder scan must be climbed one step at a time; the record is docs/Update_EXHALE.md Sect. 83.

12.9 The 2025 non-detection

Cherubim et al. detect the line in 2024 and not in 2025, at a limit of 0.6 per cent in depth. The grid below asks what fraction of the fiducial XUV reproduces that. The spectrum is the fiducial GJ 1132 proxy with its flux column multiplied by the stated factor (`sed/lhs1140_sed_gj1132_at_b_xuv*.txt`; `read_sed` recomputes L_X and L_{EUV} from the file, so the luminosity lines of `input.inp` are provenance only). The lower atmosphere is held fixed in every case (the closure loop is not re-run, and neither is the chemistry that made the profile), so the grid isolates the wind's response to the irradiation.

It is run from both lower boundaries, since the two are not interchangeable: the scalar-base solution at $\text{He}/\text{H} = 2.13$ ($EW = 1.3749\% \text{ \AA}$) and the flux-closed solution at $\text{He}/\text{H} = 9.71$ ($EW = 2.2206$). The two families are the gj1132x* groups of models/, the scalar ones with `_kzz1e9` and the profile ones with `_kzzprofile`; every point is a case in the tree and is solved by the same route as the fiducial.

Table 18: The He 10830 line against the XUV scaling, from the two lower boundaries. Bold marks the row at which each model is nearest the 0.6 per cent depth limit of the 2025 epoch. On the corrected base boundary of Sect. 2.8 four of these cases no longer solve, and they are the gap in the closed column: at 0.15, 0.20 and 0.25 of the fiducial spectrum the hydrodynamic rows of the first and last cells refuse under a composition that is still travelling, and at 0.01 neither branch has ever solved (item L17). The 0.05 rung the previous draft carried for the closed branch was solved on the earlier boundary and is not quoted here. Since 2026-09-17 all seven of these cases are solved with the Roe flux rather than HLLC, for the reason and with the systematic given after the table; rows filled from a Roe solution carry it.

$F_{\text{XUV}} / \text{fiducial}$	scalar base, $\text{He}/\text{H} = 2.13$			flux-closed, $\text{He}/\text{H} = 9.71$		
	red [%]	$EW [\% \text{ \AA}]$	$\log_{10} \dot{M}$	red [%]	$EW [\% \text{ \AA}]$	$\log_{10} \dot{M}$
1.00	4.958	1.3749	7.87	7.101	2.2206	7.90
0.33	1.020	0.2555	7.36	3.821	1.0979	7.39
0.30	0.760	0.1894	7.32	3.542	1.0097	7.35
0.25	0.279	0.0700	7.24		not solved	
0.20	0.007	0.0022	7.13		not solved	
0.15	0.001	0.0003	7.00		not solved	
0.10	0.000	0.0001	6.82	1.001	0.2674	6.84
0.01		not solved on either branch (item L17)				

The seven cases this grid is missing are, since 2026-09-17, solved with the Roe flux instead of HLLC, on the same 500-cell grid and by the same recipe in every other respect. The HLLC flux does not reach the stationary root there at all, and the reason is the grid and not the wind: the same case solved with HLLC on a grid of twice the cells does reach it, and the state it reaches is the one the Roe flux reaches on the catalog grid, the two agreeing to 1.0 per cent in $\dot{M}$ and 0.4 per cent in the equivalent width. What the change of flux costs, where both fluxes solve and the comparison is therefore possible, is 0.3 to 0.6 per cent in $\dot{M}$ and in the equivalent width, both downward, and 6 to 8 per cent in the first cell, which comes out colder and denser; the mass-flux spread does not move. Those shifts are far smaller than the spacing of the grid, so they do not change what the grid is read for, and they are quoted wherever a Roe row is.

The scalar model meets the 0.6 per cent limit at 0.28 of the fiducial XUV; the closed one has not reached it on any scaling that is solved. In the scalar model the depth crosses 0.6 between 0.25 (0.279 per cent) and 0.30 (0.760), at 0.283 read linearly in both variables. On the paper's fiducial $F_{\text{XUV}} = 33 \text{ erg cm}^{-2} \text{ s}^{-1}$ that puts the 2025 epoch at about $9 \text{ erg cm}^{-2} \text{ s}^{-1}$ or less. The closed branch is still at 1.001 per cent at 0.10, the faintest scaling it has a solution for, and the three scalings between 0.10 and 0.30 are the ones that do not solve, so *its crossing lies below 0.10 and is not bracketed*. The figure draws the gap rather than a line across it.

The two branches no longer agree on this, as the 2026-08-30 grids did (both then at about 0.33). The closed branch

makes a much stronger line at every scaling it solves, which is the same statement Table 16 makes at the fiducial, so its limit is reached at a smaller fraction of the fiducial XUV than the scalar one's: below 0.10 against 0.28, a factor of at least 2.8. How much smaller is what the three unsolved scalings would settle.

The decline is also far steeper in the scalar model than in the closed one: over the 0.33 to 0.10 interval the scalar depth falls by a factor 3×10^3 and the closed one by 3.8. The *shape* of the XUV dependence is therefore a property of the lower boundary rather than a result, and it is a much larger difference than the 2026-08-30 grids showed.

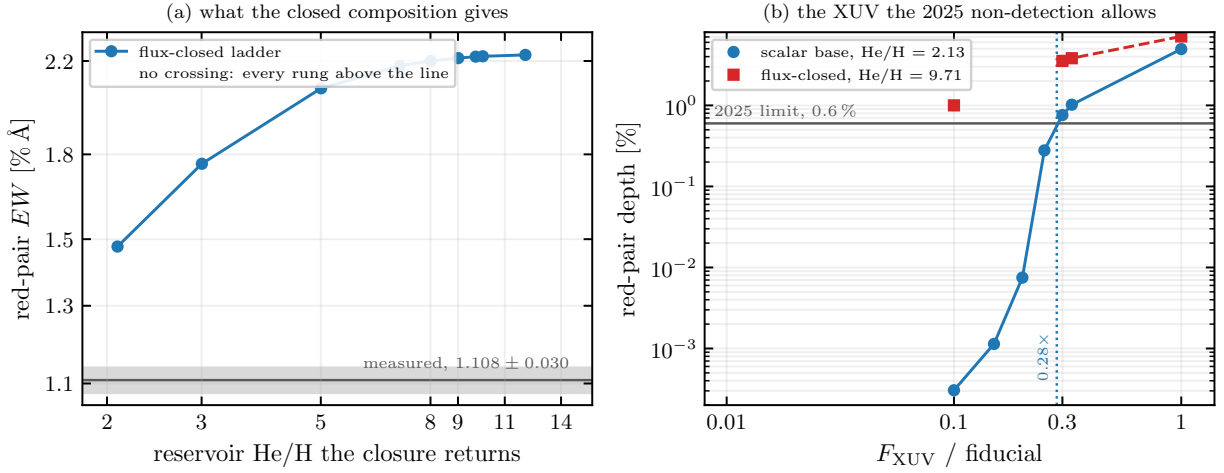


Figure 18: (a) The red-pair equivalent width of the flux-closed solution against the reservoir He/H the closure returned (Table 16), with the measured 1.108 ± 0.030 % Å as the band at the bottom of the panel; every rung stands above it, the lowest by 33 per cent, so no crossing is drawn. Every point satisfies the flux-closure condition, so the closure never picked a reservoir, and on the current code the line does not pick one either. (b) The red-pair depth against the XUV scaling (Table 18), from the scalar base at He/H = 2.13 and the flux-closed column at 9.71. The scalar model falls below the 0.6 per cent limit of the 2025 non-detection at about 0.28 of the fiducial irradiation (dotted). The closed model is drawn as one connected run over 0.30 to 1.00 and one isolated marker at 0.10, because the three scalings between them do not solve; its limit therefore lies below 0.10 and is not bracketed.

12.10 The width

Measured the way Sect. 9.2 measures it (convolve the solved profile with a Gaussian and ask which one reproduces the observed width), the lowest rung of the closure ladder asks for 22.15 km s^{-1} FWHM ($\sigma = 9.40 \text{ km s}^{-1}$) and its highest for 21.78, against 22.22 in the well-mixed limit, 22.22 with diffusion at the adopted K_{zz} , and 21.97 on the $5.1 \times$ brighter proxy. Those solutions span a factor 40 in composition and two different lower boundaries, and their kernels differ by 2 per cent. As solved, the red-pair FWHM along the ladder is 0.260–0.297 Å against the measured 0.841.

The lower-boundary axis therefore behaves like the other three: it moves the composition inference and the escape rate, and it does not move the width. It does not reach the measurement here either: at its matched kernel the lowest closed rung sits at a red depth of 1.675% against the observed $1.254^{+0.22}_{-0.23}$ and an equivalent width of 1.457 against 1.108 ± 0.030 , both above the band, because broadening conserves absorbing power and this solution has too much

of it. That is the mirror image of the p-winds case of Sect. 9.5, where the curve has too little: a convolution moves absorption in velocity and cannot change the amount.

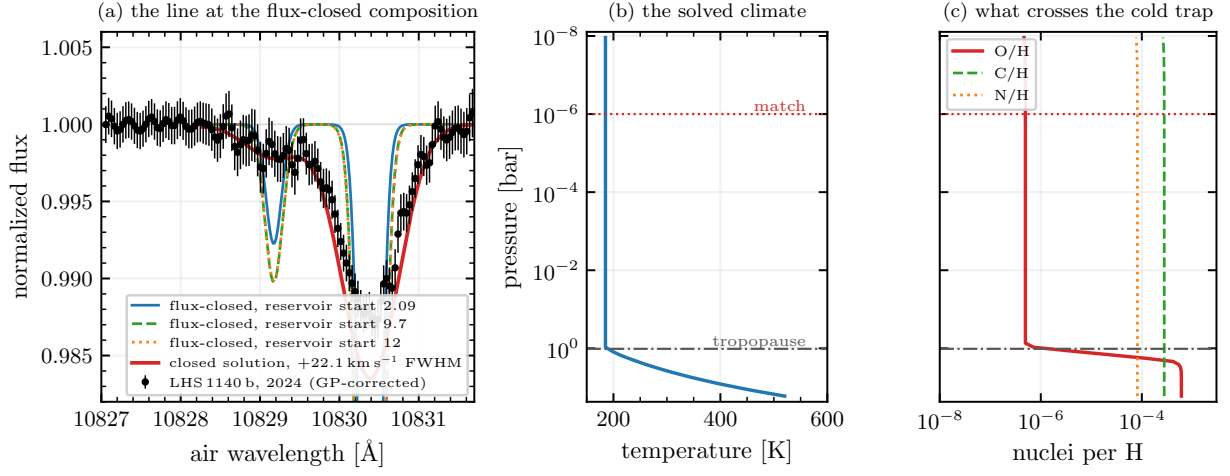


Figure 19: The flux-closed solutions and the column they stand on. (a) Three rungs of the closure ladder (started at reservoirs 2.09, 9.7 and 12) against the measurement; the thick curve is the lowest of them convolved with the 22.1 km s^{-1} FWHM Gaussian that reproduces the observed width, which then lands *below* the observed flux because the equivalent width was already 35 per cent too large. Model curves are shifted by the fitted $v_{\text{wind}} = +2.26 \text{ km s}^{-1}$, as elsewhere in this memo. (b) The solved climate of the column handed over, with the tropopause at 1.031 bar and the $1 \mu\text{bar}$ matching level marked; the stratosphere sits at the skin temperature, 185 K. (c) The elemental abundances the profile carries. Oxygen is cut by a factor 1223 at the tropopause by the water cold trap, while carbon and nitrogen cross it unchanged. On the current C, N and O cooling those elements no longer weaken the line (Sect. 2.4); on the 2026-08-30 solutions they made it $2.6\times$ weaker than the metal-free base at the same reservoir. Panels (b) and (c) place the column in the closure; every column of the file itself is in Figs. 14 and 15.

13. Is the wind collisional where it is set?

Everything above reads the solutions as continuum solutions. A flow is a hydrodynamic wind only where the gas is collisional on the scale over which it varies, and in particular through its critical point, where the topology and with it the mass flux are fixed. This section measures that. The tool is `src/utils/collisional_validity.py`, which reads an existing run directory and changes nothing; the definitions and tables are in `docs/collisional_validity.md`, and the record is `docs/Update_EXHALE` Sects. 81 and 85. The numbers below were measured on the outputs the rest of this memo uses, and nothing was re-run.

13.1 What is measured

The collisions. The four momentum-transfer limits are the ones the binary element-diffusion operator of Sect. 9.4 already uses, in the same Chapman–Enskog first approximation: rigid-sphere neutral–neutral, non-resonant ion–neutral

with the induced-dipole and rigid-core frictions added, and screened Coulomb with the Spitzer $\ln \Lambda$ for ion–ion and ion–electron. Sharing them keeps a mean free path quoted here and a diffusion coefficient used by the wind from drifting apart. The step from a binary diffusion coefficient to a collision frequency is the definition of that coefficient itself, $\nu_{st} = n_t kT / (\mu_{st} \hat{D}_{st})$ with $\hat{D} = nD$, in which the total carrier density cancels. Species are H I, H II, He I, He II, He III and e , plus the molecular carriers where a run tracks them; He I 2^3S is a level inside the He I column and is not counted twice, and metals are dropped as trace ($< 10^{-3}$ of the particles) except in the electron budget. Two channels are left out, resonant charge exchange and electron–neutral momentum transfer. Both lengthen the mean free path, so the verdict here is the conservative one.

The numbers. $\lambda_s = \bar{v}_s / \nu_s$ with the harmonic sum over partners; the local structure scale is the length over which the state vector of the local Maxwellian changes,

$$\frac{1}{L^2} = \left(\frac{d \ln \rho}{dr} \right)^2 + \left(\frac{d \ln T}{dr} \right)^2 + \left(\frac{|dv/dr|}{c_s} \right)^2 + \frac{1}{r^2}, \quad L = \max \left(\frac{1}{\sqrt{\cdot}}, \Delta r_{\text{cell}} \right), \quad (9)$$

and $\text{Kn}_s = \lambda_s / L$. The pressure is deliberately left out of it. The continuum closure expands about a local Maxwellian, which is fixed by (n, T, u) ; $p = nkT$ is a derived field, so including it would count the density and temperature gradients twice wherever they do not cancel and contribute nothing where they do. The second half of that is what matters here: across a heating front ρ falls and T rises with nearly the same logarithmic slope, so $|d \ln p / dr|$ passes through a broad near-zero while the gas is changing fast, and the earlier definition $L = \min(H_p, L_v, r)$ read a scale height fifteen times longer than the density and temperature scales there. The sum is taken in quadrature rather than as a minimum because a minimum jumps to the next length whenever the field it is taken from flattens; in quadrature a flat field contributes zero, and L still cannot exceed the shortest individual scale. The velocity enters as $|dv/dr|/c_s$ and not $|(1/v)dv/dr|$ because the term the closure drops is the viscous stress, whose size relative to the pressure is $\sim \lambda |dv/dr|/\bar{v}$: a bulk-velocity change distorts the distribution in proportion to the thermal speed, not to the local bulk speed. That removes the Mach-number floor the old L_v needed. The floor at the local cell width says that a gradient claiming structure below one cell is measuring the mesh; it binds in three cells of one of the four runs, all inside $1.002 R_p$, and in no critical region. The *exobase* is the first crossing of $\text{Kn}_{\text{bulk}} = 1$ from below, never extrapolated beyond the domain. The *critical point* is the sonic point taken with the code’s own sound speed, $c_s = \sqrt{\gamma p / \rho}$ at $\gamma = 5/3$, and not a re-definition. The *critical region* runs from the peak of volumetric heating to the sonic point. “Collisional” means $\max \text{Kn} < 0.1$ across that region, over the bulk and over every species carrying at least 10^{-3} of the mass. Where it fails the solution is reported as unvalidated rather than as overestimated, because a continuum solution that assumed collisionality away cannot decide which way a collisionless critical region moves the answer.

13.2 The result

The wind of LHS 1140 b does reach a critical point, and the domain stops short of it. All three solutions stay subsonic out to $29 R_p$, at $\mathcal{M}_{\text{max}} = 0.34\text{--}0.37$, so inside the domain the mass flux is set by the outer boundary condition and not by a critical point. Extending the domain settles what that costs (docs/lhs1140b_stationary_L8_20260913.md): at 45 and $60 R_p$ the same case crosses its critical point at 40.06 and $40.08 R_p$ and leaves the domain supersonic, and $\log_{10} \dot{M}$ is 7.8737 , 7.8740 and 7.8735 at 30 , 45 and $60 R_p$, a spread of 5×10^{-4} dex. The escape rate is therefore not an artifact

Table 19: Collisional validity of three representative LHS 1140 b solutions and of an HD 209458 b control, read from the advection-corrected profiles. Radii in R_p ; the LHS 1140 b domain ends at $29.03 R_p$, the HD 209458 b one at $4.15 R_p$. “Max Kn” is over the critical region, i.e. from peak heating to the sonic point, or to the outer boundary where there is no sonic point inside the domain. The three LHS 1140 b rows are cases of the model tree; the HD 209458 b control has no counterpart there and is the archived 2026-08-30 run.

case	critical point	exobase	Kn = 0.1 at	max Kn	verdict
well mixed, He/H = 0.55	none in domain, $\mathcal{M}_{\max} = 0.34$	above 29.0	7.30	1.03	unvalidated
diffusion, He/H = 2.13	none in domain, $\mathcal{M}_{\max} = 0.34$	above 29.0	7.44	1.00	unvalidated
flux-closed, He/H = 2.09	none in domain, $\mathcal{M}_{\max} = 0.37$	above 29.0	7.30	1.04	unvalidated
HD 209458 b control	4.085	above 4.15	-	0.018	validated

of where the boundary was put, but the profiles above $10 R_p$ are: they move by 25 per cent in temperature, 29 in density and 22 in velocity at $28 R_p$, and nothing is quoted from them.

What remains unvalidated is the *continuum* description out there. Kn_{bulk} passes 0.1 at $7.30\text{--}7.44 R_p$ and reaches 0.68–0.71 at the top of the $30 R_p$ domain, and the exobase of the extended solutions sits at $29.0 R_p$, *below* the critical point at 40.06 . The continuum equations therefore fix their own topology in gas they cannot describe. The escape rates quoted throughout this memo, $\log_{10} \dot{M} \simeq 7.8\text{--}7.9$, are unvalidated in the collisional sense, and their insensitivity to the domain size is a statement about the discrete problem rather than a physical validation.

The p-winds retrieval sits in the same regime. Its isothermal Parker solution places its sonic point at $8\text{--}9.5 R_p$, where Kn_{bulk} is 0.12–0.17, past the threshold. The critical region of LHS 1140 b is therefore *transitional*: it is not collisional, and it does not lie above the exobase either. The scale separation the continuum equations need has been lost by the radius at which either code sets the mass flux, while the gas is not yet free molecular.

Which species sets it. Neutral hydrogen decides the bulk number, and the line-forming layer is unaffected. In the critical region of the well-mixed solution the species split is H I 1.03, He I 0.73, H II 3.6×10^{-5} , He II 1.8×10^{-5} , He III 4.7×10^{-6} , e 6.5×10^{-4} (Fig. 20b). The neutrals carry the mass and the long mean free path, and the charged species are Coulomb-held. That matters for reading the rest of this memo, because the He 10830 metastable population forms at $1.1\text{--}3 R_p$, where Kn_{bulk} is $1.5\text{--}2.4 \times 10^{-4}$ at $1.1 R_p$, 0.0033–0.0036 at $2 R_p$ and 0.0110–0.0115 at $3 R_p$ in every case here. The line-forming layer is collisional by two orders of magnitude. The unvalidated quantity is the wind solution’s mass flux. The absorption that Sects. 8.–9. measure, and the 22 km s^{-1} of missing width, are asked for at radii where the continuum description is secure.

The energy equation. The single-temperature energy equation is not where the description fails. Electron–ion energy coupling is fast wherever it matters: the largest $\tau_{ei}^E / \tau_{\text{heat}}$ over the critical region is 0.027, 0.024 and 0.026 in the three LHS 1140 b solutions and 8.3×10^{-5} in the control. Momentum coupling times are likewise short compared with the flow time. That is a different test, not a contradiction: a strongly subsonic flow gives hundreds of collisions per flow time even at $\text{Kn} \sim 1$. The Knudsen number is what decides continuum validity, because the terms the closure drops

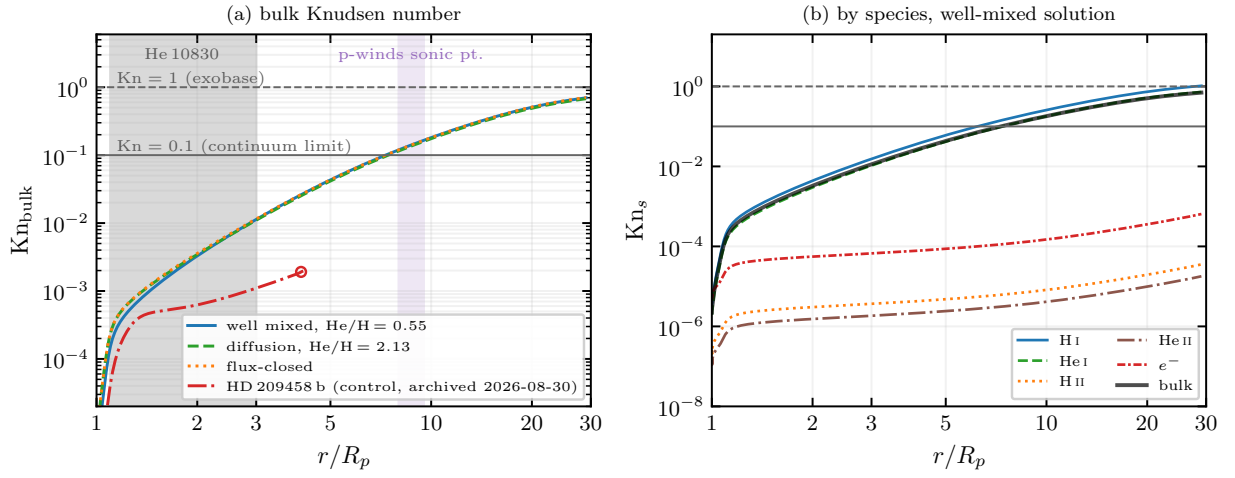


Figure 20: Where the continuum description holds. (a) The bulk Knudsen number of the three representative LHS 1140 b solutions and of the HD 209458 b control, whose sonic point at $4.085 R_p$ is the open circle; the shaded bands are the He 10830 line-forming region, $1.1\text{--}3 R_p$, and the $8\text{--}9.5 R_p$ at which the p-winds retrieval places its sonic point. The LHS 1140 b solutions cross $Kn = 0.1$ at $7.30\text{--}7.44 R_p$ and reach $0.68\text{--}0.71$ at the top of the domain, while the control stays below $Kn_{\text{bulk}} = 3 \times 10^{-3}$ everywhere and is collisional through its critical point, over which its largest Kn is 0.018 against $1.00\text{--}1.04$ for the LHS 1140 b solutions, a factor of about 55. (b) The species split of the well-mixed solution: neutral hydrogen and neutral helium set the number, while the ions are held by Coulomb collisions four to five orders of magnitude more tightly and the electrons three.

(viscous stress, heat flux, the non-Maxwellian tail) scale with the gradient over a mean free path rather than with the collision count.

The kinetic scale. Jeans escape through the top of the domain (Chamberlain & Hunten 1987, eq. 7.2.5) gives $4.5\text{--}4.7 \times 10^7 \text{ g s}^{-1}$, a factor 1.5–1.7 below the continuum $\dot{M}$. That is a *scale*, not a bound. Jeans escape bounds a continuum rate from below only when the exobase is well bound, and here $\lambda_J = 0.46\text{--}0.48$: the outer atmosphere is barely gravitationally bound, the Jeans integral is most of the Maxwellian rather than its escaping tail, and the atmosphere is in blow-off. What the number does say is that the continuum result sits within a factor of two of what a collisionless outer atmosphere at the same density and temperature would lose, so the failure of the assumption is not hiding an order of magnitude. On HD 209458 b, where $\lambda_J = 10.6$ and the solution is validated through its critical point, the hydrodynamic rate exceeds Jeans by 10^4 : the signature of a driven wind rather than an evaporating atmosphere.

13.3 What this changes, and what it leaves open

- The domain stops inside the critical point, and the region it stops in is not collisional. The extended-domain runs put the critical point at $40.06 R_p$ and the exobase at 29.0 , so the topology that sets the mass flux is fixed in gas the continuum equations do not describe. That the flux itself moves by 5×10^{-4} dex between 30 , 45 and $60 R_p$ bounds the discretization error and not the physical one. Any $\dot{M}$ quoted for LHS 1140 b from these solutions has to carry that warning, and so does the p-winds number, whose sonic point sits at $\text{Kn} = 0.12\text{--}0.16$.
- The outer profiles are not usable, and the residual cannot say so. At $28 R_p$ the $30 R_p$ solution stands 25 per cent too hot, 29 per cent too dense and 22 per cent too slow against the converged answer, at a certified residual. Nothing above about $10 R_p$ in these profiles is quoted.
- The rate itself is not settled here, and cannot be. Settling it would take a kinetic or transitional-flow calculation (DSMC, or a 13-moment closure carried through the transitional region), and this repository contains neither. It is recorded as open, not estimated.
- The line analysis stands. The He 10830 layer is collisional by two orders of magnitude, so the equivalent-width matching, the composition inference conditional on it, and the invariant 22 km s^{-1} width deficit are unaffected by this section. It constrains what the escape rate may be quoted as, not what the line says.

14. Status against the goal

The line is not reproduced. What stands:

- Amount of absorption: matched over a prescribed metal-free scalar base, and no longer matched over a solved photochemical column. Over the scalar base the answer is $\text{He}/\text{H} = 1.5161$ at the adopted $K_{zz} = 10^9 \text{ cm}^2 \text{ s}^{-1}$, and inside that branch it is further conditional: 0.0526 well mixed on the GJ 699 proxy, 0.4044 well mixed on the GJ 1132 proxy, 3.10 wherever $K_{zz} \leq 10^5$ leaves the molecular coefficient alone. Every one of those is read inside a bracket of two solved rungs. Over the photochemical column every closed rung from $\text{He}/\text{H} = 2.09$ to 12.01

stands *above* the measurement, the lowest by 33 per cent, so that branch has no crossing (Sect. 12.7). The match, where it exists, is conditional: it constrains the composition only as far as the assumptions behind it are secure.

- Everything here was re-solved twice, and one of the two branches reversed. Against 2026-08-30 the prescribed-base crossings moved by 2 to 19 per cent and $\dot{M}$ by about 0.1 dex, and the solved-column branch went from making the line $2.6\times$ weaker than the metal-free base at the same composition to making it 7.4 per cent stronger; the cause is the C, N and O line cooling, which was evaluated in the coronal limit above the critical density of the levels it excites and is now the CHIANTI statistical equilibrium at the cell's own electron density (Sect. 2.4). The first re-solve was needed because the 2026-08-30 states were never stationary below $1.1R_p$, and it was possible because of the well-balanced reconstruction. The second was needed because the base boundary chose its entropy branch on the cell-1 velocity rather than on a flux, so the campaign of 2026-09-14 carried a 418 K first cell where base.inp states 226 K (Sect. 2.8). That correction takes the base cell down by a median 46 per cent in temperature, $\dot{M}$ down by 2 per cent everywhere, and the crossings *up* by 1 to 14 per cent.
- Line shape: not matched, by a factor 3.3 in depth and width alike, and not fixable by composition. The missing ingredient is 22.22 km s^{-1} FWHM of velocity spread ($\sigma = 9.44 \text{ km s}^{-1}$), measured in Sect. 9.2 and Fig. 5; supplied as one Gaussian it puts depth, blue depth and red/blue ratio inside the observed 1σ interval at once, but its physical origin is unidentified. The demand is unchanged under diffusion and the adopted K_{zz} : measured the same way on the EW-matched diffusive solution it is the same 22.22 to four digits (Sect. 9.4); it is 22.15 km s^{-1} at the lowest flux-closed rung (Sect. 12.7); and it is 21.97 km s^{-1} on the $5.1\times$ brighter proxy spectrum (Sect. 11.), 1.1 per cent of spread over four solutions that differ by a factor 40 in composition. The adiabatic metastable bump, which would have supplied it without a free parameter, is present and measured (Sect. 9.6) but sits too slow to contribute.
- The width shortfall is not a measurement artifact, and it is not this code's alone. Audited term by term, the instrument profile, the triplet splitting, stellar rotation, planetary rotation and orbital smearing account for at most 0.4 km s^{-1} of it; deconvolving both sides from their own instrument kernels puts the like-for-like requirement at 22.00 km s^{-1} FWHM (Sect. 9.3, docs/lhs1140b_width_measurement_audit.md). The paper's own p-winds fit, measured the same way, is itself 1.5 times narrower than the width printed in the same paper, so the shortfall appears to be a property of one-dimensional steady-state wind models rather than of this one.
- Escape rate: an output, not a fit, and set by the SED. $\log_{10}\dot{M} = 7.84\text{--}7.87$ on the GJ 1132 proxy, a factor 2.7 below the value the retrieval imposes, and nearly independent of composition, of diffusion, of the eddy coefficient and now also of what is placed at the lower boundary (the closed ladder gives 7.90–7.91). On the GJ 699 proxy, $5.1\times$ brighter in XUV, it is 8.55–8.57: within 0.06 dex of the value the paper retrieves on that same proxy (Sect. 11.).
- That escape rate is unvalidated in the collisional sense; the line is not. The solutions are the subsonic part of a transonic escape whose critical point sits at $40.06R_p$, outside the $30R_p$ domain, and whose exobase sits at $29.0R_p$, below the critical point. Moving the boundary to 45 and $60R_p$ changes $\log_{10}\dot{M}$ by 5×10^{-4} dex and the equivalent width by at most 1.7 per cent, but changes the profiles at $28R_p$ by 22–29 per cent, so nothing above about $10R_p$ is quoted. The p-winds sonic point at $8\text{--}9.5R_p$ sits at $\text{Kn} = 0.12\text{--}0.17$, so the critical region of this

planet is transitional for both codes. The He 10830 layer at $1.1\text{--}3R_p$ is collisional by two orders of magnitude ($\text{Kn} \leq 0.012$), and an HD 209458 b control validates cleanly at a factor 56 lower Kn (Sect. 13.). What the true rate is needs a kinetic or transitional-flow calculation, which this repository does not contain.

- Composition: limited by the base and by the SED, and weakly limited by K_{zz} . In the well-mixed limit the equivalent-width crossing is $\text{He}/\text{H} = 0.4044$ on one proxy spectrum and 0.0526 on the other, helium-rich against helium-poor, with the measurement unable to separate them. With binary H/He diffusion active the crossing rises, and the whole K_{zz} axis has been walked: $\text{He}/\text{H} = 3.10$ on the plateau at $K_{zz} \leq 10^5$, then 2.89, 2.54, 1.97, 1.52, 1.16 and 0.89 at 10^6 through $10^{11} \text{ cm}^2 \text{ s}^{-1}$ (Table 5, Fig. 6), a factor 1.7 across the range the literature leaves open and 3.5 across the entire axis, against the factor 7.7 the SED carries. So any composition read off this line, the paper's included, is strongly conditional on the assumed XUV spectrum, and weakly conditional on the eddy coefficient, as $K_{zz}^{-0.115}$ wherever the eddy term acts at all.
- The lower boundary is still an axis, and it now points the other way. With the lower atmosphere solved as a photochemical column and the elemental fluxes iterated to continuity across the $1 \mu\text{bar}$ match, He/H at the match is a solution rather than an input, and at $K_{zz} = 10^9$ that solution is the well-mixed value because the eddy term reaches the match (Sect. 12.). What the column brings with it is C, N and O, and on the current cooling those elements cost the wind 164 K in T_{max} and the line 0.20 per cent of its equivalent width at fixed composition (Table 15). The closed ladder therefore sits above the measurement everywhere, and the composition this base would need lies below every rung solved.
- The 2025 non-detection sits at about 0.28 of the fiducial XUV over the scalar base; over the photochemical column it is not yet bracketed. Scaling the irradiation with the lower atmosphere held fixed, the red-pair depth falls below the 0.6 per cent limit of the 2025 epoch at $\simeq 0.279$ of fiducial in the scalar-base model, about $9 \text{ erg cm}^{-2} \text{ s}^{-1}$ (Sect. 12.9, Table 18). The flux-closed grid is still at 1.001 per cent at 0.10, the faintest scaling it solves, so its limit lies below 0.10 and is not bracketed: on the corrected base boundary the three scalings between 0.10 and 0.30 no longer solve, and the 0.01 points of both branches have never solved (item L17).
- Not verified: the second-chemistry-code cross-check. VULCAN did not converge on this cold helium-rich column, its convergence measure rose against the criterion with the timestep pinned, the limiting cell being H_2O at the cold trap, so no composition from a second code, and no cross-code spread, is quoted (Sect. 12.).
- Partly solved: the molecular lower boundary. Three groups of the model tree solve the molecular layer in EXHALE itself rather than taking it from the column, and two of their cases now certify, both with element diffusion at $K_{zz} = 10^9$: $\text{He}/\text{H} = 2.13$ at $\log_{10} \dot{M} = 7.91$ with a red-pair $EW = 1.5758 \% \text{ \AA}$, and 9.7 at 7.95 with $2.4212 \% \text{ \AA}$. Both stand above the measurement, as the flux-closed rungs do. The other rungs of those groups are uncertified, a marching stop, or in flight at the time of writing, so no crossing is read from a molecular group here. What they wait on is the H_2 carrier balance of item L7e.

Four axes have now been walked: the assumed XUV spectrum, the mixing assumption with its eddy coefficient, composition itself, and what is placed at the lower boundary. Each separates the two questions rather than answering them

together. All four move the composition the equivalent width implies, two of them by a large factor, and none of them moves the width. The possible sources of the missing 22 km s^{-1} are therefore unchanged: velocity fields that are three-dimensional, or heating a 1D hydrodynamic energy equation does not carry. The adiabatic-cooling metastable bump, the candidate that would have needed no free parameter, has been looked for on both spectra and does not supply it for this host (Sect. 9.6).

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